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Review article

Pre-combustion fuel reforming in gas turbines: Hydrocarbon and carbon-neutral fuel pathways

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ABSTRACT

Gas turbines are a cornerstone of modern power generation, providing reliable and dispatchable electricity across the globe. However, their significant contribution to global CO_2 emissions has become increasingly untenable under international climate agreements such as the Paris Agreement. This creates a critical imperative to develop technologies that can improve the efficiency and sustainability of existing and future gas turbine fleets without compromising their operational flexibility. While the long-term vision is a fully renewable energy system, the intermittent nature of sources like wind and solar necessitates the continued use of thermal power for grid stability in the foreseeable future. Within this context, the transition to low-carbon and carbon-neutral fuels – such as hydrogen, ammonia, and methanol – has emerged as a key decarbonization pathway. However, the direct use of these fuels in conventional gas turbines is hampered by challenges related to combustion stability, flame speed, NO_x formation, and, in some cases, fuel infrastructure and storage.

Pre-combustion fuel reforming offers a transformative solution to these challenges. By harnessing waste heat from the turbine exhaust – or external, low-grade heat sources like solar thermal energy – to drive endothermic reforming reactions, primary fuels are converted into hydrogen-rich gas prior to combustion. This process of the use of exhaust heat to drive pre-combustion fuel reforming, known as thermochemical recuperation (TCR), achieves a dual benefit: it upgrades low-grade thermal energy into high-value chemical energy stored in the reformate, thereby enhancing the overall cycle efficiency, and it produces a fuel with superior combustion properties, including higher flame speed and wider flammability limits. Gas turbines implementing this strategy are termed Chemically Recuperated Gas Turbines (CRGTs).

While CRGT upgrades fuel via waste heat recuperation – thereby increasing the share of work output from the more efficient gas turbine cycle – an alternative hybrid pathway integrates a fuel cell with a gas turbine (FC-GT), where the fuel cell converts the majority of the fuel's chemical energy electrochemically. This fundamental difference enables FC-GT systems to achieve even higher electrical efficiencies (60%–70%). This review provides a comparative analysis between FC-GT systems and CRGT, focusing on their primary constraints.

This paper presents a comprehensive and critical review of the potential of pre-combustion reforming to redefine the role of gas turbines in a low-carbon energy future. The analysis is structured around three primary fuel pathways: natural gas, where steam methane reforming offers a route to significant efficiency improvements; ammonia, where pre-combustion cracking is critical for achieving stable and efficient combustion with a carbon-free footprint; and methanol, which acts as a convenient hydrogen carrier and offers a carbon-neutral pathway when produced from renewable sources. For each fuel, the review examines the underlying thermodynamics, catalyst development, reformer design, combustion characteristics of the resulting hydrogen-rich gas, and integration into various gas turbine cycles — from simple and reheat configurations to advanced combined cycles.

Furthermore, the work explores the innovative integration of concentrated solar thermal energy to drive the pre-combustion fuel reforming process, creating solar-hybrid gas turbine systems that offer a dispatchable, partially renewable power generation solution. Key technical hurdles are identified and discussed, including

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catalyst durability under moderate operating conditions, combustion dynamics and NO_x control for hydrogen-rich fuels, materials challenges in high-temperature reformers, and the techno-economic viability of scaling such integrated solar-gas turbine systems. By synthesizing decades of research from fundamental kinetics to system-level analysis, this review aims to provide a clear roadmap for future development. It concludes that pre-combustion fuel reforming is not merely an incremental improvement but a pivotal enabling technology that can accelerate the decarbonization of the power sector by making gas turbines compatible with a sustainable, fuel-flexible, and high-efficiency future.

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1. Introduction

Global power generation is at a pivotal juncture, tasked with reconciling rising energy demand with the urgent need to decarbonize [1,2]. Gas turbines, which currently form the backbone of reliable electricity supply and grid stability in many regions, are responsible for a significant portion of power sector emissions [3]. According to the International Energy Agency (IEA), electricity and heat generation accounted for approximately 40% of global energy-related CO_2 emissions in 2022, with natural gas-fired power plants being a major contributor [4]. The continued operation of these assets is critical for energy security, yet their environmental impact is untenable under stringent climate targets, such as the Paris Agreement's goal to limit global warming to 1.5°C [5]. This creates a powerful imperative to develop and deploy transformative technologies that can drastically improve the efficiency and sustainability of existing and future gas turbine fleets without compromising their operational flexibility and reliability.

While the transition to a fully renewable energy system is the ultimate goal, the intermittency of sources like solar and wind necessitates the continued use of dispatchable thermal power for the foreseeable future [6–8]. Two main strategies for reducing CO_2 emissions in existing and newly designed power plants based on gas turbines can be highlighted. The first is the improvement of overall thermal efficiency, as an increase in efficiency directly reduces CO_2 emissions per watt of produced electricity. The second is the use of either carbon-free or low carbon-footprint fuels – such as ammonia, hydrogen, methanol, ethanol, and e-methane – in gas turbines [9–12].

A promising pathway to address both of the aforementioned strategies is pre-combustion fuel reforming in gas turbines. This process involves the thermochemical conversion of primary fuels (e.g., natural gas, methanol, or ammonia) into a hydrogen-rich gas prior to combustion [13–15]. The energy required for this endothermic conversion can be sourced from the gas turbine's own exhaust heat [16,17] or from external, low-grade heat sources such as solar thermal energy [18,19], thereby enabling novel, hybrid energy cycles. The integration of pre-combustion fuel reforming can enhance gas turbine operation through several key mechanisms:

- Thermal efficiency gains: The endothermic reforming reaction recuperates thermal energy that would otherwise be lost, improving the overall cycle efficiency of the gas turbine system.
- Combustion efficiency and stability: The high reactivity and fast flame speed of hydrogen improve combustion efficiency and lean blow-out limits, which is particularly beneficial for stabilizing combustion when using challenging carbon-free fuels like ammonia.
- Fuel flexibility: Reforming provides a pathway to utilize a wider range of fuels efficiently by converting them into a more reactive hydrogen-rich mixture.

This review provides a comprehensive overview of the potential of pre-combustion reforming to redefine the role of gas turbines in a low-carbon energy future, with a specific focus on its impact on thermal efficiency across various gas turbine cycles. The analysis centers on three primary fuels: natural gas, where reforming offers a route to significant efficiency improvements; ammonia, where it is critical for achieving stable and efficient combustion with a carbon-free footprint when produced from renewable sources (e.g. "green" ammonia); and methanol, which can act as a hydrogen carrier and, when green, offers a carbon-neutral pathway. By examining the integration of reforming technologies – utilizing both exhaust gas and external heat sources – this work aims to synthesize the state-of-the-art in leveraging fuel reforming to push the boundaries of gas turbine performance and sustainability.

The scope of this review is defined by the following criteria. First, the review focuses exclusively on pre-combustion fuel reforming processes that are thermally driven by waste heat from the gas turbine exhaust or by external heat sources (e.g., solar thermal energy). Second, only endothermic reforming reactions that produce hydrogen-rich gas streams suitable for direct combustion in gas turbine combustors are considered, specifically steam methane reforming, ammonia cracking, and methanol reforming. Third, the review covers system-level integration of reformers with gas turbine cycles, including simple, recuperated, combined, and solar-hybrid configurations, but does not address in detail the design of individual reactor components or combustion chamber aerodynamics. Fourth, the review is limited to gaseous and liquid fuels that can be reformed at temperatures compatible with gas turbine exhaust heat (500–800°C), namely natural gas, ammonia, and methanol. Finally, the review emphasizes thermodynamic performance, combustion characteristics, and system-level efficiency, while catalyst development and reactor design are discussed only insofar as they impact these system-level metrics. This scope definition provides the reader with a clear rationale for the selection of topics covered and technologies analyzed in the review.

2. Challenges for gas turbines

2.1. Thermal efficiency

The thermal efficiency of a gas turbine is highly dependent on its configuration, placing it in a unique position relative to other heat engines [20,21]. In a simple cycle configuration (where the exhaust is vented directly), a modern heavy-duty gas turbine achieves an efficiency of 35%–44%, which is generally lower than large marine diesel engines (~50%) and supercritical coal power plants (40%–45%) [22–25]. This lower simple-cycle efficiency is primarily due to the significant energy rejected as high-temperature exhaust gas, a fundamental characteristic of the open Brayton cycle. However, this apparent weakness becomes its greatest strength. By channeling the hot exhaust into a Heat Recovery Steam Generator (HRSG) to power a steam turbine, a combined cycle gas turbine (CCGT) plant can achieve efficiencies of 60%–64%, making it the most efficient large-scale heat engine for stationary power generation available today [26–28].

Despite the progress in the development of Combined Cycle Power Plants (CCPP), it is important to note that the exhaust heat from the gas turbine is utilized in a steam turbine cycle, which inherently has a lower thermal efficiency than the gas turbine cycle itself. A promising direction for improving overall efficiency involves recovering this exhaust heat back into the gas turbine cycle [29]. Modern gas turbines achieve high thermal efficiency by operating at turbine inlet temperatures (TIT) of approximately 1600°C [30]. However, these high TITs also result in high turbine exhaust gas temperatures (TEGT), which can exceed 600°C in advanced machines. The thermal performance parameters of several modern gas turbine types are summarized in Table 1.

The conventional exhaust heat recovery methods [31], e.g., air preheating before combustion, are not appropriate for gas turbines because the air temperature after the compressor is high and the temperature difference between hot exhaust gases and compressed air is minimal, which leads to an extremely large heat transfer area [32,33]. The use of exhaust heat to preheat fuel before combustion also has low practical implementation due to the low mass flow rate of fuel compared to the mass flow rate of exhaust gases [34,35].

A highly effective method for achieving a high degree of waste heat recovery in gas turbines is thermochemical exhaust heat recuperation (TCR) [36–39]. In this approach, exhaust heat drives the endothermic pre-combustion reforming of the primary fuel. Gas turbines implementing this strategy are known as chemically recuperated gas turbines (CRGTs) [40–42].

2.2. Transition to fuels with low carbon footprint

The transition to low-carbon fuels like hydrogen (H_2), ammonia (NH_3), and methanol (CH_3OH) is a critical pathway for decarbonizing gas turbine power generation [9,43–45]. Each presents a unique blend of promise and challenge. Hydrogen is the most prominent, offering the ultimate goal of carbon-free combustion when produced from renewable sources. However, its use requires significant modifications to combustion systems to manage high flame speeds and flashback risks, and it can increase NO_x emissions due to high flame temperatures [46]. Ammonia is a carbon-free hydrogen carrier but faces major combustion challenges, including low reactivity, difficult ignition, and high fuel-bound NO_x production, requiring advanced combustor designs [47]. Methanol, a liquid fuel, offers easier handling and storage but still produces some CO_2 emissions (though less than conventional fuels) and can lead to incomplete combustion and formaldehyde formation [48].

The overarching challenge for gas turbine operators is to redesign fuel injection, combustion, and fuel delivery systems to accommodate these fuels' vastly different chemical and combustion properties while maintaining stability, minimizing emissions, and ensuring hardware durability. Despite these hurdles, extensive research and development, including GT testing by major manufacturers, is underway, positioning these fuels as essential components for a future where flexible, dispatchable power generation can coexist with deep decarbonization goals.

Hydrogen

Hydrogen holds one of the most immediate promises for carbon-free power, and its use is being actively demonstrated. Major manufacturers like GE Vernova, Siemens Energy, and Mitsubishi Power have GTs capable of operating on high-hydrogen blends [49–51]. For instance, the list of existing hydrogen-fueled gas turbines:

Table 1
Heavy-duty industrial gas turbine specifications.

Manufacturer & Class	TIT	TEGT	Thermal efficiency
General Electric			
F-Class	1350–1430°C (2460–2600 °F)	590–620°C (1100–1150 °F)	38%–40%
H-Class	1430–1500°C (2600–2730 °F)	610–640°C (1130–1185 °F)	40%–42%
HA-Class	~1600°C (~2910 °F+)	610–640°C (1130–1185 °F)	43%–44%+
Siemens Energy			
H-Class (SGT5/6-8000H)	~1500°C (~2730 °F)	620–640°C (1150–1185 °F)	40%–42%
Adv. H-Class (SGT5-9000HL)	>1500°C (>2730 °F)	620–640°C (1150–1185 °F)	42%–43%
Mitsubishi Power			
G-Class	~1500°C (~2730 °F)	~610°C (~1130 °F)	39%–41%
J-Class	~1600°C (~2910 °F)	630–650°C (1165–1200 °F)	43%–44%
Ansaldo Energia			
GT26	>1350°C (1st) >1500°C (2nd)	620–640°C (1150–1185 °F)	40%–41.5%
GT36	>1600°C (2nd) (>2910 °F)	620–640°C (1150–1185 °F)	43%–44%

Note: Thermal efficiency values represent simple cycle operation. Combined cycle efficiencies are significantly higher (typically 60%–64% for modern H/J/HA-class turbines). Actual efficiency depends on specific model, ambient conditions, and operational parameters.

- Ohio, USA: A Long Ridge Energy Center turbine has successfully demonstrated operation on a 38% hydrogen (by volume) blend with natural gas [44];
- France: Siemens Energy SGT-400 gas turbine has successfully demonstrated operation on 100% green hydrogen in an industrial power-to-hydrogen-to-power plant [44];
- Kobe, Japan: A Kawasaki Heavy Industries 1 MW gas turbine has successfully demonstrated operation on 100% hydrogen for urban combined heat and power generation [44];
- Leipzig Süd CHP, Germany: The plant is equipped with Siemens Energy SGT-800 turbines that are 100% hydrogen-ready for future operation [52].

However, the widespread adoption of pure hydrogen faces a operational challenge: the lack of a pervasive hydrogen supply infrastructure [53,54]. Producing green hydrogen via electrolysis requires massive renewable capacity, while transporting and storing gaseous hydrogen demands new, dedicated pipelines and facilities, which are capital-intensive [55–57]. Consequently, the most common and practical near-term strategy is the use of hydrogen-natural gas blends [58–60]. This approach allows for a gradual reduction in carbon emissions while utilizing existing natural gas infrastructure, with the hydrogen content typically limited to 20%–30% by volume to avoid major modifications. However, hydrogen and natural gas – which consists mainly of methane – have different physical and thermochemical properties. Consequently, the addition of hydrogen to methane results in a nonlinear change in the properties of the resulting hydrogen-enriched natural gas, as illustrated in Fig. 1.

Ammonia and methanol

Ammonia and methanol are attractive as hydrogen carriers due to their easier liquefaction and transport compared to pure hydrogen. Moreover, CH_3OH and NH_3 can be considered as gas turbine fuels with low carbon footprint [2,61]. For instance, the list of existing ammonia and methanol fueled gas turbines:

- Germany (ammonia): Turboden ammonia-fired gas turbinewas installed in a cogeneration plant in Germany in 2018. It has a capacity of 15 MW and is designed to operate on pure ammonia or a mixture of ammonia and natural gas. The turbine was manufactured by Turboden, an Italian company that specializes in organic Rankine cycle (ORC) and gas turbine technologies [9];
- Japan (ammonia): Kawasaki Heavy Industries has developed a 1 MW ammonia-fired gas turbine that is currently being tested in Japan. The turbine is designed to operate on pure ammonia or a mixture of ammonia and natural gas. Kawasaki Heavy Industries is a Japanese company that produces a wide range of industrial equipment, including gas turbines [9];
- Sweden (ammonia): Siemens has developed an ammonia-fired gas turbine that is currently being tested in Sweden. The turbine has a capacity of 2 MW and is designed to operate on pure ammonia or a mixture of ammonia and natural gas. Siemens is a German company that produces a wide range of industrial equipment, including gas turbines [9];
- Israel (methanol): The Israel Electric Company converted a 50 MW Pratt & Whitney turbine to run on methanol in 2014, citing cost-effectiveness and reduced emissions compliance [62];
- Germany (methanol): A joint project is demonstrating the load-following operation of a power-to-methanol technology at the Lünen power plant, with full commissioning scheduled for 2017, according to Carbon Recycling International [63].

However, the combustion properties of ammonia and methanol present severe challenges for direct use in standard gas turbines:

- Ammonia's combustion is hampered by a low laminar flame speed and a large quenching distance, making the flame unstable and difficult to sustain, leading to low combustion efficiency and a high risk of flame-out. Furthermore, its fuel-bound nitrogen is a direct source of fuel-NO_x, which is challenging to control [64,65].
- Methanol, while easier to burn than ammonia, can suffer from incomplete combustion at lower temperatures, leading to the

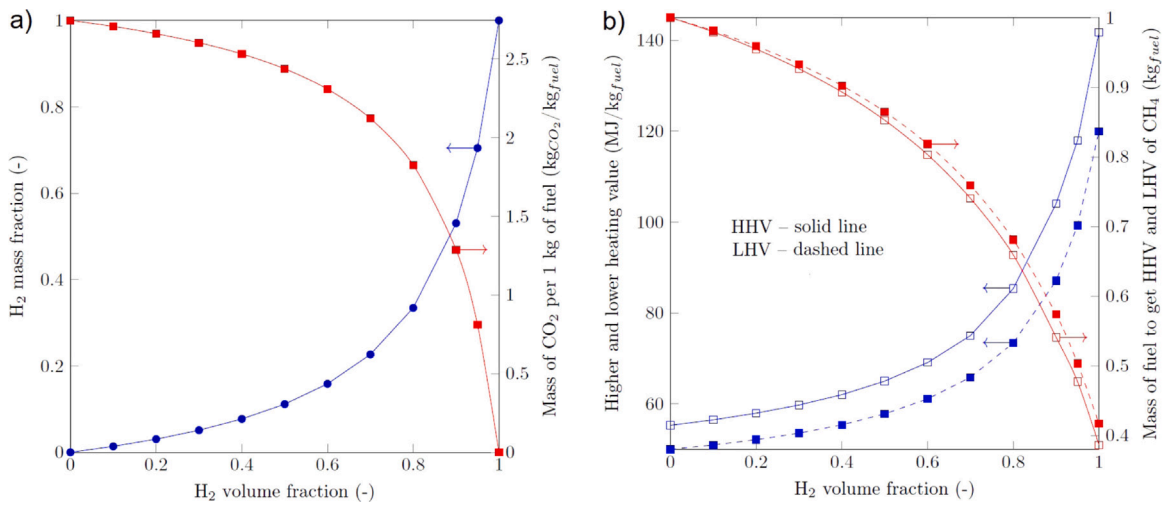


Fig. 1. Heat of combustion and CO_2 emissions of hydrogen-rich fuel: (a) H_2 mass fraction and CO_2 mass per kg of fuel as functions of the H_2 volume fraction; (b) Higher (HHV) and lower (LHV) heating values per kg of fuel, and the mass of fuel required to match the HHV (55.5 MJ/kg) and LHV (50.0 MJ/kg) of 1 kg of methane.

formation of toxic by-products like formaldehyde and carbon monoxide [66,67].

A promising solution to these combustion issues is the addition of hydrogen to create hydrogen-rich blends [68–70]. Hydrogen's high flame speed and reactivity can effectively anchor the flame and stabilize the combustion of ammonia or methanol. For example, research indicates that blending 20%–30% hydrogen with ammonia can make it a viable gas turbine fuel. However, this approach introduces a new layer of complexity, requiring a dual-fuel supply infrastructure with separate storage, handling, and injection systems for both the primary fuel (e.g., ammonia) and the hydrogen stabilizer.

Fig. 2 compares the laminar flame speeds of various fuels, including hydrogen, methane, methane-hydrogen blends, methanol, carbon monoxide, and ammonia. The flame speeds of gaseous fuels derived from chemical processes are also presented, specifically for reformate gas obtained from the full decomposition of methanol, reformate from steam methane reforming, and the product of partially cracked ammonia [71]. The experimental data by Pater and Shah shows that even minor addition of hydrogen has visible effect on combustion process, e.g. for methane-air combustion [72]. The results demonstrate that pre-combustion fuel reforming significantly increases the laminar flame speed, which is primarily due to the high hydrogen content in the resulting reformate. For instance, while the laminar flame speed of pure ammonia is approximately 8 cm/s, the flame speed of the gas mixture produced from 20% ammonia cracking is comparable to that of methane [71]. Furthermore, complete decomposition processes – such as full ammonia cracking, methanol reforming, and steam methane reforming (SMR) – yield reformates with laminar flame speeds substantially higher than those of their parent fuels.

At the same time, for both existing and developing ammonia- and methanol-fueled gas turbines, improving thermal efficiency is a crucial challenge. Pre-combustion thermochemical fuel transformation presents a highly attractive solution, offering significant benefits for both overall thermal efficiency and combustion performance. The pre-combustion reforming of ammonia or methanol, when driven by external heat (e.g., from turbine exhaust or a separate source), yields a hydrogen-rich gas mixture containing up to 75% hydrogen by volume. Consequently, this pre-combustion reforming process can be considered an on-board hydrogen production method, thereby eliminating the need to create a separate hydrogen storage and supply infrastructure [73,74].

On-board H_2-rich fuel production

A highly promising pathway to circumvent dual-infrastructure needs is onboard hydrogen-rich gas production from the primary fuel via thermochemical transformation [75]. In this system, a reformer is integrated with the gas turbine to convert the primary fuel before combustion. For example, ammonia can be cracked (decomposed) into hydrogen and nitrogen, or methanol can be reformed into hydrogen and carbon monoxide. This same approach can also be applied to natural-gas-fueled gas turbines, where natural gas in the presence of steam is converted into a hydrogen-rich gas via a thermochemical reforming process (steam methane reforming – SMR) prior to combustion, absorbing a significant amount of heat [76,77].

The required heat for this endothermic reaction can be supplied by the gas turbine's own high-temperature exhaust gases, effectively upgrading exhaust heat into a high-value chemical (hydrogen-rich gas) and simultaneously solving the combustion stability problem [17]. Alternatively, low-grade external heat sources, such as solar thermal energy, could also drive this process, further reducing the carbon footprint [18,19]. This method of on-board hydrogen-rich gas production presents a pathway to efficiently use challenging carbon-neutral fuels like ammonia and methanol without the need for a separate hydrogen delivery infrastructure, representing a synergistic solution to the dual challenges of efficiency and fuel flexibility.

Pre-combustion fuel reforming increases the lower heating value (LHV) per unit mass (e.g., 1 kg) or per mole of the primary fuel [78–80]. The fuel transformation reactions are endothermic; according to the first law of thermodynamics, the heat supplied to the reformer to drive these reactions is converted into the chemical energy of the reformate (hydrogen-rich gas), manifesting as an increase in its lower heating value. Table 2 provides a comparison of various endothermic processes for the thermochemical transformation of primary fuels.

The following notations are used in Table 2:

- LHV_{pf} [MJ/kg] – lower heating value of the primary fuel;
- ΔH^0_{298} [kJ/mol] – standard enthalpy of the reforming reaction;
- LHV_{syngas} [MJ/kg_{pf}] – lower heating value of the synthesis gas produced from 1 kg of primary fuel;
- η_q – heat transformation coefficient (HTC), defined as the ratio LHV_{syngas}/LHV_{pf}.

Table 2 illustrates that the transformation of methane, methanol, and ammonia into a hydrogen-rich gas is a strongly endothermic process. The reforming of one mole of these primary fuels can absorb

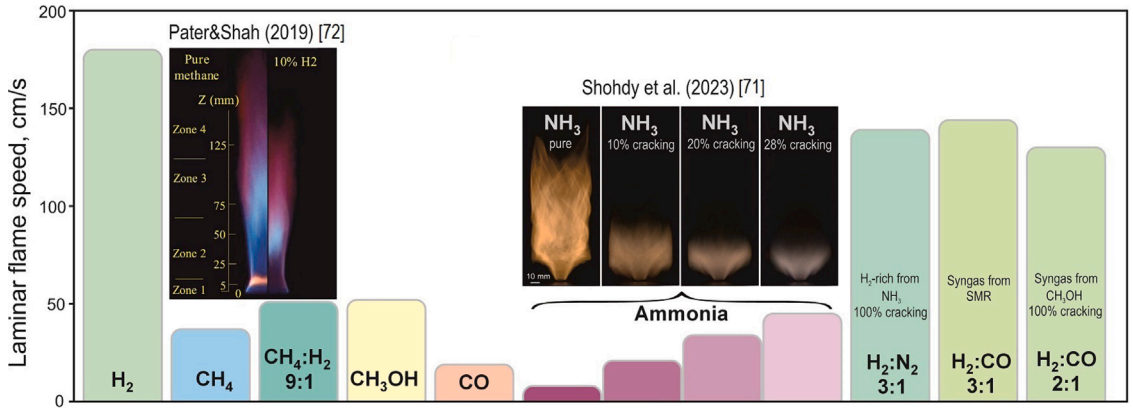


Fig. 2. Laminar flame speeds of various fuels under standard conditions.

Table 2
Comparison of various endothermic processes of pre-combustion fuel reforming.

Chemical reaction	LHV_{pf}, MJ/kg	ΔH^0_{298}, kJ/mol	LHV_{syngas}, MJ/kg_{pf}	η_q
1. $CH_4+H_2O \Leftrightarrow CO+3H_2$ [81]	50.009	206.1	62.854	1.2568
2. $CH_4+CO_2 \Leftrightarrow 2CO+2H_2$ [82]	50.009	247.3	65.410	1.3070
3. $CH_3OH \Leftrightarrow CO+2H_2$ [83]	19.930	91.0	22.772	1.1426
4. $CH_3OH+H_2O \Leftrightarrow CO_2+3H_2$ [84]	19.930	49.7	21.348	1.0711
5. $2NH_3 \Leftrightarrow N_2+3H_2$ [85]	18.600	91.8	22.325	1.2003

a significant amount of thermal energy, equivalent to up to 30% of the fuel's own heat of combustion. Consequently, these endothermic reactions serve a dual purpose: they not only produce a hydrogen-rich gas but also effectively convert low-grade thermal energy into high-value chemical energy stored in the reformate.

2.3. Summary of Section 2

Therefore, it is concluded that pre-combustion fuel reforming using either exhaust heat or external heat sources, e.g. solar thermal energy, in gas turbines offers the following advantages:

- Enhanced thermal efficiency: This is achieved by utilizing exhaust heat or external sources to drive the endothermic reforming process. This strategy simultaneously recovers exhaust heat and upgrades the fuel's chemical energy (lower heating value), leading to greater cycle efficiency;
- Improved combustion performance: Replacing the primary fuel with a hydrogen-rich reformate results in superior combustion characteristics, including higher flame speed, a wider flammability range, and greater stability. This is particularly beneficial for enabling the efficient combustion of challenging low-reactivity fuels like ammonia.

3. Pioneering studies and initial concepts

The concept of pre-combustion fuel reforming for energy systems, now widely studied for gas turbines, has deep roots that extend back to the 1960s. The initial development of this technology was not for gas turbines, but for internal combustion engines (ICEs) and industrial furnaces, driven by the goals of reducing emissions and improving efficiency.

3.1. Industrial furnaces

The conceptual origins of pre-combustion fuel reforming for industrial applications can be traced to a 1961 patent by Professor Schultz [86]. The invention detailed a system for a continuous billet-heating furnace, where waste heat from exhaust gases was used to drive the steam reforming of natural gas. The resulting synthesis gas was then utilized directly as the furnace's fuel, establishing the core principle of converting waste heat into chemical energy. This process was later termed thermochemical exhaust-heat recuperation (TCR), a method where exhaust gases provide the heat for pre-combustion fuel reforming.

This line of research was subsequently advanced in the USSR, focusing on enhancing the energy efficiency of glass-melting furnaces. A dedicated research group led by Pereletov at the Moscow Power Engineering Institute, formed in 1965, successfully developed and implemented an industrial-scale TCR system [87]. Their work demonstrated that using exhaust gas heat to reform natural gas significantly improved exhaust heat recuperation compared to conventional methods via air preheating before combustion. The schematic diagram of Pereletov's glass-melting furnace with thermochemical exhaust heat recuperation is presented in Fig. 3.

The schematic in Fig. 3 illustrates a hybrid heat recuperation system, integrating TCR with conventional air preheating. In this configuration, the hot flue gases are split into two separate streams. The first stream supplies heat to the reformer for natural gas reforming with steam, after which the cooled gases preheat the inlet natural gas feed. It should be noted that the steam required for reforming is generated externally to this system. The second flue gas stream passes through a two-stage air heater. The synthesis gas produced by the reformer is then combusted in the glass-melting furnace alongside the preheated air.

Pereletov and colleagues quantified the performance improvement, reporting a reduction in specific fuel consumption per kg of charge from 0.74 m^3 to 0.524 m^3 [88]. This constitutes a 29.2% decrease, equivalent to a 1.41-fold improvement in fuel economy. Experimental data for fiberglass production showed a similar trend, with consumption falling from 1.34 m^3/kg to 0.994 m^3/kg, a 25.8% reduction (1.35-fold improvement). This performance corresponded to a maximum heat recuperation degree of 0.621 for the TCR-equipped furnace. A comparative summary of the industrial trials, with and without TCR, is provided in Table 3.

The key findings from the industrial investigation of TCR in the glass-melting furnace were as follows:

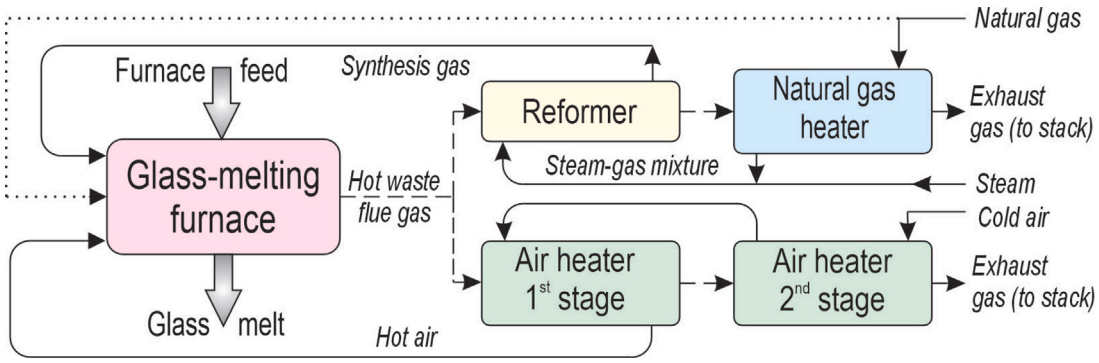


Fig. 3. Schematic of the first industrial glass-melting furnace with pre-combustion natural gas reforming using waste flue gas heat [88,89].

Table 3
Comparison of industrial glass-melting furnace performance with and without thermochemical recuperation (TCR) [88,89].

Parameter	Unit	Without TCR	With TCR		
			Test 1	Test 2	Test 3
Gas consumption	m^3/h	140	115	116	102
Air consumption	m^3/h	1500	1185	1280	1150
Steam consumption	kg/h	–	162	190	157
Hot air temperature	°C	725	365	425	610
Steam-air mixture temperature	°C	–	380	340	310
Syngas temperature	°C	–	917	883	858
Methane conversion	%	–	90	79	65
Waste-heat recuperation degree	–	0.332	0.550	0.558	0.621

- The implementation of pre-combustion fuel reforming (via thermochemical recuperation) led to a significant reduction in fuel consumption.
- The waste-heat recuperation degree was nearly doubled in the furnace with TCR compared to the conventional furnace.
- For high-temperature flue gases, TCR systems can be effectively combined with conventional air preheating recuperators.

A comprehensive summary and analysis of most industrial and laboratory-scale implementations of pre-combustion fuel reforming via thermochemical recuperation in industrial gas furnaces is provided in Pashchenko's previous review [36].

3.2. Internal combustion engines

Concurrently, initial research into pre-combustion fuel reforming in internal combustion engines (ICE) began in the United States in the late 1960s, primarily aimed at reducing pollutant emissions from spark-ignition engines [90]. A foundational study by Lee and Wimmer [90] demonstrated that a steam-reformed liquid fuel, which they termed "fuel gas", possessed superior lean-burn characteristics and yielded lower emissions compared to conventional fuels like propane and methane. This early work catalyzed further investigation into various reforming pathways throughout the 1970s and 1980s, with research encompassing steam reforming (SR), thermal decomposition, partial oxidation, and exhaust gas fuel reforming [91].

A significant conceptual advance was made by Martin [92], who was the first to model the reformer and internal combustion engine as an integrated energy system. His work proposed using engine waste heat to drive the endothermic catalytic steam reforming of gasoline at approximately 893 K. Martin's theoretical analysis identified a dual mechanism for efficiency gains: first, an increase in the specific heating value of the reformate, and second, a reduction in pumping losses. This second benefit arises from the wide flammability limits of the hydrogen-rich reformate, which enables stable combustion under ultra-lean conditions. Consequently, the engine can operate at part-load with a wide-open throttle (WOT), eliminating the need for intake throttling and its associated parasitic losses. Fig. 4 shows Martin's original system flow diagram, which constitutes the first documented proposal for integrating pre-combustion fuel reforming based on thermochemical recuperation within an internal combustion engine.

Recently, Tartakovsky and Sheintuch published a comprehensive review on fuel reforming in internal combustion engines [93], systematically analyzing various approaches to pre-combustion fuel reforming. The authors concluded that this technology significantly improves the thermal efficiency of internal combustion engines while simultaneously reducing pollutant emissions.

3.3. Gas turbines

The application of TCR to gas turbines was first proposed in 1970s by Olmsted and Grimes [94]. Their seminal work suggested using gas turbine exhaust heat to drive the steam methane reforming process, noting that the process provided both thermal and chemical recuperation of waste heat, offering a higher potential for efficiency gains than conventional air preheating.

In the 1980s, Davis et al. published a comprehensive study on pre-combustion fuel reforming for alcohol-fueled gas turbines [95]. Their work describes a high-efficiency system where the alcohol fuel is catalytically reformed using exhaust gas waste heat in an endothermic process. The paper includes a performance analysis of catalytic reformers for methanol, methanol/water, and ethanol/water mixtures. System performance predictions for a Westinghouse 501D (100 MWe) gas turbine indicated significant improvements in efficiency and power output over conventional distillate fuel operation, alongside potential reductions in NOx emissions. An economic comparison with distillate fuel also identified the conditions under which a reformed-alcohol gas turbine could become feasible (see Fig. 5).

Davis et al. concluded that gas turbines utilizing pre-combustion fuel reforming achieve higher overall efficiency than those operating without it [95]. This improvement is attributed to the recuperation of exhaust gas heat into the fuel cycle, which upgrades the energy in both chemical (reformate enthalpy) and sensible forms. The authors also demonstrated the economic viability of implementing pre-combustion fuel reforming in alcohol-fueled gas turbines.

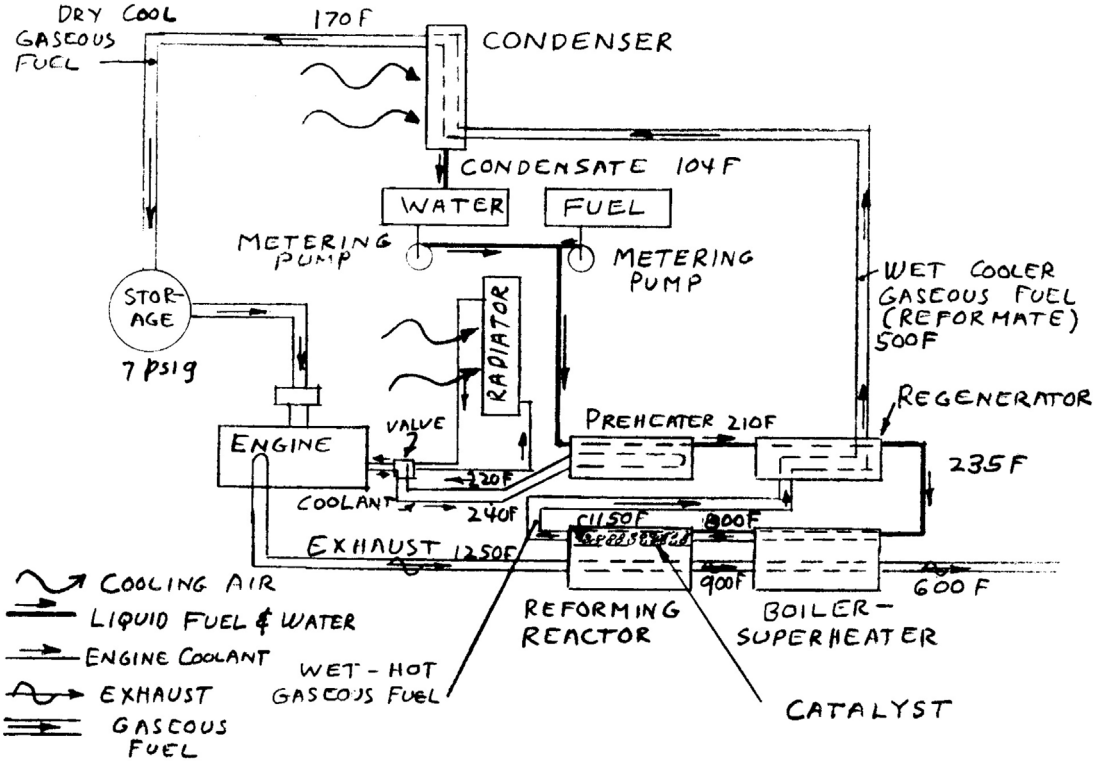
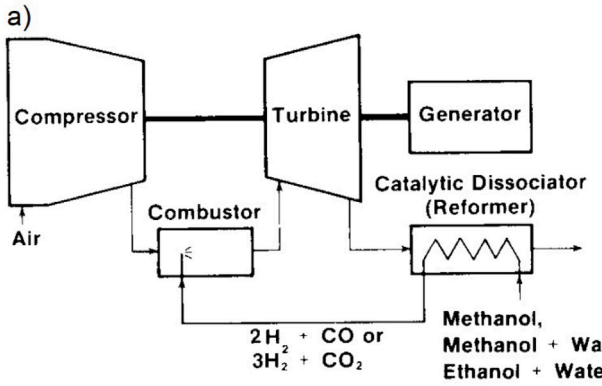


Fig. 4. Martin's original system flow diagram, which constitutes the first documented proposal for integrating pre-combustion fuel reforming within ICE [92].
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b)

Fuel	Efficiency
Distillate	0.320
Liquid methanol	0.340
Liquid ethanol	0.342
Dissociated methanol	0.358
Steam-reformed methanol	0.352
Combined-cycle liquid methanol	0.336

Fig. 5. Original re-print from paper by Davis et al. [95]: (a) Schematic of a reformed alcohol combustion gas turbine; (b) Thermal efficiency of Westinghouse 501D gas turbine with various combinations of pre-combustion fuel reforming.
Source: Reproduced with permission from American Society of Mechanical Engineering.

In summary, the foundational era of pre-combustion fuel reforming occurred in the 1960s and 1970s, with pioneering experimental and industrial results established for internal combustion engines, industrial furnaces, and gas turbines, using exhaust heat for the primary fuel reforming. The following decades have built upon this robust historical foundation, with continued global research interest focused on overcoming technical challenges and realizing the full potential of pre-combustion fuel reforming.

3.4. Summary of Section 3

The pioneering era of pre-combustion fuel reforming, spanning the 1960s to the 1980s, established the foundational principles of thermochemical recuperation (TCR) across three major energy conversion platforms. The concept originated not in gas turbines, but in industrial furnaces, with Schultz's 1961 patent for a billet-heating furnace that used exhaust heat to reform natural gas into a more valuable syngas. This principle was successfully scaled and validated by Pereletov's group in the USSR, who demonstrated a 25%–30% reduction in fuel consumption in an industrial glass-melting furnace by integrating TCR with conventional air preheating, nearly doubling the waste-heat recuperation degree.

Concurrently, research for internal combustion engines (ICEs) began in the late 1960s, initially focused on emission reduction. A key conceptual leap was made by Martin, who first modeled the reformer and engine as an integrated system. He theorized that using waste heat to reform gasoline would yield a dual efficiency benefit: an increase in the fuel's heating value and a reduction in pumping losses due to the superior combustion properties of the hydrogen-rich reformate.

The application of pre-combustion fuel reforming to gas turbines was first proposed in the 1970s by Olmsted and Grimes, who identified

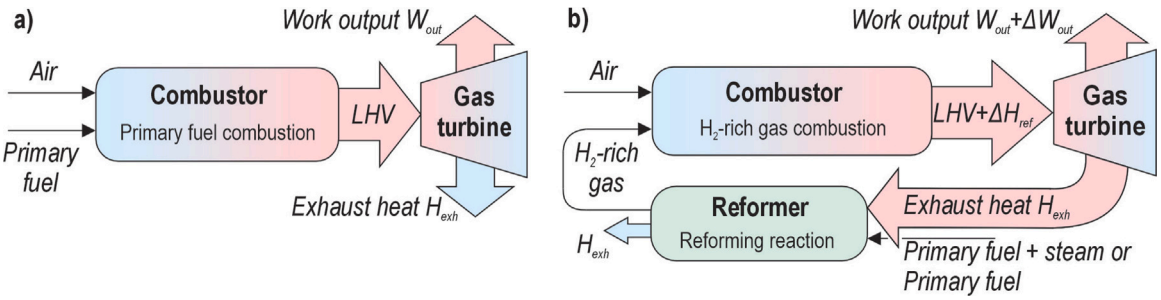


Fig. 6. Comparison of energy conversion pathways: (a) direct combustion in a traditional gas turbine and (b) the two-stage process in a chemically recuperated gas turbine.

its superior potential for waste heat recovery compared to conventional methods. This was followed in the 1980s by Davis et al.'s comprehensive study on alcohol-fueled turbines, which confirmed significant improvements in efficiency and power output, while also establishing the economic viability of the approach. In conclusion, this foundational period demonstrated that pre-combustion reforming, by converting waste heat into chemical energy, provides a universal mechanism for enhancing efficiency and reducing emissions across diverse thermal systems.

4. Chemically recuperated gas turbines: Fundamentals

A chemically recuperated gas turbine (CRGT) is defined by its use of turbine exhaust heat to drive the pre-combustion reforming of the primary fuel (typically natural gas, methanol, or ammonia). The core concept relies on harnessing this waste energy for the endothermic reactions that convert primary fuels into a hydrogen-rich gas. A fundamental distinction of the CRGT is that the transformation of the fuel's chemical energy into heat occurs in two distinct stages [17,42,96]. In a conventional gas turbine, this transformation is a single-stage process involving direct fuel combustion. In a CRGT, however, it is divided into two stages. The first stage is the endothermic reforming of the primary fuel using exhaust heat, which produces reformate with a higher heat of combustion (per 1 kg of the primary fuel) than the original fuel. The second stage is the combustion of these reforming products as the final fuel. The comparison of the conversion of chemical energy of fuel into heat for conventional gas turbines (a) and chemically recuperated gas turbines (b) is presented in Fig. 6.

This pre-combustion reformation provides two major benefits. First, the chemical energy content (lower heating value) of the reformed fuel is higher than that of the original fuel, as the exhaust heat has been "upgraded" and stored within the chemical bonds of the hydrogen-rich gas. Second, the hydrogen-rich gas exhibits superior combustion properties, such as higher flame speed and wider flammability limits, which can lead to more stable combustion and lower emissions. By combusting this hydrogen-rich gas instead of the original fuel, the CRGT achieves a higher overall cycle efficiency than would be possible with traditional thermodynamic recuperation alone.

The schematic diagram and corresponding thermodynamic cycle of a chemically recuperated gas turbine are presented in Fig. 7. As shown in the system schematic (Fig. 7a), the exhaust gas heat is utilized in a reformer to drive the pre-combustion reforming of the primary fuel via the chemical reactions listed in Table 2. Consequently, a portion of the exhaust heat (q_{rec}, representing the recuperated heat) is effectively recuperated into the thermodynamic cycle, as illustrated in the T-s diagram of Fig. 7b.

According to the principle of energy conservation, the recuperated heat equals the sum of the standard enthalpy of the endothermic reforming reaction and the change in sensible enthalpy of the reacting mixture. This can be expressed as:

$$\Delta H_{rec} = \Delta H^0_{298} + (H_{H_2 rich} - H_{pf}) = H_4 - H_5, \quad (1)$$

where ΔH^0_{298} is the standard enthalpy of the primary fuel reforming reaction; $H_{H_2 rich}$ and H_{pf} are the enthalpies of the H_2-rich gas after the reformer and the primary fuel mixture before the reformer, respectively; H_4 and H_5 are the enthalpies of the exhaust gas stream before and after the reformer, respectively.

While the preceding analysis provides a first-law energy balance, a complete thermodynamic assessment requires an exergy-based perspective to account for energy quality. In the CRGT concept, the endothermic reforming reaction acts as an "exergy pump" that upgrades low- and medium-grade thermal energy into high-quality chemical exergy stored in the hydrogen-rich reformate [97]. The exergy balance of the reforming process can be expressed as:

$$\Delta Ex_{fuel} = Ex_{prod} - Ex_{fuel} = Ex_q - Ex_{dest}, \quad (2)$$

where ΔEx_{fuel} is the increase in chemical exergy of the fuel stream, Ex_{prod} and Ex_{fuel} are the chemical exergies of the reformate and primary fuel, respectively, Ex_q is the thermal exergy of the exhaust heat supplied to the reformer, and Ex_{dest} represents the exergy destroyed due to irreversibilities in the heat transfer and reaction processes.

A key performance indicator is the coefficient of exergy transformation (CET) [97], defined as:

$$k_{ex,q} = \frac{\Delta Ex_{fuel}}{Ex_q}. \quad (3)$$

For ammonia decomposition at 700°C, analysis shows that the CET can exceed unity ($k_{ex,q} \approx 1.275$), indicating that the chemical exergy gained by the fuel is larger than the thermal exergy of the supplied heat [97]. This apparent amplification does not violate the second law, because the baseline exergy of the primary fuel serves as a "carrier" that enables the exergy upgrade, analogous to a heat pump where work input transfers more thermal energy than the work consumed. The upgraded exergy is realized after combustion, when the reformate releases heat at a higher flame temperature compared to the primary fuel. Thus, the CRGT effectively functions as a thermochemical "exergy pump", converting low-quality waste heat into high-quality chemical energy and improving the overall thermodynamic performance of the gas turbine cycle. This exergy-based perspective complements the first-law analysis presented earlier and provides deeper insight into the qualitative transformation of energy in CRGT systems.

It should be noted that for high-temperature gas turbines, the exhaust gas leaving the reformer still possesses significant thermal potential for further use, for instance, in steam generation. This steam can be partially utilized in a bottoming cycle and to drive the steam reforming reaction of fuels like methane or methanol. Combined cycle power plants integrating chemically recuperated gas turbines are discussed in detail in the following sections.

The research scope for chemically recuperated gas turbines (CRGTs) encompasses thermodynamic analysis, heat and mass transfer within the reformer, and combustion characteristics, as outlined in Fig. 8. Thermodynamic analysis of the endothermic reforming processes is essential for determining the relationship between reformate composition, reaction enthalpy, and key operational parameters such as

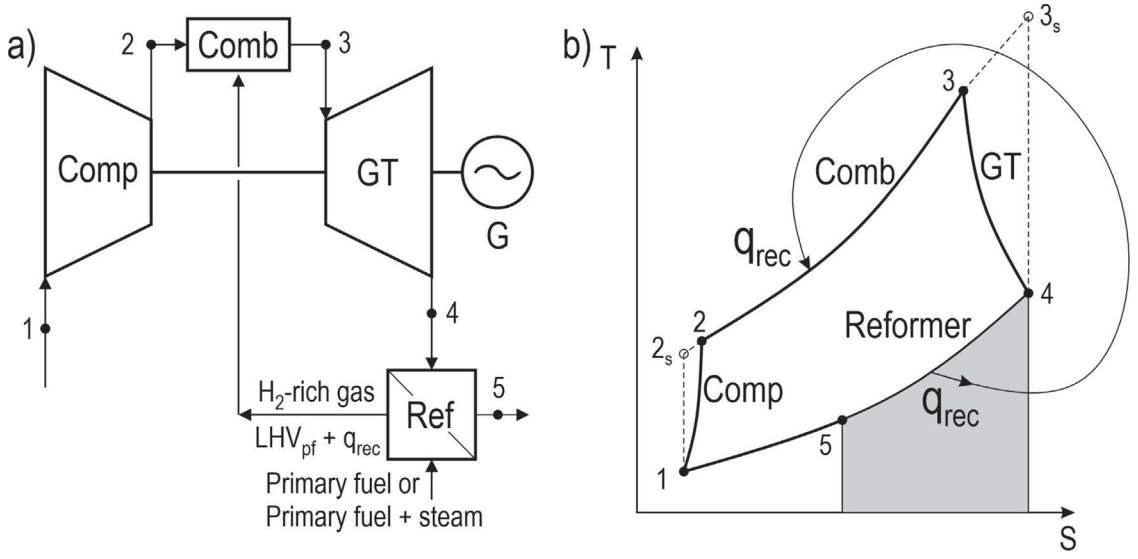


Fig. 7. Schematic of a chemically recuperated gas turbine (a) and its thermodynamic cycle (b): 1 – inlet air; 2 – compressed air; 3 – working gas; 4 – hot exhaust gas; 5 – cooled exhaust gas; Comp – compressor; Comb – combustor; GT – gas turbine; G – generator; Ref – reformer.

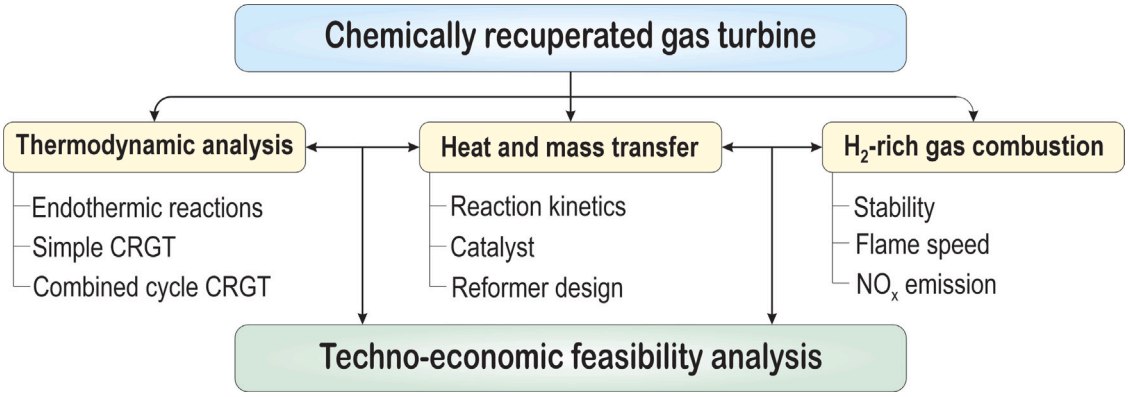


Fig. 8. Research task flowchart for study of chemically recuperated gas turbines.

temperature, pressure, and inlet mixture composition. Furthermore, thermodynamic modeling of both simple and combined-cycle CRGT configurations will establish the thermal efficiency of these systems for a given set of gas turbine initial conditions.

The study of heat and mass transfer in the reformer is critical for defining the reactor's design parameters. These transport phenomena are highly dependent on the catalyst properties, including its composition, shape, and the type of support used. Finally, a major research focus for CRGT development is the combustion of hydrogen-rich gas, with particular emphasis on combustion stability, flame speed, and NO_x emissions control.

This section has provided a comprehensive overview of the fundamentals, research scope, and specific characteristics of natural gas-fueled chemically recuperated gas turbines (CRGTs). The core principle of a CRGT is the thermochemical recuperation of exhaust heat through an endothermic reforming process, which distinguishes it from conventional thermodynamic recuperation. This two-stage energy conversion – first into chemical energy of a hydrogen-rich syngas and then into heat via combustion – enables superior performance.

5. Natural gas-fueled CRGT

Natural gas is the predominant fuel for stationary gas turbines in power generation and industrial applications, primarily due to its widespread availability, established infrastructure, and relatively clean combustion profile compared to liquid fossil fuels. Its composition consists mainly of methane, typically comprising 90%–95% of its volume, with smaller amounts of other hydrocarbons like ethane and propane, as well as inert gases such as nitrogen and carbon dioxide. The high hydrogen-to-carbon ratio of methane results in lower carbon dioxide (CO_2) emissions per unit of energy released, while its simple molecular structure enables stable, efficient combustion with minimal particulate matter and unburned hydrocarbon emissions, making it a favorable choice from both operational and environmental perspectives.

For advanced cycles like the chemically recuperated gas turbine, natural gas serves as the primary feedstock for the production of syngas. The principal method for this conversion is steam methane reforming (SMR), an endothermic process represented by the reaction (1) from Table 2. This reaction efficiently transforms the chemical energy of methane and the thermal energy of exhaust heat into a hydrogen-rich syngas, thereby upgrading the fuel's energy content (per 1 kg of mass of original methane) and enabling higher cycle efficiency. The widespread use of SMR in industrial hydrogen production underscores its technological maturity and relevance for gas turbine fuel processing [55,98].

In contrast to methane, alternative carbon-neutral fuels like ammonia and methanol offer different pathways for hydrogen production [99–102]. A key operational advantage of these fuels is their ability to undergo decomposition into syngas without requiring an additional reactant like steam. Ammonia can be thermally cracked via reaction

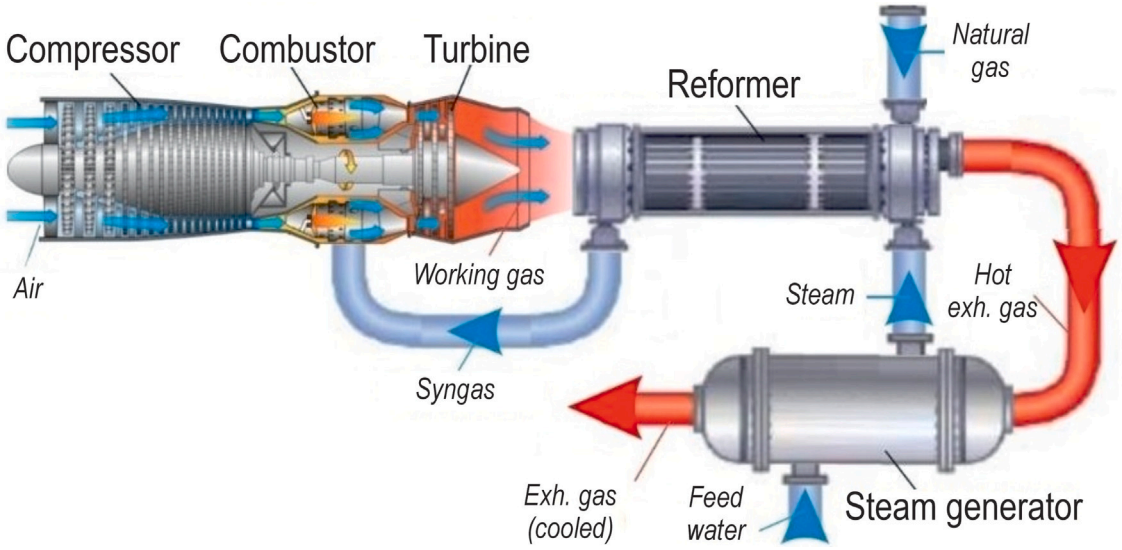


Fig. 9. Concept of natural gas-fueled chemically recuperated gas turbine.

(5) (Table 2), and methanol can be decomposed via reaction (3) (Table 2) using heat alone. This characteristic simplifies the fuel reforming system by eliminating the need for a separate steam supply, presenting a potential technical simplification compared to the steam reforming process necessary for natural gas.

The schematic concept of natural gas-fueled chemically recuperated gas turbine is presented in Fig. 9. In this diagram, the cycle begins with ambient air being compressed and then directed to the combustor. Simultaneously, the natural gas is not sent directly to the combustor but is first mixed with steam. This fuel-steam mixture is then fed into a catalytic reformer, which is thermally integrated with the turbine's exhaust stream. The hot exhaust gases, after expanding through the turbine, pass through the reformer, providing the necessary endothermic heat for the steam methane reforming (SMR) reaction and sensible heat for the heating of steam-methane mixture. This process thermochemically recuperates the exhaust heat by converting it into the chemical energy of the resulting hydrogen-rich syngas, which is then combusted in the main combustor, and increasing the enthalpy of reacting mixture.

This configuration represents the internal generation of steam required for the reforming reaction, creating a self-sustaining loop within the recuperation system. After the exhaust gases transfer their heat to the SMR reaction inside the reformer, they are not simply vented. Instead, these partially-cooled gases are routed to a heat recovery steam generator (HRSG). In the HRSG, the residual thermal energy in the exhaust is used to vaporize feed water, producing the steam needed for the reforming process. This steam is then mixed with the incoming natural gas, closing the loop. This design ensures that the waste heat is utilized twice: first for the endothermic chemical reaction and subsequently for the steam generation that enables that very reaction, thereby maximizing the overall thermal efficiency of the cycle.

Over recent decades, numerous studies have explored a wide range of theoretical and practical aspects of chemically recuperated gas turbines (CRGTs) fueled by natural gas. This body of research can be broadly categorized into two main directions: simple CRGT cycles and combined cycle CRGT configurations. The latter, in particular, demonstrates significant potential for achieving exceptionally high overall thermal efficiency in power generation, building upon the inherent advantages of natural gas as a fuel.

5.1. Overview

5.1.1. Steam reforming

One of the first works on the detailed analysis of natural gas-fueled chemically recuperated gas turbines is a 1990 report by Jack Janes of the California Energy Commission, who proposed a new cycle promising NO_x levels as low as 1 ppm [103]. Janes proposed using gas turbine exhaust heat for the pre-combustion reforming of natural gas – a system he termed the chemically recuperated gas turbine (CRGT). Similar to other pioneering studies on fuel reforming, the primary objective of Janes' proposal was to achieve low-emission combustion. Concurrently, he demonstrated that the CRGT offers a threefold advantage: superior emission control, high thermal efficiency, and reduced capital expenditure. Performance projections in the report, based on General Electric gas turbine data, indicated that a CRGT derived from the CF6-80C2 aircraft engine with intercooling and reheat could achieve an efficiency of up to 59%. This finding was corroborated by a separate analysis from the Pacific Gas and Electric Company, which, using the same GE data, estimated a comparable efficiency of 56%. This comparison is for chemically recuperated gas turbine (CRGT) and steam injected gas turbine (STIG) [104].

One year late, Alistair Lloyd published their report on thermodynamics of chemically recuperated gas turbines [105]. The author considered the chemically recuperated gas turbine as an advanced evolution of the steam-injected gas turbine (STIG) cycle. In a CRGT, the exhaust heat is used not merely for steam generation but to drive the endothermic steam methane reforming reaction. This process converts natural gas and steam into a hydrogen-rich syngas, effectively upgrading waste heat into chemical energy. This fundamental difference confers two primary advantages. First, the presence of hydrogen in the fuel gas enables combustion at lower flame temperatures, leading to projected NOx emissions as low as 1 ppm, which allows it to meet the strictest current emissions standards without secondary cleanup systems like selective catalytic reduction. Second, thermodynamic analyses indicate that a simple CRGT cycle is more efficient than a STIG cycle due to reduced irreversibilities in heat recovery, achieving a maximum efficiency gain of approximately half a percentage point, albeit at a roughly 20% lower power output for a given turbine inlet temperature.

Lloyd reported that from a performance perspective, CRGT cycles show significant promise, with modeled efficiencies for configurations featuring intercooling and reheat estimated to exceed 50%–53% on a lower heating value basis. The author concluded that the combination of very high efficiency, ultra-low emissions, and potentially lower capital costs continues to fuel significant research interest in the CRGT as a highly flexible and environmentally favorable power generation technology.

Lloyd conducted a comprehensive comparative thermodynamic analysis of various gas turbine cycles, with the primary objective of

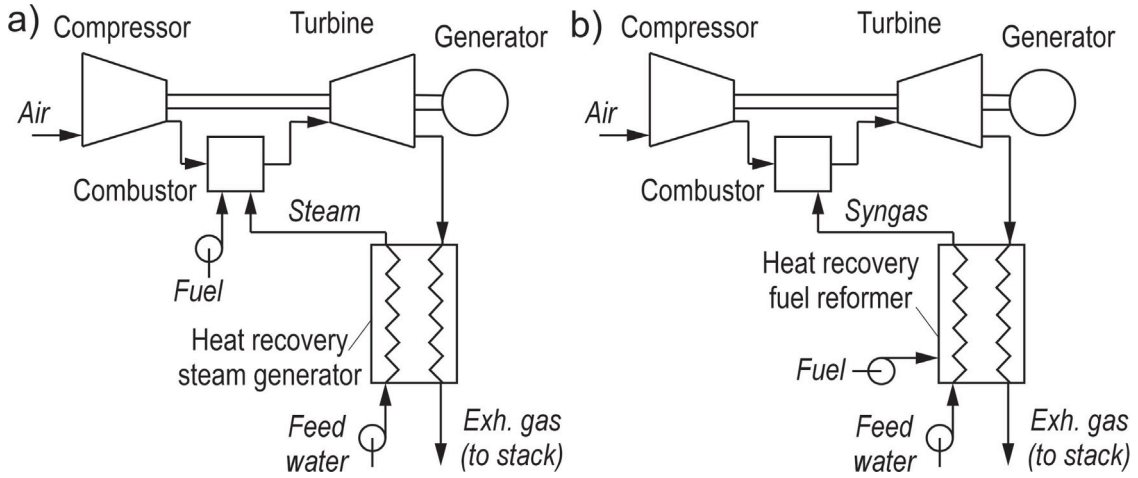


Fig. 10. Schematic diagrams of the steam-injected gas turbine (a) and the chemically recuperated gas turbine (b) [105].

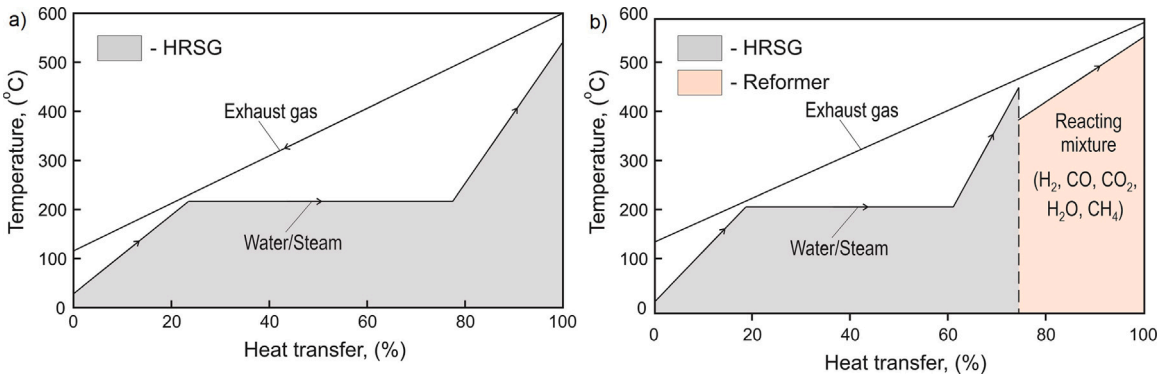


Fig. 11. Heat recovery profiles for STIG and CRGT cycles from Fig. 10.

evaluating the performance of chemically recuperated (CRGT) configurations against other systems. This work provided the first direct comparison between the CRGT and steam-injected gas turbine (STIG) cycles, the schematic diagrams of which are presented in Fig. 10.

Lloyd showed that significant part of exhaust heat is absorbed in the reforming to drive steam methane reforming process. The hear recovery profiles for STIG and CRGT are shown in Fig. 11. Both cycles are based on a heavy-duty gas turbine operating at a turbine inlet temperature of 1250°C and a pressure ratio of 15. The temperature profiles in Fig. 11 correspond to unrestricted STIG and CRGT cycles, where the exhaust gas temperature at the stack is identical for both systems. Under this condition, the mass flow rates of steam differ between the STIG and CRGT cycles. Another configuration is the restricted STIG and CRGT cycle, defined by an equal mass flow rate of steam generated in the heat recovery steam generator for both systems. In this case, the exhaust gas temperatures at the stack are different.

The summary of performance analysis of unrestricted and restricted STIG and CRGT cycles is provided in Table 4. The performance advantage of CRGT over STIG cycle depends significantly on the comparison basis. Under restricted conditions with equal steam flow, the CRGT achieves a 2.3 percentage point efficiency gain with no power loss, as its superior heat recovery leads to a lower stack temperature. In contrast, an unrestricted comparison shows a smaller efficiency gain but a 5% power reduction. This demonstrates that the primary benefit of chemical recuperation is not increased steam production, but a more effective utilization of exhaust heat, even at low methane conversion rates.

Concluding the overview of Lloyd's report, it is crucial to highlight that it contains the first published attempt to estimate the capital

Table 4
Comparison of restricted and unrestricted STIG and CRGT cycles [105].

	Unrestricted		Restricted	
	STIG	CRGT	STIG	CRGT
Specific work (kJ/kg$_{air}$)	596	507	399	393
LHV efficiency (%)	45.6	47.6	37.9	40.2
Steam production (kg/kg$_{air}$)	0.217	0.143	0.050	0.050
Stack temperature (°C)	108	110	439	384
Methane conversion (%)	n/a	37.3	n/a	18.0

cost of a CRGT system. Lloyd's analysis, based on early 1990s pricing, calculated the specific capital cost for a CRGT at approximately 456 $/kW. This was significantly lower than the costs for a steam-injected gas turbine (840 $/kW) and an intercooled steam-injected gas turbine (474 $/kW). Therefore, the CRGT was shown to offer not only higher thermal and combustion efficiency but also a substantially lower capital cost per kilowatt.

A 1994 study by Kesser et al. [40] involved the development of a computational model to assess the performance of a basic chemically recuperated gas turbine (CRGT) against established cycles. The analysis, conducted under consistent assumptions, compared the CRGT's thermal efficiency and specific work to that of steam-injected (STIG), combined cycle (CC), and simple cycle (SC) systems. Their model incorporated a steam methane reformer that utilized turbine exhaust heat to produce a hydrogen-rich syngas. Simulation results demonstrated that the basic CRGT configuration achieved higher thermal efficiency than both the STIG and simple cycles, though it did not surpass the performance of the combined cycle. The summary results of their work are provided in Table 5. The results are derived from two reference

Table 5
Summary of results for cases 1 and 2 (combustor outlet temperature T_3 = 1225°C).

	π_s	η_{th} (%)	$\dot{W}_{net}/\dot{m}_{air}$ (kJ/kg)	CAF_T	SA
Reference Case 1					
CRGT	30.0	47.0	398	0.21	0.087
STIG	30.0	46.7	419	0.22	0.101
Combined cycle	30.0	47.9	351	0.16	0.077
Simple cycle	30.0	39.5	292	0.15	N/A
Reference Case 2					
CRGT	15.0	47.3	468	0.21	0.138
STIG	15.0	45.9	527	0.23	0.186
Combined cycle	15.0	49.7	451	0.14	0.126
Simple cycle	15.0	36.0	330	0.13	N/A

CAF_T – total cooling air fraction; SA – pinch point limited steam-air ratio.

cases established to ensure a consistent comparison between the CRGT, STIG, simple cycle, and combined cycle power plants. Reference Case 1 models a high-performance GE LM5000 aeroderivative gas turbine with a pressure ratio of 30.0, while Reference Case 2 represents a heavy-duty industrial turbine with a lower pressure ratio of 15.0. The input parameters, including key temperatures and cooling flow data, were primarily sourced from a proprietary engineering report on the LM5000.

The performance comparison of the four cycles at maximum steam production revealed that the combined cycle achieved the highest thermal efficiency (η_{th}), followed by the Basic CRGT, the STIG cycle, and finally the simple cycle. In terms of specific work ($\dot{m}_{air}$), however, the STIG cycle was superior, with the Basic CRGT, combined cycle, and simple cycle ranking subsequently.

Analysis of the Basic CRGT demonstrated attractive thermal efficiencies of approximately 47.0% and 47.3% for compressor pressure ratios of 30 and 15, respectively. These promising results suggest that constructing a demonstration plant based on the simpler Basic CRGT configuration, which forgoes intercooling and reheat, could be a strategically valuable initiative. Such a project would entail significantly lower cost and development time compared to an advanced CRGT, while providing early validation of the core concept and identifying potential operational challenges.

A further contribution to the development of chemically recuperated gas turbines (CRGT) was made by de Souza-Santos in 1997 [106]. The author conducted a thermodynamic study of advanced power generation systems based on the principle of thermochemical recuperation (TCR), specifically analyzing and comparing two sophisticated configurations: a thermochemically recuperative cycle with reheating (TCRR) and a more complex system incorporating both reheating and intercooling (TCRRI). These systems were modeled using a computer code that rigorously computes physicochemical properties and simulates the performance of all cycle components, with a particular focus on the kinetics and equilibrium within the reformer to prevent issues like catalyst-fouling carbon deposition.

The comparative analysis reveals that the configuration has a significant impact on overall performance. Under a set pressure ratio of 34 and a turbine inlet temperature of 1700 K, the TCRRI system achieved a global efficiency of 58.80%, which was almost a full percentage point higher than the 57.93% efficiency of the TCRR system. The TCRRI cycle also produced a higher net power output per unit mass flow of inlet air. The study concludes that while both TCR configurations are highly promising for future high-temperature gas turbines, the addition of intercooling to a reheated cycle (TCRRI) offers a measurable advantage in thermodynamic performance, establishing it as the most efficient system analyzed.

de Souza-Santos reported that many chemical reactions can occur in a steam methane reformer and stated that the accurate prediction of heat and mass transfer within it is crucial for the efficient development of CRGT. Progress in specialized tools for the thermodynamic analysis of various chemical and physical processes has significantly intensified the study of chemically recuperated gas turbines. Among the first to use such specialized software for a CRGT study were Harvey and Kane [107]. In their paper, they presented the results of a thermodynamic analysis of a reheat gas turbine cycle with chemical recuperation, which was modeled after the ABB GT26 heavy-duty gas turbine and simulated using ASPEN+ software. The authors highlighted several key challenges, including the need for high turbine exhaust temperatures – achieved through reheat combustion – and the significant combustor redesign required to handle the low heating value and high flow rates of the reformed fuel. Furthermore, they noted that the reformer operates at relatively low temperatures and high pressures, leading to only partial methane conversion (27%), which limits potential efficiency gains.

When compared to a state-of-the-art combined cycle, the reheat CRGT cycle achieves a thermal efficiency of 54.8% and a net power output of 383 MW, slightly lower than the 58.5% efficiency reported for a triple-pressure reheat combined cycle based on the same turbine. The authors attribute this efficiency gap to significant exergy losses, particularly in the combustors, due to the mixing of steam-rich reformed fuel with air. Although chemical recuperation reduces exergy losses in the reformer compared to a conventional steam generator, these benefits are offset by mixing irreversibilities and the injection of excess steam into the combustion process. The study suggests that further optimization – such as adjusting the pressure ratio and adopting dual-pressure heat recovery – could narrow this gap, but also highlights other advantages of CRGT cycles, including ultra-low NO_x emissions and potential for better part-load performance.

In 2001 Abdallah and Harvey conducted thermodynamic analysis of Chemically Recuperated Gas Turbine (CRGT) cycles, comparing their performance to other advanced cycles like the Steam Injected Gas Turbine (STIG), Humid Air Turbine (HAT), and Combined Cycle Gas Turbine (CCGT) [41]. The authors use detailed simulation models that account for critical factors such as turbine blade cooling and chemical kinetics in the methane steam reformer. Their analysis reveals that while CRGT cycles achieve higher efficiency than STIG cycles – primarily due to reduced exergy losses in the combustor and heat recovery system – their performance is ultimately limited by the thermodynamic constraints of methane reforming. Even with advanced configurations featuring intercooling and reheat, the maximum efficiency achieved is around 54%, which falls short of the 60% efficiency offered by state-of-the-art combined cycles. A summary of performance and exergic comparison of CRGT and STIG cycles is provided in Table 6.

Abdallah's and Harvey's study concludes that the most viable application for CRGT technology is in intermediate-load power generation using aeroderivative gas turbines, where its efficiency is comparable to that of a CCGT based on the same engine [41]. A key advantage identified is the cycle's potential for ultra-low NO_x emissions, owing to the hydrogen-rich, steam-diluted reformed fuel. However, the authors caution that future development must focus on overcoming significant technical and economic hurdles. These include the development of cost-effective, low-temperature reforming catalysts and a thorough evaluation of the costs associated with retrofitting or developing new turbine designs optimized for CRGT operation, especially when weighed against the rapid advancements in dry low-NO_x combustors for conventional cycles.

Further development of CRGT systems was pursued by Carapellucci and Milazzo [42]. In their work, the authors focused on the thermodynamic optimization of a reheat CRGT based on a commercially available ABB GT24 engine, with a specific emphasis on optimizing the heat recovery scheme through multi-stage pressure levels. The authors propose a slight modification to the original reheat design – lowering the pressure at which the reheat combustor is located – to raise the turbine exhaust temperature and thereby improve the

Table 6
Performance and exergic comparison of CRGT and STIG cycles [41].

	CRGT	STIG		Exergy losses [%]	
				CRGT	STIG
Pressure ratio β	15	15	Combustor	27.48	28.55
TIT [K]	1575	1575	Heat exchangers	7.45	8.84
η_{el} [%]	49.86	47.88	Stack gas	6.89	7.11
W [kJ kg_{air}^{-1}]	565	633	Turbine	5.06	4.79
Reformed gas pressure	18	–	Compressor	2.45	2.08
Steam-to-methane ratio	6	–	Auxiliaries	2.43	2.45
Methane conversion [%]	40.5	0	Exergy efficiency	48.05	46.18
m_{H_2O} [% of m_{air}]	14.37	19.8			

methane conversion rate in the steam reformer. They systematically compare a single-pressure heat recovery system against a dual-pressure system, demonstrating that while a single-pressure reformer offers a good balance of performance and simplicity, a dual-pressure system with independently optimized steam-to-methane ratios for the high-pressure and low-pressure reformers allows for superior performance. This optimized dual-pressure configuration achieves a more balanced energy recovery, higher methane conversion, and a significant boost in both net power output (292.9 MW) and efficiency (53.8%) by effectively matching the different fuel pressure requirements of the two combustors and minimizing exergetic losses. The efficiency of this configuration is 2.4% higher than that achieved with a single-pressure reformer of CRGT.

The previously discussed studies primarily focused on integrating a thermochemical recuperation (TCR) system with a specific gas turbine, characterized by fixed parameters such as turbine inlet temperature and pressure ratio, which consequently define the exhaust gas temperature. However, the equilibrium methane conversion – a critical factor for TCR performance – is itself a function of operational conditions, namely temperature, pressure, and the inlet steam-to-methane ratio (the influence of these design and operational parameters on steam methane reforming is detailed in subsequent sections).

To address this, the author of the present review, Dmitry Pashchenko, conducted an energy optimization analysis of a TCR system for a gas turbine unit [96]. This work systematically investigated the impact of turbine exhaust gas temperature, system pressure, and the steam-to-methane ratio on the exhaust heat recovery rate, with key results presented in Fig. 12. The heat recuperation rate is defined as:

$$R = \frac{\Delta H_{\text{rec}}}{H_{\text{exh}}}, \tag{4}$$

where ΔH_{rec} is the heat recuperated in the TCR system; H_{exh} is the enthalpy of the exhaust gas after the gas turbine.

As Fig. 12 clearly illustrates, the recuperation rate is highly sensitive to these operational parameters and exhibits a distinct maximum within the temperature range of 900–1000 K. Furthermore, the figure demonstrates that increasing the steam content of the inlet reaction mixture beyond a steam-to-methane ratio (β) of 2.0 leads to a decrease in the recuperation rate, primarily due to the increased energy expenditure for steam generation.

In 2022 Pashchenko and his graduate students published a paper on the effect of operational parameters on the efficiency of CRGT [17]. The computational scheme is presented in Fig. 13. The study aims to determine how variables such as gas turbine inlet temperature (800–1500°C), pressure (5–25 bar), and the steam-to-methane ratio (1–3) influence the overall system efficiency and heat recuperation potential.

The methodology combines thermodynamic cycle analysis with a detailed examination of the SMR process using Aspen HYSYS simulation software. The Gibbs free energy minimization method is employed to model the reforming reactions and determine the equilibrium composition and enthalpy of the syngas. Key performance metrics, including the heat recuperation rate (HRR) and gas turbine unit (GTU) efficiency, are calculated across a wide range of conditions. The results demonstrate that increasing the turbine inlet temperature and the steam-to-methane ratio significantly enhances both the system efficiency and the amount of waste heat recuperated, with the HRR reaching up to 63% and GTU efficiency approaching 49% under optimal high-temperature conditions as shown in Fig. 14.

The findings confirm that CRGT systems offer superior performance compared to conventional steam-injected gas turbines (STIG), primarily due to the additional chemical energy recovery from the reforming process. The analysis also reveals that while higher pressure improves the gas turbine cycle efficiency, it can negatively impact the reforming process enthalpy, indicating a need for parameter optimization. The study concludes that thermochemical recuperation is most effective for high-temperature gas turbines (1200–1500°C) and highlights the potential for integrating TCR with other waste heat recovery methods to further improve performance and reduce emissions.

5.1.2. Combined cycle with CRGT

Many authors demonstrated that increasing the steam content in the steam-methane mixture fed to the reformer enhances methane conversion. However, this increase in steam concentration also raises the energy loss associated with steam generation. For instance, in an unrestricted CRGT cycle, where exhaust gas heat is used solely for steam generation and to drive the steam methane reforming process, the steam-to-methane ratio can exceed 6.0 [41,107]. A promising solution for improving CRGT efficiency is to combine it with a bottoming steam turbine cycle, where a portion of the steam is used in a Rankine cycle and another portion is diverted for steam methane reforming. The first patent for a combined cycle chemically recuperated gas turbine (CCCRGT) was granted to Basu and Rajagopal in 2001. In their patent, they describe a power generation method using an Advanced Thermal Recuperation Cycle that integrates a thermochemical recuperator (TCR) with a bottoming Rankine cycle [108]. This patent will be overviewed later in the present paper, as the authors proposed using methanol as the fuel for such a system.

In addition, Pashchenko et al. established that at low steam-to-methane ratios, the exhaust gas retains significant temperature potential [17]. This heat can be utilized, e.g., for steam generation in a bottoming steam turbine cycle. As shown in Fig. 15, the exhaust gas temperature after the TCR system remains above 400°C, confirming the potential for combining a CRGT with a steam turbine cycle to form a combined cycle chemically recuperated gas turbine.

The first attempt to analyze the integration of a gas turbine cycle, a steam methane reforming system, and a bottoming steam turbine cycle was presented in a 2004 paper by Verkhivker and Kravchenko [109]. The authors analyzed a combined cycle power plant where the exhaust gas heat from the gas turbine is used to drive the steam methane reforming process, and the remaining exhaust heat after the reformer is used to produce steam for a Rankine cycle. In their proposed system, the steam for the SMR is supplied from the flow exiting a high-pressure steam turbine. The authors confirmed the high energy efficiency of this configuration. However, they proposed not to use the produced syngas as fuel in the gas turbine, but rather to supply it to an external consumer.

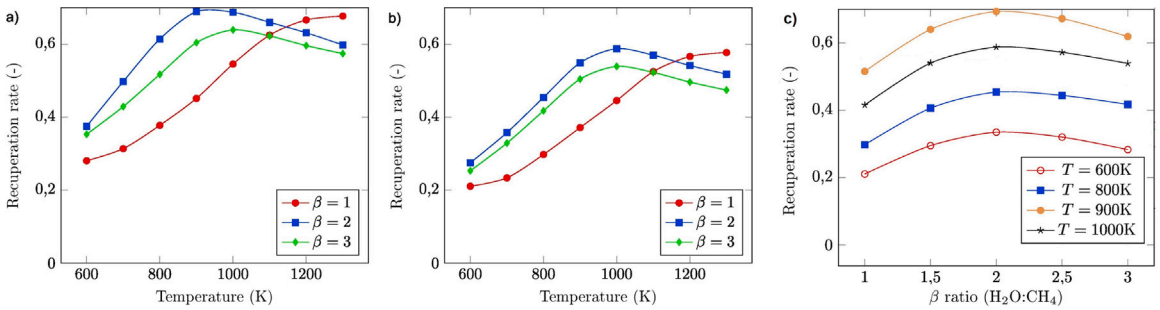


Fig. 12. Recuperation rate as a function of turbine exhaust gas temperature for various inlet steam-to-methane ratios β at pressure ratios of 5 (a) and 10 (b); recuperation rate as a function of steam-to-methane ratio (c) [96].

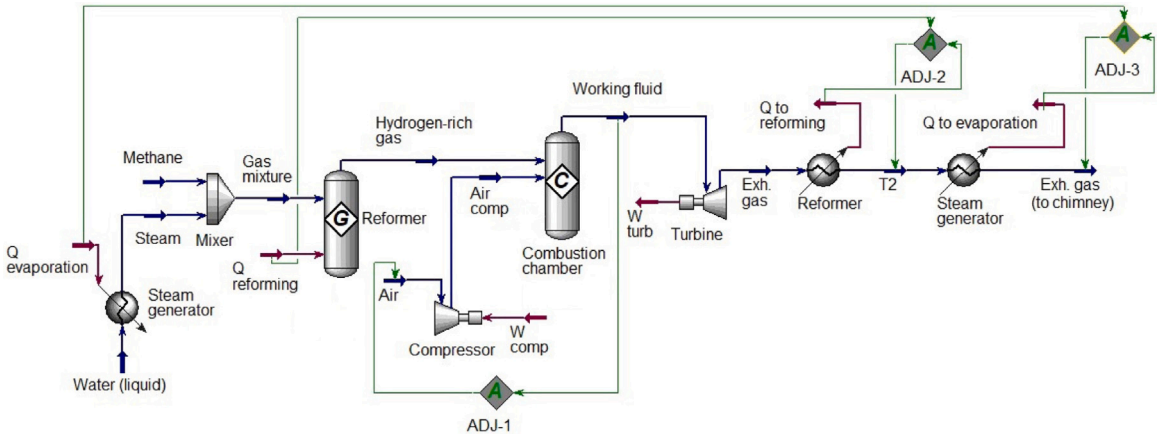


Fig. 13. The computational scheme of methane-fueled CRGT in Aspen HYSYS [17].

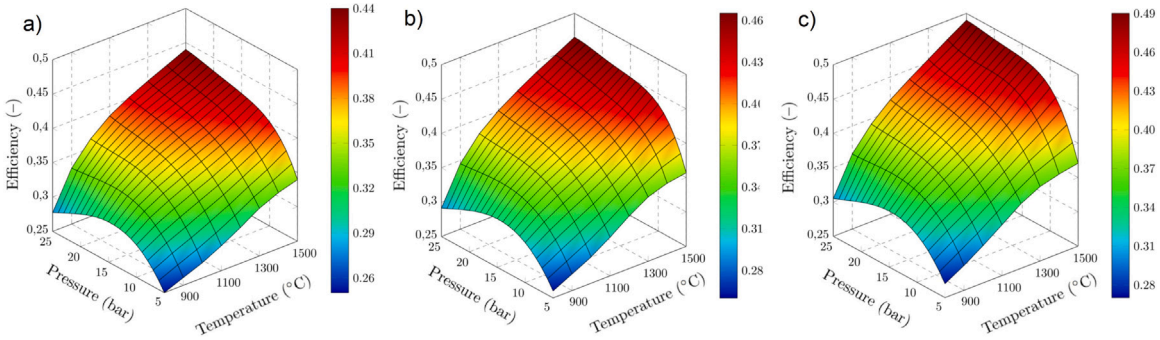


Fig. 14. The thermal efficiency of CRGT as a function of turbine inlet temperature and pressure at the steam-to-methane ratios 1.0 (a), 2.0 (b), 3.0 (c).

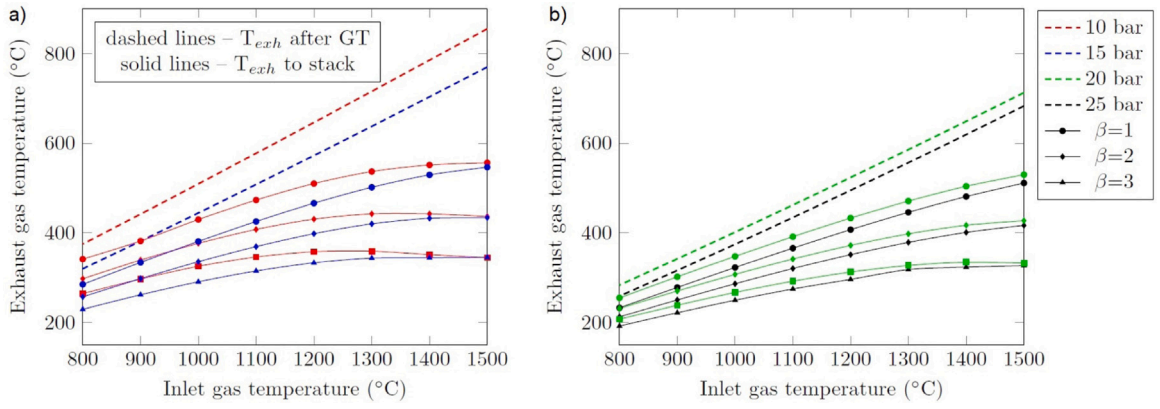


Fig. 15. Exhaust gas temperature after the gas turbine and the TCR system as a function of turbine inlet temperature (TIT) for various pressures and steam-to-methane ratios. Solid lines represent the temperature after the recuperation system (to stack); dashed lines represent the temperature after the gas turbine [17].

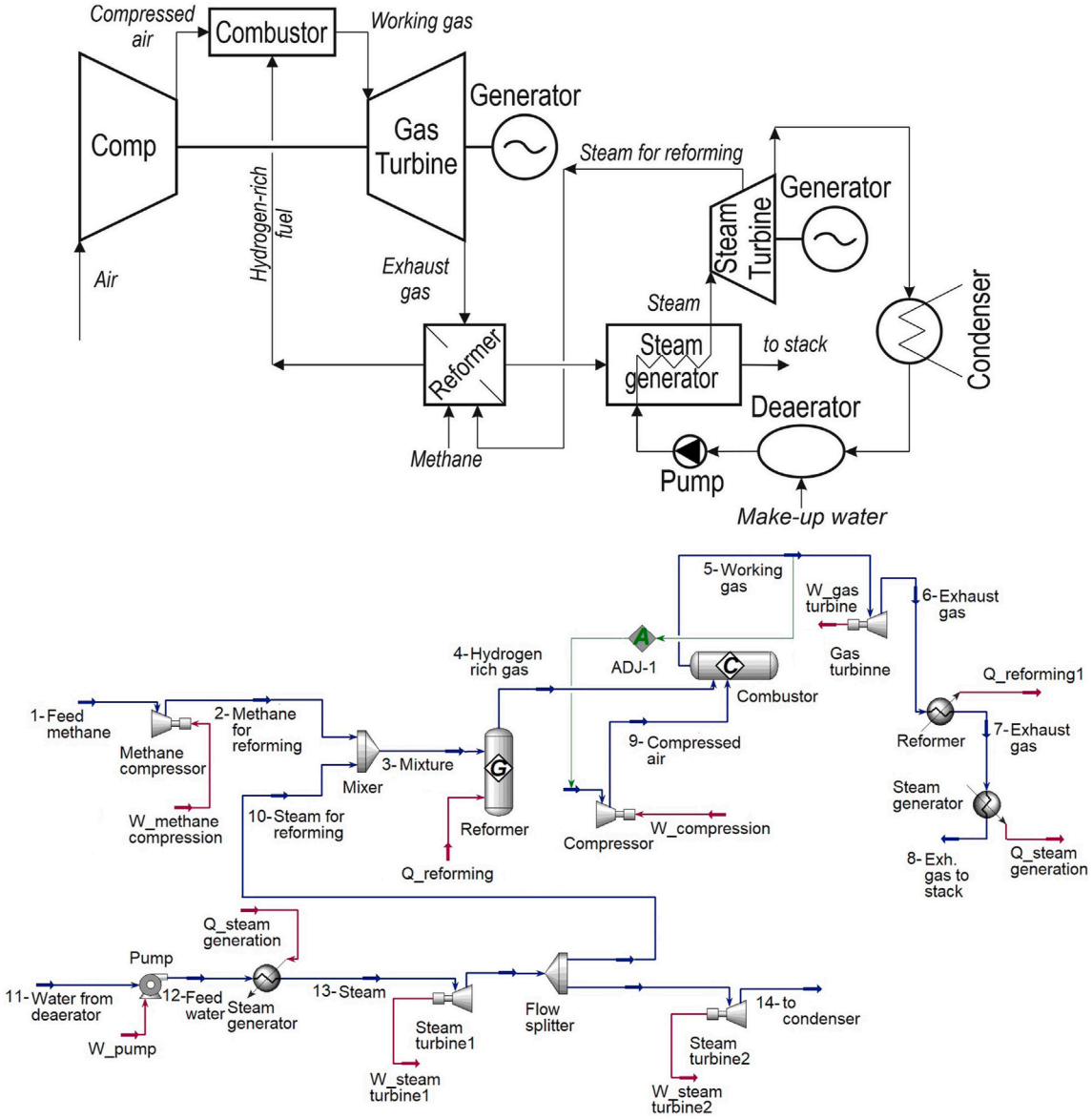


Fig. 16. Schematic diagram of a combined cycle with a chemically recuperated gas turbine (CCCRGT) and the computational scheme for its thermodynamic analysis in Aspen HYSYS [110].

The first papers on the comparative analysis of a conventional combined cycle power plant (CCPP) and a combined cycle with a chemically recuperated gas turbine (CCCRGT) were published by Pashchenko [110,111]. In these papers, a thermodynamic analysis of the CCCRGT was conducted over a wide range of operational parameters via simulation in Aspen HYSYS, using the computational scheme presented in Fig. 16.

To better understand the efficiency of the CCCRGT, its performance was compared against a conventional CCPP, where all exhaust heat is used for steam generation in the bottoming cycle. Fig. 17 shows the efficiency of the CCCRGT (a) and the resulting efficiency increase compared to the conventional CCPP (b). The analysis reveals that the efficiency of a CCPP with CRGT is 4%–7% higher than that of a conventional CCPP without TCR. This improvement is attributed to the use of exhaust heat to drive the endothermic reforming process, which transforms methane into a hydrogen-rich syngas. This new fuel has a higher combustion heat value per unit of initial methane, thereby recuperating waste energy back into the cycle.

The performance of the CCCRGT system was found to be highly dependent on key operational parameters: the gas turbine inlet temperature (T_{in}), pressure, and the steam-to-methane ratio (β). The maximum overall efficiency of 0.6412 (64.12%) was achieved at a high turbine inlet temperature of 1600°C, a pressure of 23 bar, and a steam-to-methane ratio of 3.0. The results indicate that the benefits of TCR are most pronounced in high-temperature gas turbines; for inlet temperatures below 1200°C, the efficiency gain from TCR was minimal (less than 2%). Furthermore, increasing the steam-to-methane ratio generally boosts efficiency, but this is counterbalanced by a reduction in steam turbine power due to steam extraction for the reforming process.

The main reason for these results is the utilization of exhaust heat within the gas turbine cycle, which operates at a higher efficiency than the steam turbine cycle. Consequently, the share of the overall work output from the gas turbine increases, while the share from the steam turbine cycle decreases. This shift results in a higher overall efficiency for the CCCRGT system compared to a traditional CCPP system.

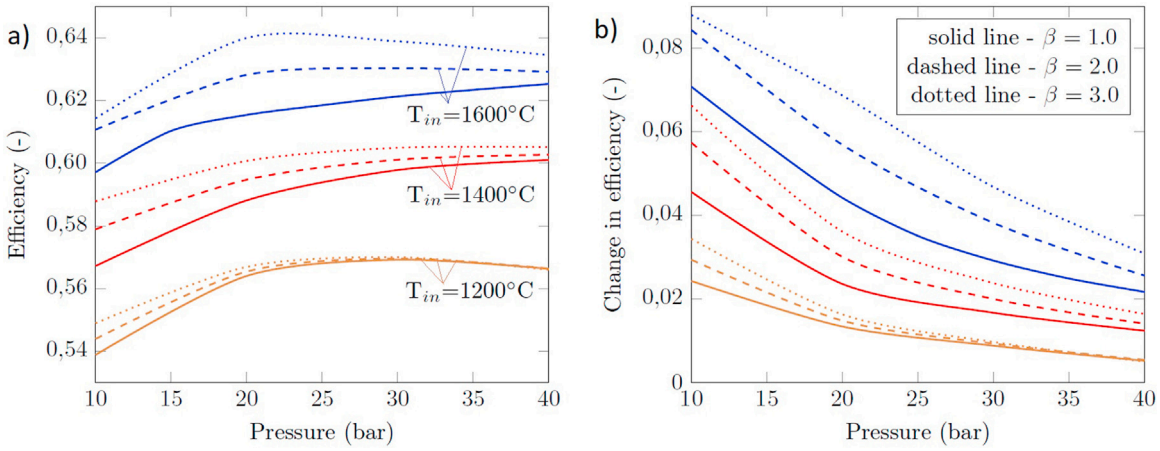


Fig. 17. (a) Efficiency of the combined cycle power plant (CCPP) with a chemically recuperated gas turbine (CRGT) as a function of operating parameters such as temperature, pressure, and inlet steam-to-methane ratio; (b) Change in the efficiency of CCPP with CRGT in comparison with CCPP without CRGT: solid lines – $\beta = 1.0$; dashed lines – $\beta = 2.0$; dotted lines – $\beta = 3.0$; blue lines – $T_{in} = 1600°C$; red lines – $T_{in} = 1400°C$; brown lines – $T_{in} = 1200°C$ [110].

An analysis of the energy flows, illustrated by a Sankey diagram for an optimal case (T_{in} = 1600°C, p = 20 bar, β = 2.0), showed that the CRGT cycle is responsible for 85.7% of the total power generation, while the steam turbine cycle contributes only 14.3%. This highlights that optimizing the chemically recuperated gas turbine cycle is more critical for the overall system performance than optimizing the steam cycle.

5.1.3. CO_2 reforming in CRGT

According to Table 2, not only steam methane reforming reaction can be considered as an endothermic process for methane reforming. Dry methane reforming has also found an research interest to be applied for chemically recuperated gas turbine. Cao and Zheng have published a set of papers where thermochemical exhaust heat recuperation in gas turbines was realized via CO_2 methane reforming process [112,113].

Cao and Zheng in their paper [113] conducted a comprehensive thermodynamic analysis of a novel zero-emission power cycle that integrates a chemically recuperated gas turbine with an ammonia absorption refrigeration system. The proposed system utilizes an O_2/CO_2 cycle where carbon dioxide serves as the working fluid and oxygen as the oxidizer. The key innovation lies in using CO_2 reforming of methane (dry reforming) for chemical recuperation, where waste heat from the turbine exhaust drives the endothermic reforming reaction, converting methane and CO_2 into syngas (CO and H_2). This approach simultaneously enhances cycle efficiency while enabling CO_2 capture.

The performance analysis revealed that the system achieves remarkable efficiency levels under optimized conditions. With a turbine inlet temperature of 1573 K and a pressure ratio of approximately 8, the cycle reaches a net power generation efficiency of 49.6% based on the lower heating value, with an exergy efficiency of 47.9%. For a methane feedstock of 1 kg s^{-1}, this translates to net electric power output of 24.799 MW while capturing 2.743 kg s^{-1} of liquid CO_2. The system also produces 0.609 MW of export chilled load, demonstrating effective utilization of waste heat across multiple temperature levels.

The distribution of waste heat recovery between the reformer and refrigeration system was carefully analyzed. At the optimal pressure ratio (π = 8), approximately 92.9% of the recovered waste heat is utilized in the CO_2-CH_4 reformer, while the remaining 7.1% drives the ammonia absorption refrigeration cycle. This distribution maximizes the conversion of low-quality heat into chemical energy through the reforming process, where the resulting CO-rich syngas releases more energy during high-temperature combustion compared to the original methane fuel.

The parametric study showed that system performance is highly sensitive to the pressure ratio and inlet gas cooling temperature. Both the net electric-power generation efficiency and exergy efficiency initially increase with pressure ratio, reaching maximum values at $\beta \approx 8$ before declining. Lowering the turbine inlet gas cooling temperature from 303 K to 273.15 K improves efficiency by 3.3% and 2.4% for power generation and exergy efficiency, respectively, demonstrating the significant benefit of the integrated refrigeration system.

The proposed cycle offers multiple technical advantages including near-zero emissions of both CO_2 and NO_x, high net cycle efficiency comparable to advanced combined cycles, and minimal water consumption. The system achieves 88.7% recovery of turbine exhaust heat, with only 11.3% remaining waste heat discharged to the environment. The integration of chemical recuperation through dry reforming with waste-heat driven refrigeration represents a sophisticated approach to energy integration that simultaneously addresses efficiency improvement and environmental concerns in power generation.

Another Cao's and Zheng's paper [112] focused on the detailed exergy analysis of novel CRGT via CO_2 methane reforming using an Energy Utilization Diagram (EUD) to reveal the thermodynamic principles behind the efficiency gains. The analysis demonstrated that reforming methane with CO_2 into syngas (CO and H_2) significantly improves the combustion process; compared to burning pure methane, the combustion of syngas releases 37.8% more heat and 30.6% more exergy while simultaneously reducing the exergy loss of the combustion itself by 25.5%. Furthermore, the system achieved an overall exergy regeneration ratio of 84.2% for the turbine exhaust heat, with the high-temperature segment of the heat recovery in the reformer alone achieving an impressive exergy regeneration ratio of 91.7%.

The integrated system also effectively utilizes lower-temperature waste heat, producing a net export chilled load of 1.070 MW. The inlet cooling of the working fluid, which consumed 2.156 MW of this chilled load, was found to reduce the power consumption of the CO_2 compressor by 1.674 MW, enhancing the net power generation efficiency by 3.3%. Crucially, the cycle achieves its environmental goal by capturing 2.743 kg s^{-1} of liquid CO_2 without consuming extra power, resulting in a design with zero CO_2 emissions.

Further development of chemically recuperated gas turbine where methane reforming using CO_2 as an oxidant has not garnered widespread interest among researchers. The following key reasons can be identified:

- The cost of CO_2 is significantly higher than that of steam.
- Achieving high methane conversion requires a higher temperature level compared to steam methane reforming. Consequently, the application of CO_2 methane reforming is limited by the temperature of the exhaust gas from the gas turbine.

5.2. Natural gas reforming in CRGT

Natural gas reforming is a well-studied process, as it is a key pathway for hydrogen and synthesis gas production. Industrial large-scale steam reforming of natural gas occurs at high temperatures (above 1000°C), high pressure, and a high steam-to-methane ratio, primarily to achieve high methane conversion and to suppress carbon formation and deposition. The distinguishing feature of using steam methane reforming in a chemically recuperated gas turbine is the moderate temperature level of the exhaust gas (approximately 600–700°C), pressure not exceeding 20–30 bar, and a moderately low steam-to-methane ratio (typically no more than 3.0). Therefore, the key focus for the application of steam methane reforming in this context should be directed towards the following points:

- achieving high methane conversion under moderately constrained operating parameters;
- minimizing carbon formation and deposition on catalytic surfaces to enhance the stability of the steam methane reforming process.

Natural gas reforming is a catalytic and endothermic process. The composition and enthalpy of the reforming products depend on both operational parameters – such as temperature, pressure, and the inlet steam-to-methane ratio – and design parameters, including the type of catalyst, method of heat supply, and reformer configuration. Thermodynamic analysis establishes the relationship between operational parameters and the equilibrium composition and enthalpy of the process. To account for the effects of design parameters, experimental and numerical studies are required. Therefore, although natural gas reforming is a well-studied process, there remains significant interest in reviewing papers focused on its thermodynamic analysis, as well as associated heat and mass transfer studies, specifically for application in thermochemical recuperation systems for gas turbines.

5.2.1. Thermodynamic analysis

The thermodynamic analysis of steam methane reforming (SMR) via the Gibbs free energy minimization method is a powerful approach for predicting the equilibrium composition of the product syngas without requiring detailed reaction kinetics. The fundamental principle is that a chemical system at equilibrium minimizes its total Gibbs free energy, G^{sys}, for a given temperature and pressure.

The total Gibbs free energy of the system is the sum of the chemical potentials of all components:

$$G^{sys} = \sum_{i=1}^{N} n_i \mu_i, \tag{5}$$

where n_i is the number of moles of component i and μ_i is its chemical potential.

The chemical potential for each component is given by:

$$\mu_i = \Delta_f G_i^0 + RT \ln\left(\frac{\hat{f}_i}{f_i^0}\right) \tag{6}$$

where $\Delta_f G_i^0$ is the standard Gibbs free energy of formation of species i, R is the universal gas constant, T is the system temperature, $\hat{f}_i$ is the fugacity of species i in the mixture, f_i^0 is the standard state fugacity (typically 1 bar).

For an ideal gas mixture, the fugacity can be replaced by the partial pressure, $\hat{f}_i = y_i P$, where y_i is the mole fraction and P is the total pressure. The minimization is performed subject to the constraints of elemental mass balance for atoms of C, H, and O.

The equilibrium condition is mathematically expressed as:

$$(dG^{sys})_{T,P} = 0 \tag{7}$$

This non-linear constrained optimization problem is typically solved using numerical algorithms, such as the method of Lagrange multipliers, to find the set of mole numbers n_i that satisfy this condition. This method directly yields the equilibrium concentrations of all species (e.g., CH_4, H_2O, CO, CO_2, H_2) and allows for the calculation of key performance metrics like methane conversion and the enthalpy of the reforming process.

Table 7
Possible chemical reactions in the reformer [114].

No.	Reaction	ΔH_{298}, (kJ/mol)
	Methane oxidation	
1	$CH_4 + H_2O \Leftrightarrow CO + 3H_2$	206.1
2	$CO + H_2O \Leftrightarrow CO_2 + H_2$	−41.1
3	$CH_4 + CO_2 \Leftrightarrow 2CO + 2H_2$	247.3
	Carbon formation	
4	$CH_4 \Leftrightarrow C + 2H_2$	74.8
5	$2CO \Leftrightarrow C + CO_2$	−173.3
6	$CO + H_2 \Leftrightarrow C + H_2O$	−90.13
7	$CO_2 + 2H_2 \Leftrightarrow C + 2H_2O$	−131.3

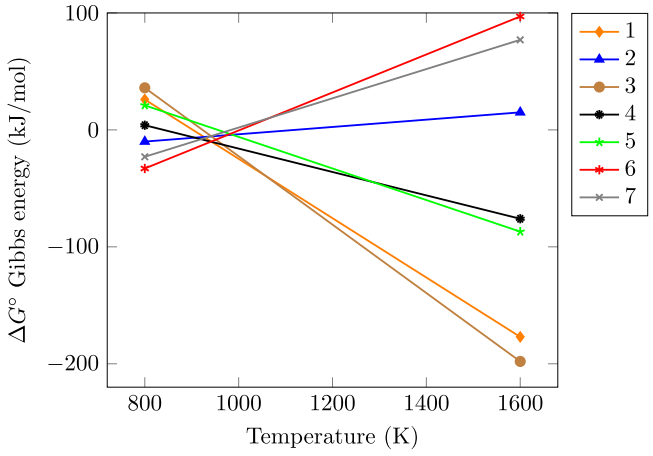


Fig. 18. Change of Gibbs free energy as a function of temperature for reactions in Table 7.

The steam methane reforming (SMR) process takes place in a thermochemical reformer-recuperator. The overall reaction to produce a new synthetic fuel, primarily composed of hydrogen (H_2) and carbon monoxide (CO), can be described by several chemical equations, as listed in Table 7 [81,114]. The reactions in Table 7 are reversible, and the change in Gibbs free energy (ΔG) indicates the thermodynamically favorable direction of each reaction at equilibrium. As shown in Table 7, side reactions leading to carbon formation are also possible. To clarify the reaction directions, the change in Gibbs energy as a function of temperature was calculated and is presented in Fig. 18.

The reaction numbers in Table 7 correspond to those in Fig. 18. Fig. 18 clearly shows that across the studied temperature range, the chemical reactions for methane reforming are thermodynamically more favorable than the carbon formation reactions. Furthermore, the change in Gibbs energy for the water-gas shift reaction (Reaction 2 in Table 7) is minimal, indicating that its overall contribution to the material and energy balance of the system is negligible.

Thermodynamic analysis enables the determination of equilibrium methane conversion and the resolution of the heat balance. Using Aspen HYSYS, the effect of operational parameters on composition and heat balance of thermochemical exhaust heat recuperation system of gas turbine was determined under the following initial conditions: $t^{in}_{CH4} = t^{in}_{fw} = 25°C$; $p^{in}_{CH4} = p^{in}_{fw} = p_{ref}$; $t_{steam} = t_{s,p}$.

The effect of temperature, pressure, and the inlet steam-to-methane ratio on the equilibrium CH_4 conversion is presented in the heatmap in Fig. 19. This figure enables a preliminary assessment of the feasibility of using thermochemical exhaust heat recuperation in a gas turbine, based on its exhaust gas temperature. The equilibrium conversion represents

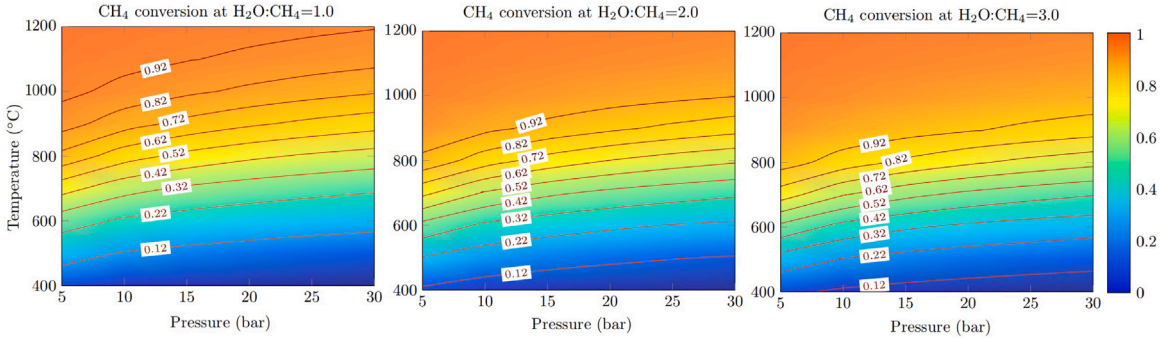


Fig. 19. Equilibrium CH_4 conversion as a function of temperature and pressure for various H_2O:CH_4 ratio (heatmap diagram where value of CH_4 conversion is represented as colors in a grid).

the maximum achievable methane conversion for a given set of these operational parameters.

Fig. 19 shows clear effect of temperature, pressure and steam-to-methane ratio on CH_4 conversion. The steam methane reforming process is highly endothermic, meaning it requires a substantial input of heat to proceed. According to Le Chatelier's principle, a system at equilibrium will shift to counteract any imposed change. Therefore, an increase in temperature provides energy that the reaction absorbs by favoring the forward reaction — the production of hydrogen and carbon monoxide. This shift directly results in a higher equilibrium conversion of methane. In industrial practice, this is why SMR is conducted at high temperatures, typically above 700–750°C. Operating at these elevated temperatures is essential to achieve economically viable conversion levels, as lower temperatures would leave a significant amount of unreacted methane, reducing process efficiency and hydrogen yield.

In contrast to temperature, the reforming reaction leads to a large increase in the number of gas molecules (one mole of CH_4 and one mole of H_2O produce four moles of products). Le Chatelier's principle dictates that an increase in pressure will shift the equilibrium towards the side with fewer gas molecules to relieve the stress. Consequently, higher pressure negatively impacts the equilibrium conversion of methane by favoring the reverse reaction. From a purely equilibrium standpoint, the ideal pressure for maximizing conversion would be as low as possible. However, practical engineering and economic considerations make operating at elevated pressure (typically 10–30 bar) necessary. High pressure reduces the size of equipment for a given throughput, lowers the cost of compressing the final hydrogen product, and provides a more favorable driving force for the subsequent water-gas shift reaction. Thus, a compromise is struck between the unfavorable equilibrium and the significant operational advantages of pressurized systems.

The steam-to-carbon (H_2O:CH_4) ratio is a crucial operating variable that can be used to manipulate the reaction equilibrium without the negative trade-offs associated with pressure. An increase in H_2:CH_4 ratio means supplying more steam relative to methane. Le Chatelier's principle states that increasing the concentration of a reactant will shift the equilibrium to the right to consume the added reactant. Therefore, a higher H_2O:CH_4 ratio directly drives the reforming reaction forward, leading to higher equilibrium methane conversion. Furthermore, excess steam helps suppress carbon formation (coking) on the catalyst, which is a major cause of deactivation. While a higher H_2:CH_4 ratio improves conversion and catalyst stability, it also comes with an energy penalty, as more steam must be heated to the high reaction temperature, increasing operational costs. Typically, steam methane reformers operate with an H_2O:CH_4 ratio between 2 and 3 to balance high methane conversion with energy efficiency.

The operational parameters also have a significant effect on the heat balance of the steam methane reforming process. Fig. 20 shows the absolute component contributions (upper plots) and their relative percentage contributions (lower plots) to the overall heat balance at 15 bar for various steam-to-methane ratios as a function of temperature. The following energy components are considered: the enthalpy for steam generation (latent heat), the sensible heat for raising the temperature of the steam-methane mixture to the reforming temperature, and the reaction enthalpy for the reforming process itself.

Fig. 20 shows that the total heat requirement for the steam methane reforming process increases significantly with both temperature and the steam-to-methane (S/C) ratio. As temperature rises from 400°C to 1200°C, the total enthalpy load grows substantially for all S/C ratios, driven by the highly endothermic nature of the reforming reaction. Furthermore, at any given temperature, a higher S/C ratio demands more energy. For instance, at 800°C, the total heat required jumps from 3.8 MJ/kg (S/C = 1.0) to 5.8 MJ/kg (S/C = 2.0) and 7.7 MJ/kg (S/C = 3.0), highlighting the considerable energy penalty associated with using excess steam.

The contribution of each component to the overall heat balance shifts dramatically with temperature. At lower temperatures (e.g., 400–500°C), the largest energy demand comes from steam generation, which is a fixed value for each S/C ratio. However, as temperature increases, the endothermic heat of the SMR reaction itself becomes the dominant contributor. For example, at S/C = 2.0 and 1200°C, the reaction enthalpy (3.88 MJ/kg) is more than double the energy needed to heat the steam (1.81 MJ/kg). The energy required to heat the methane-steam mixture also rises steadily with temperature but consistently remains a secondary factor compared to the SMR reaction enthalpy at high temperatures. This analysis underscores that while steam pre-heating is a major initial energy sink, the reforming reaction is the primary driver of the total energy demand at the high operating temperatures required for significant methane conversion.

Thermodynamic analysis is essential for predicting and preventing carbon formation within thermochemical recuperators, a critical issue often overlooked in studies of steam methane reforming (SMR). By evaluating the chemical equilibria of reactions listed in Table 7, it is possible to define operational boundaries that suppress carbon deposition. These conditions are expressed through the following inequalities governing partial pressures (P) and equilibrium constants ($K_{eq,i}$), where the subscript i corresponds to the reaction numbers in Table 7:

- For temperatures above ~950 K:

$$\frac{P_{H_2}^2}{P_{CH_4}} > K_{eq,4}; \quad \frac{P_{CO_2}}{P_{CO}^2} > K_{eq,5}; \tag{8}$$

- For temperatures below ~950 K:

$$\frac{P_{CO} \cdot P_{H_2}}{P_{H_2O}} < K_{eq,6}; \quad \frac{P_{CO_2} \cdot P_{H_2}^2}{P_{H_2O}^2} < K_{eq,7}. \tag{9}$$

Fig. 21 illustrates the minimum operating temperature required to avoid carbon formation as a function of the steam-to-methane ratio (β) at different pressures [115,116]. The region to the left of each curve

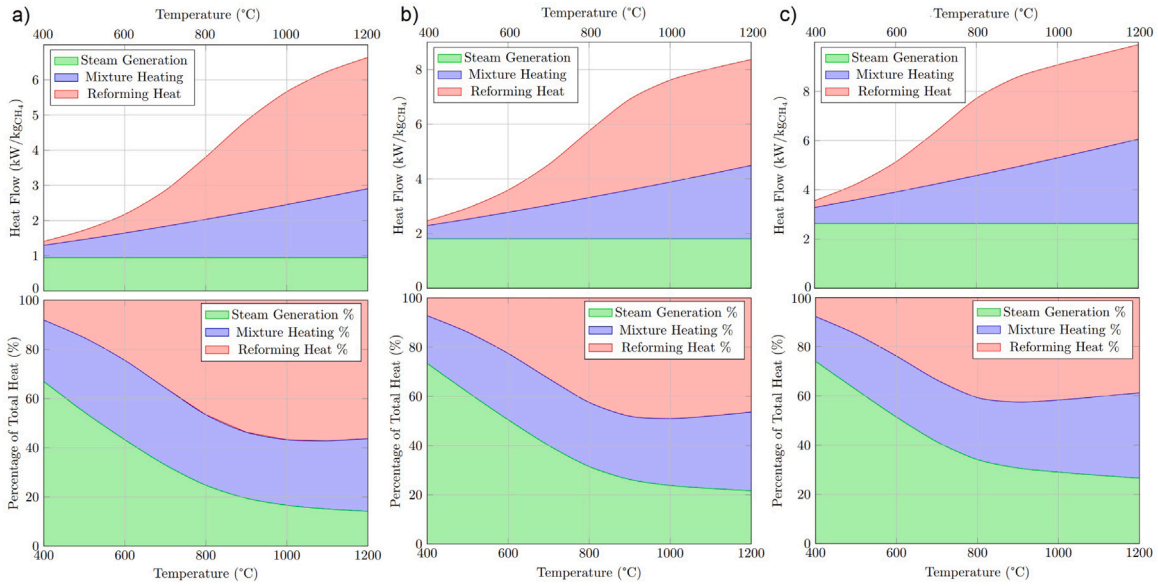


Fig. 20. Heat balance of the steam methane reforming process at p = 15 bar. The upper plots show the absolute component contributions per 1 kg of inlet methane; the lower plots show the corresponding percentage contributions for: (a) H_2O:CH_4 = 1.0, (b) H_2O:CH_4 = 2.0, (c) H_2O:CH_4 = 3.0.

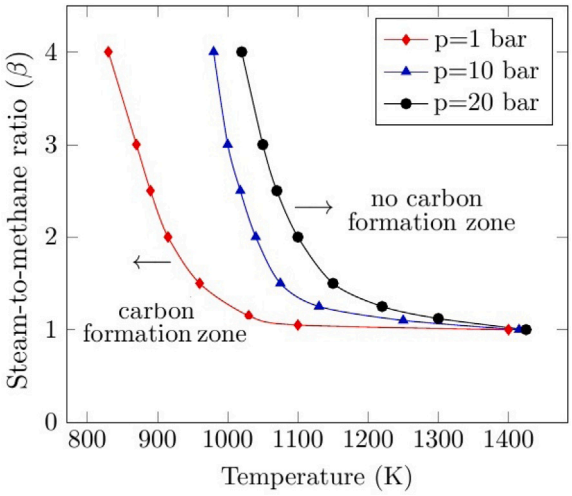


Fig. 21. Boundary conditions for carbon formation at the chemical equilibria.

represents conditions where carbon deposition is thermodynamically favored. Conversely, operating at temperatures and steam-to-methane ratios to the right of the curve for a given pressure ensures a carbon-free regime. The graph demonstrates that increasing the system pressure shrinks the carbon formation zone, which explains why industrial-scale hydrogen production via SMR typically operates above 10 bar [117].

5.2.2. Catalyst

The distinctive feature of the use of steam methane reforming process for thermochemical recuperation in gas turbines is different goal comparing to the use of steam methane reforming, e.g. for hydrogen production. Thus, despite the SMR is well studied process for hydrogen production, in this section, the main focus will be forwarded on the use of SMR for thermochemical recuperation under moderate operating conditions and large scale production.

Nickel-based catalysts are a prevalent, cost-effective choice for large-scale steam methane reforming, typically supported on oxides like Al_2O_3, MgO, CeO_2, or La_2O_3. However, a primary challenge is their susceptibility to deactivation via high-temperature sintering and carbon deposition. Previous author's researches show that Ni-based catalyst provides an opportunity to obtain high methane conversion [115,118,119]. At the same time, research indicates that carbon formation is strongly linked to nickel particle size, with a critical threshold near 2 nm below which carbon accumulation is significantly mitigated [120–122]. To enhance stability, various strategies have been explored. These include employing advanced supports such as anatase TiO_2 nanosheets or dealuminated zeolites [123–126], improving metal dispersion, adding promoters, and developing novel catalyst configurations like Ni-vanadate oxides or TiN-promoted systems [127, 128].

The steam methane reformer with a catalyst packed bed is a widely used, large-scale technique for steam methane reforming. The author's previous paper presents an experimental investigation into the effects of catalyst particle shape on the performance of such a reformer, focusing on both methane conversion and pressure drop [129]. The study tests four distinct catalyst shapes – simple cylinders, Raschig rings, 7-hole cylinders, and 7-hole spheres – using a Ni-αAl_2O_3 catalyst in a cylindrical packed bed reactor of 100 mm diameter. The research aims to address a gap in the literature by simultaneously examining reforming characteristics and flow dynamics, which are often studied separately. By systematically varying operational parameters such as temperature, steam-to-methane ratio, and residence time, the study provides a comprehensive comparison of how different catalyst geometries influence reactor efficiency and hydraulic performance.

A key finding is that catalyst shape significantly impacts methane conversion 22, with the 7-hole spheres achieving the highest conversion rates up to 23% greater than simple cylinders under identical conditions. The authors attribute this improvement to the larger geometric surface area of the 7-hole spheres, which enhances reactant-catalyst contact. To better evaluate catalyst effectiveness, the study introduces two novel parameters: weight residence time (kg_{cat} min mol^{-1}) and surface residence time (m^2 min mol^{-1}). The results indicate that, when compared at the same surface residence time, the catalytic activity per unit surface area is nearly identical across all shapes. This suggests that increasing the geometric surface area of the catalyst bed – for instance, by using smaller or more complex particles – can improve reforming performance without altering the intrinsic activity of the catalyst material.

Pashchenko's research demonstrates that optimizing catalyst shape can enhance reformer efficiency by maximizing surface area and min-

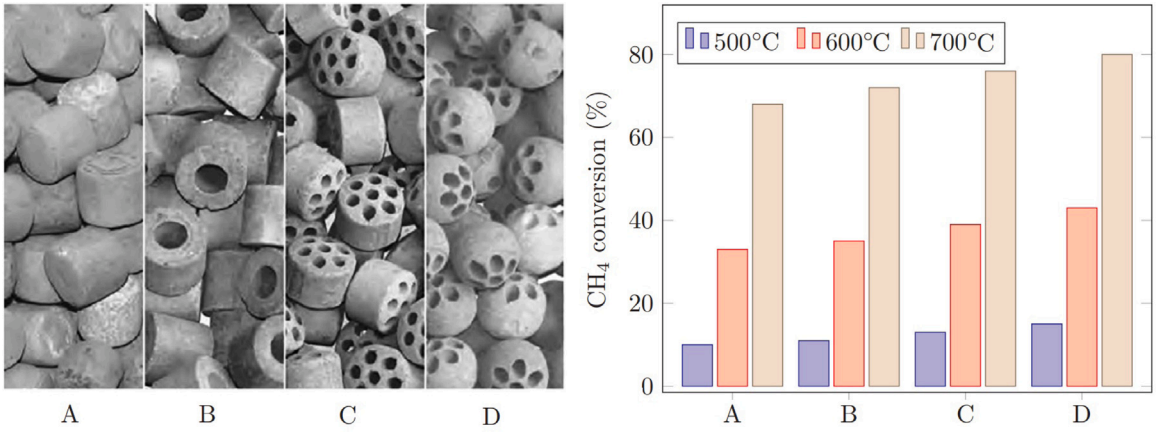


Fig. 22. Photo of packed bed filled with various Ni-αAl_2O_3 catalyst particles: A – cylinders; B – Raschig rings; C – 7-hole cylinders; D – 7-holes spheres. Methane conversion for various operating parameters: temperature of 500–700°C; H_2O:CH_4 = 1 : 1, p = 3 bar [129].

Table 8
Summary of some papers on study of steam methane reforming in packed bed.

Catalyst shape	Catalyst properties	Operating conditions	X_{CH4}	Ref.
Cylinder Raschig ring 7-hole cylinder 7-hole spheres	NiO = 14.5% SiO_2 = 0.2% Support CaO–MgO–La_2O_3–αAl_2O_3	T = 700°C P = 0.3 MPa S/C = 1 Res.time = 8 kg_{cat}min/mol	62 69 75 81	[129]
1-hole cylinder	Ni = 20 wt%, CaO–Al_2O_3 = 80 wt%	Equilibrium X_{CH4} reached at GHSV = 2 l_{CH4}/(h g_{cat}), T > 800°C, P = 1 MPa, S/C = 3		[130]
Cylinder	E-5Ni E-3.5Ni E-3.5Ni-5Mg	T = 700°C S/C = 3 WHSV = 3000 mL/(g_{cat} h)	96 96 95	[131]
Plate	NiO = 15.87 wt%	Equilibrium X_{CH4} reached at T = 900°C, GHSV = 240,000 h^{-1} P = 2 MPa, S/C = 3	[132]	

imizing pressure losses. The 7-hole spherical particles offer the best balance between high methane conversion and manageable flow resistance. These insights are valuable for the design of more efficient steam methane reforming systems, especially in applications where energy efficiency and operational costs are critical. The study provides practical guidance for selecting catalyst geometries and supports the use of surface-based residence time as a key parameter for evaluating packed-bed reactor performance.

In the context of applying steam methane reforming (SMR) to chemically recuperated gas turbine (CRGT) cycles, achieving high methane conversion under moderate operating conditions is crucial for overall system efficiency. Table 8 summarizes research findings on methane conversion X_{CH4} in reformers utilizing a packed-bed configuration. The data indicate that with modern industrial Ni-based catalysts, methane conversion approaching the thermodynamic equilibrium can be attained at temperatures as low as 700°C and with low steam-to-carbon (S/C) ratios.

These moderate conditions are particularly significant for integration with gas turbines. Modern gas turbines operate with high turbine inlet temperatures, resulting in exhaust gas temperatures of approximately 700°C. This thermal energy in the exhaust is capable to drive the endothermic SMR reaction directly. Therefore, the advanced design of large-scale packed-bed reformers, filled with high-performance Ni-based catalysts, enables highly efficient pre-combustion fuel reforming. This process facilitates thermochemical exhaust gas recuperation, where waste heat is upgraded to the chemical energy of the reformed fuel (a hydrogen-rich syngas), thereby enhancing the cycle's efficiency and reducing its carbon intensity.

5.3. Summary of Section 5

The primary advantages of natural gas-fed CRGT systems are twofold. First, they achieve significant thermodynamic efficiency gains by upgrading waste heat into chemical fuel energy, often outperforming steam-injected cycles. Second, the hydrogen-rich reformed fuel enables ultra-low NO_x emissions due to favorable combustion properties and the potential for steam dilution. Research has evolved from analyzing simple cycles to sophisticated combined-cycle configurations (CCCRGT), with the latter demonstrating the potential for overall efficiencies exceeding 60%, as the recuperated heat is utilized within the higher-efficiency gas turbine loop.

However, the technology presents some challenges. A key challenge is the inherent complexity and energy penalty associated with generating the steam required for the steam methane reforming (SMR) reaction, which is subsequently discharged to the atmosphere after combustion. Furthermore, system performance is highly sensitive to operational parameters like pressure, temperature, and steam-to-methane ratio, requiring careful optimization to balance high methane conversion against reformer efficiency and to avoid catalyst deactivation through carbon formation. The successful implementation of CRGTs hinges on advanced catalyst development for moderate operating conditions and the integration of efficient, large-scale reformers with the gas turbine heat recovery system.

6. Ammonia-fueled CRGT

A wide discussion of the use of ammonia as a fuel, including in gas turbines, has been started in the recent decade. Despite the ammonia

production technology is well established and well known, the use of ammonia as a fuel for power generation is moderate new research topic. Recently, Valera-Medina et al published a comprehensive overview for the use of ammonia for power generation [2]. In addition, the interest for ammonia was significantly increased when NH_3 was realized as a nice hydrogen carrier at the beginning of epoch of hydrogen as an energy carrier. One molecule of ammonia contains 3 atoms of hydrogen without carbon atoms. Moreover, ammonia makes it possible to solve two main challenges in hydrogen as a fuel story — storage and transportation systems. Ammonia can be easily transported similarly to natural gas by gas carrier ships, pipelines, etc.

Despite its advantages, using ammonia directly as a combustion fuel, including for gas turbines, faces several significant challenges:

- Low reactivity and combustion instability:
  - Low laminar burning velocity: Ammonia's flame speed is very low (around 8 cm/s, more than four times slower than methane as shown in Fig. 2), making it difficult to stabilize flames and leading to incomplete combustion.
  - High ignition energy: The energy required to ignite an ammonia-air mixture is 2–3 orders of magnitude higher than for common hydrocarbons.
  - Narrow flammability limits: This further complicates stable combustion across a wide range of operating conditions.
- High nitrogen oxide (NO_x) emissions:
  - This is the primary environmental challenge. The fuel-bound nitrogen in ammonia can easily be converted into various NO_x species during combustion, which are major air pollutants.
  - Controlling these emissions is a central focus of research, with strategies like operating under fuel-rich conditions to promote the recombination of NO_x with unburned ammonia.
- Production of unburned ammonia and nitrous oxide (N_2O):
  - The slow combustion can lead to significant slip of unburned ammonia into the exhaust, which is a toxic gas.
  - Nitrous oxide (N_2O), a potent greenhouse gas, can also be formed as an intermediate species.
- System modifications and dual-fuel requirements:
  - Due to its poor combustion properties, pure ammonia often requires blending with more reactive "combustion promoters" like hydrogen (which can be cracked from ammonia itself) or methane. This adds complexity to the fuel system.
  - Existing combustion devices (engines, turbines) designed for hydrocarbons need significant modification to run efficiently on ammonia or its blends.

In summary, ammonia is a promising carbon-free energy vector because it contains no carbon and can be produced renewably. However, its direct use in gas turbine is hampered by its inherently slow and unstable combustion characteristics, which lead to challenges in maintaining efficiency while controlling the emission of nitrogen-based pollutants like NO_x and unburned NH_3.

6.1. Overview

The promising concept of using ammonia as a gas turbine fuel involves its pre-combustion decomposition into a hydrogen-rich gas. As shown in Fig. 2, even 10% ammonia decomposition significantly increases the flame speed. A distinguishing feature of ammonia decomposition is its lower required temperature compared to the steam methane reforming reaction. Furthermore, it does not produce side reactions that generate solid carbon particles, which are a primary cause of catalyst deactivation.

The author of this paper was among the first to propose and analyze using gas turbine exhaust gas heat for ammonia decomposition and subsequently using the decomposition products as fuel for the gas turbine. Thus, Pashchenko is a pioneer of the ammonia-fueled chemically recuperated gas turbines. A schematic diagram of this concept is presented in Fig. 23. In this configuration, exhaust gas heat is used to drive the ammonia decomposition reaction. The product of this reaction is a hydrogen-rich gas, which is then used as the fuel in the gas turbine.

The concept of an ammonia-fueled chemically recuperated gas turbine is based on the principles of thermochemical fuel transformation using exhaust heat. In this system, exhaust heat is converted into the chemical energy of the ammonia decomposition products. In traditional ammonia-fueled gas turbines, the chemical energy of ammonia is converted into heat in a single stage: direct combustion. In the chemically recuperated cycle, this conversion can be divided into two virtual stages, as presented in Fig. 24. The first stage is the thermochemical transformation of primary ammonia into a hydrogen-rich gas. This stage can be described as a "chemical pump" because it converts the heat of the exhaust gas into the chemical energy of a new, more reactive fuel. The second stage is the combustion of this hydrogen-rich fuel, which has a higher heating value per kilogram of primary ammonia than the original ammonia. A comparison of the chemical energy conversion for traditional (a) and chemically recuperated (b) gas turbines is presented in Fig. 24.

Pashchenko's research group was the first who published a paper on thermodynamic analysis of ammonia-fueled chemically recuperated gas turbine [133]. The study demonstrates that integrating the thermochemical recuperation system significantly enhances the cycle's thermal efficiency compared to a conventional ammonia-fired gas turbine without recuperation. The efficiency gain is particularly pronounced at higher turbine inlet temperatures (above 800°C), as the system more effectively converts waste heat into chemical energy, thereby increasing the heat of combustion of the fuel entering the combustor. The computational scheme of thermodynamic analysis of ammonia-fueled CRGT realized via Aspen HYSYS is presented in Fig. 25.

It was established that the thermal efficiency of the ammonia-fueled CRGT is highly dependent on the operating pressure and turbine inlet temperature, with a distinct optimal pressure for each temperature to achieve maximum efficiency. For instance, the maximum efficiency occurs at 9 bar for a 700°C inlet temperature, shifting to 15 bar for a 900°C inlet. The analysis reveals that while increasing pressure and temperature generally improves efficiency, the gains diminish above 15 bar for temperatures over 1000°C. The maximum CRGT efficiency reported for the analyzed parameters is 44% at 1300°C and 21 bar, substantially higher than a baseline gas turbine without TCR as shown in Fig. 26. As shown in Fig. 26, the thermochemical exhaust recuperation system increased the gas turbine's thermal efficiency by up to 9% across the turbine inlet temperature range of 800–1300°C.

Additionally, Pashchenko et al. reported [133] that the thermochemical recuperation system itself can recover up to 43% of the exhaust heat, with this recovered energy split approximately equally between the energy required for the decomposition reaction (in the reformer) and the sensible/latent heat for preheating and regasifying the ammonia (in the heater and regasifier). The study concludes that the high temperature (300–450°C) of the exhaust gases even after the TCR system indicates further potential for efficiency improvements through the integration of additional bottoming cycles, such as steam cycles.

In 2023 Su et al. proposed a novel ammonia-driven chemically recuperated gas turbine (CRGT) cycle based on a dual-fuel mode [134]. The central innovation is the use of a portion of the turbine's exhaust heat to drive a partial ammonia cracking process, producing a hydrogen-rich gas. This gas is then mixed with the remaining ammonia to create a dual fuel (NH_3/H_2) for combustion. This design directly addresses

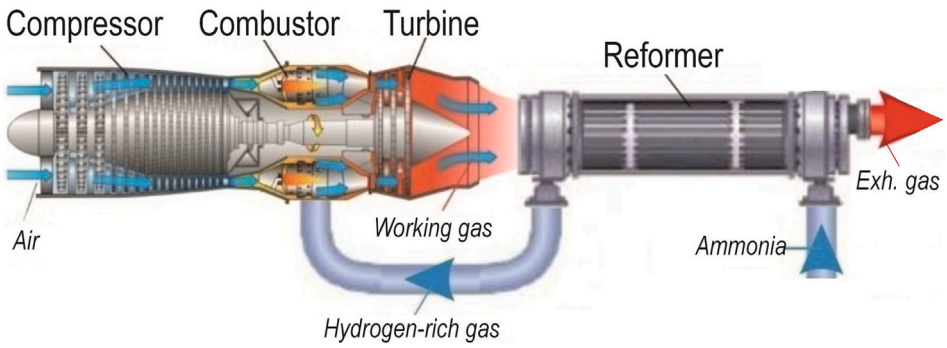


Fig. 23. Schematic diagram of ammonia-fueled chemically recuperated gas turbine.

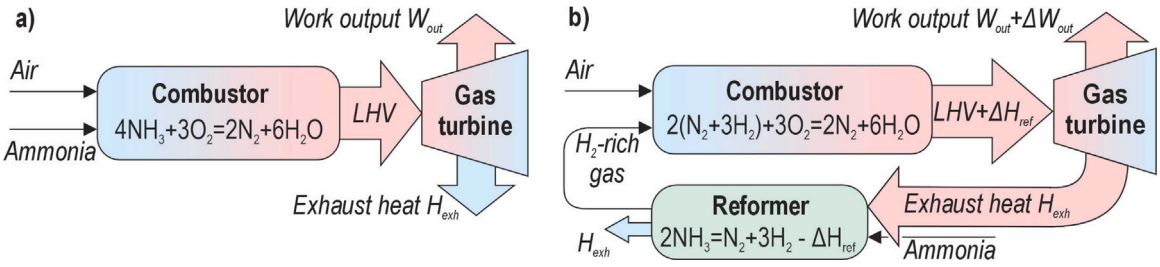


Fig. 24. Conversion of chemical energy of ammonia into heat: (a) traditional gas turbine; (b) chemically recuperated gas turbine.

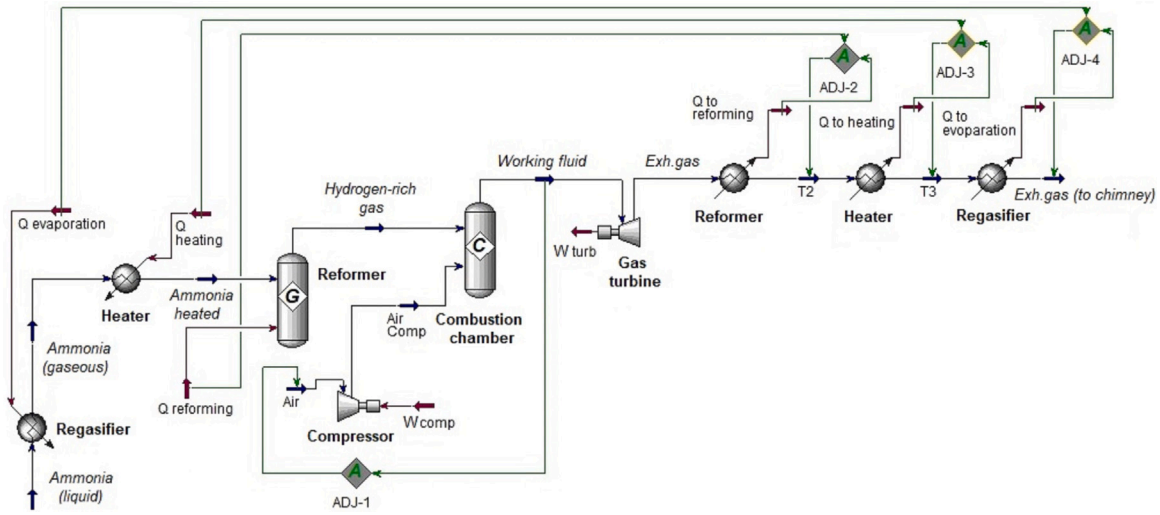


Fig. 25. Computational scheme of ammonia-fueled chemically recuperated gas turbine in Aspen HYSYS [133].

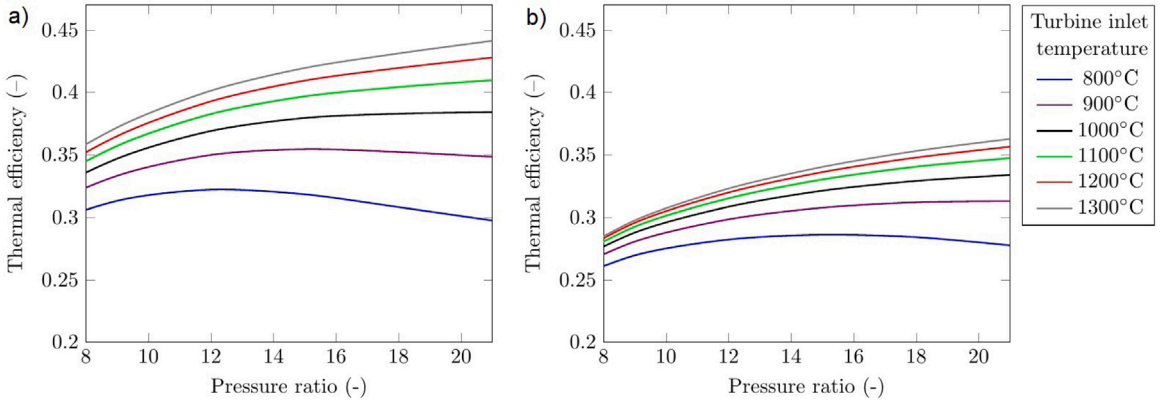


Fig. 26. Thermal efficiency of ammonia-fueled gas turbine with (a) and without (b) thermochemical exhaust heat recuperation system for various turbine inlet temperatures.

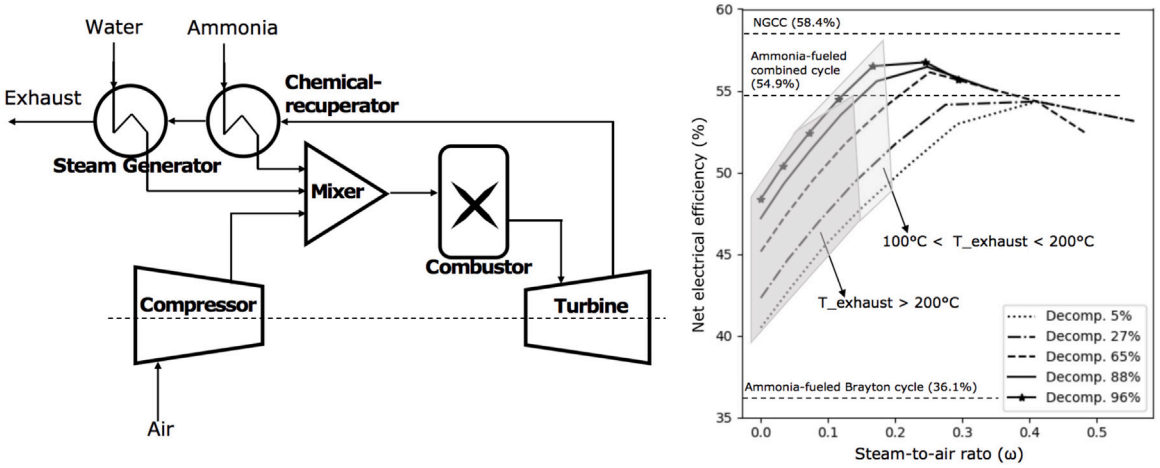


Fig. 27. The schematic diagram of a novel power cycle concept termed the chemical-recuperated and humidified gas turbine (CHGT) and its efficiency [135]. (Reprinted from [135] with permission from Energy journal, Elsevier.)

the key weaknesses of using pure ammonia (low reactivity, low flame speed leading to incomplete combustion) and pure hydrogen (safety hazards, combustion instability) by leveraging their complementary characteristics.

The study demonstrates that this system achieves a significant boost in thermal performance. Under the base case, the new design achieves a thermal efficiency of 35.48%–36.09%, which is higher than a reference system using pure ammonia (29.16%–31.79%). The efficiency gain is attributed to two main factors: the thermochemical upgrade of waste heat into chemical energy (hydrogen) and the enhanced combustion efficiency from the hydrogen-enriched fuel. The optimal performance was found when all ammonia is cracked (Ammonia Split Ratio = 1), resulting in a fuel blend with a 62.80%–64.38% hydrogen molar ratio and a maximum power generation efficiency of 43.55%–43.90% at a higher turbine inlet temperature. The paper concludes that this cycle provides an efficient and safe pathway for ammonia utilization in carbon-free power generation.

In 2023 Shen et al. introduced a novel power cycle concept termed the chemical-recuperated and humidified gas turbine (CHGT), which uniquely integrates thermochemical recuperation (TCR) with steam injection to optimize waste heat recovery in an ammonia-fueled gas turbine [135]. The core novelty of the schematic diagram lies in the synergistic combination of these two processes, creating a cascade waste heat recovery system. The exhaust gas from the turbine is first used to drive the endothermic decomposition of ammonia into a hydrogen-nitrogen mixture in a chemical recuperator, effectively upgrading waste heat into chemical energy. The exhaust gas, still carrying substantial thermal energy, is then passed to a steam generator to produce steam, which is subsequently injected into the combustor. The schematic diagram of the novel proposed gas turbine system and its efficiency is shown in Fig. 27.

Fig. 27 shows that this integrated configuration allows the system to overcome the inherent limitation of standalone TCR, where only a portion of the exhaust heat can be recovered due to the finite amount of fuel available for decomposition. By adding a steam injection loop, the cycle can utilize a much larger fraction of the waste heat, increasing the mass flow through the turbine and thereby boosting power output without increasing compressor work. The study demonstrates that this combined approach yields a remarkable peak net electrical efficiency of 56.7%, significantly outperforming a standard ammonia-fueled Brayton cycle (by 20.6 percentage points) and even surpassing a conventional ammonia-fired combined cycle. The paper concludes that this CHGT schematic presents a highly efficient and less complex alternative for carbon-free power generation.

As demonstrated in the discussion of chemically recuperated gas turbines (CRGTs) for combined cycle power plants, steam injection can significantly reduce exergy destruction. Unlike processes like steam methane reforming, the ammonia decomposition reaction does not require a supplementary oxidizer, such as steam or carbon dioxide, for its thermochemical transformation. Furthermore, as shown in Pashchenko's work [133], the exhaust gas exiting the reformer retains a significant temperature potential, making it suitable for further heat recovery. Consequently, integrating an ammonia-fueled CRGT with a bottoming steam cycle represents a highly promising development path.

Waluyo et al. presented a comprehensive process analysis of a chemically recuperated ammonia-fired combined cycle power generation system [136]. The primary novelty of the proposed schematic is the tight integration of ammonia cracking within a conventional combined cycle structure, which includes a gas turbine cycle, a steam turbine cycle (bottoming cycle), and Exhaust Gas Recirculation (EGR). Unlike simpler cycles, this design uses waste heat in a cascading manner: high-temperature exhaust from the gas turbine is first used to drive the endothermic ammonia cracking reaction, then passes through a Heat Recovery Steam Generator (HRSG) to produce steam for the bottoming cycle, and finally provides low-grade heat for ammonia regasification. The integration of EGR is a key feature to control the turbine inlet temperature (TIT) and manage NOx emissions without diluting the fuel mixture.

The study demonstrates that this integrated system achieves high thermal efficiencies while operating within strict NOx emission and flame stability constraints. The analysis identifies optimal operating points, with one mode emphasizing high efficiency achieving a combined cycle efficiency of 48.1%–49.3%. This is accomplished with a fuel mixture containing 90% ammonia and 10% hydrogen (from cracking) and an EGR rate of 40%. The work highlights that while chemical recuperation (cracking) slightly reduces the system's maximum potential efficiency compared to a pure-ammonia baseline, it is essential for widening the operational range and ensuring stable, low-emission combustion, making the high efficiency practically attainable.

Zhou and Duan published a paper on a comprehensive analysis of using partially cracked ammonia (PCA) as a carbon-free fuel in gas turbine combined cycle (GTCC) power plants [137]. The research focuses on overcoming key technical challenges and evaluating the economic viability of this transition. The study proposes and assesses a novel Partially Cracked Ammonia Combined Cycle (PCACC) system, with a core innovation being a multi-stage reciprocating heat decomposition (RHD) method that uses waste heat from the gas turbine exhaust to drive the ammonia cracking process efficiently. The goal is to leverage hydrogen's superior combustion properties while maintaining ammonia's advantages for storage and transport.

Regarding thermal efficiency, the study's key findings show a promising technical performance. The reference Natural Gas Combined Cycle (NGCC) system serves as a baseline with a net plant efficiency of 56.3%. The method used to vaporize liquid ammonia significantly impacts the PCACC's efficiency; using flue gas for heating (FGHV) achieves an efficiency of 56.7%, while using extracted steam from the steam cycle (ESHV) is more efficient, achieving 58.2%, as it better utilizes low-grade waste heat. The design of the ammonia cracker is also crucial, with the proposed multi-stage reciprocating heat adiabatic packed column (RHAPC) configuration performing best. It achieves the highest system energy efficiency of 55.9% and a net plant efficiency of 59.5%. Furthermore, these multi-stage RHD configurations reduce the cracker's physical scale and catalyst usage by nearly 90% compared to conventional designs. Increasing the Ammonia Cracking Ratio (ACR) from 10% to 40% also improves system efficiency, raising it from 58.2% to 59.5%. In conclusion, the integration of advanced thermal management strategies allows PCACC systems to achieve thermal efficiencies that are comparable to or even surpass the baseline NGCC system, demonstrating strong technical feasibility.

In terms of economic efficiency, the analysis reveals significant current challenges. The Levelized Cost of Electricity (LCOE) for the baseline NGCC system is €107.2/MWh, while the LCOE for PCACC systems is substantially higher, ranging from €210.6/MWh to €222.8/MWh. The cost breakdown shows that in PCACC systems, fuel cost is the dominant factor, accounting for 68.2% to 70.1% of the total LCOE, which is drastically higher than the 29.2% share in the NGCC system. It is important to note that the capital expenditure (CAPEX) for PCACC is only 0.6% to 5.3% higher than for NGCC, demonstrating that the multi-stage RHD design successfully minimizes equipment costs. The study identifies several key economic barriers and levers. First, Ammonia Price is the most critical barrier; a decrease from €500/t to €200/t would reduce the LCOE by ~43%, and the technology becomes competitive with carbon capture when green ammonia prices fall to about €180/t. Second, Scale offers benefits, as scaling up the plant capacity from 50 MW to 830 MW reduces the LCOE by 23.4%. Finally, increasing the Cracking Ratio (ACR) from 10% to 40% only slightly reduces LCOE by 1.4%, but increases the fuel supply equipment cost by 35%. The economic conclusion is that while the technology is thermodynamically sound, the PCACC system is not currently economically competitive with natural gas due to the high cost of green ammonia.

In overall conclusion, this paper demonstrates that Partially Cracked Ammonia Combined Cycles are a technically viable pathway for zero-carbon power generation, with thermal efficiencies that can exceed those of natural gas plants. The innovative multi-stage reciprocating heat decomposition process is key to this technical success. However, widespread economic competitiveness is currently hampered by high ammonia fuel costs. The economic future of PCACC is directly tied to the expected decline in green ammonia prices, and once ammonia becomes cheaper, this technology is positioned to become a competitive alternative to fossil-fueled plants with carbon capture.

6.2. Ammonia decomposition

Ammonia decomposition is a well-studied process, as it is a promising pathway for carbon-free hydrogen production and hydrogen-rich gas generation. Industrial large-scale cracking of ammonia typically occurs at high temperatures (above 700°C) and in the presence of active catalysts, primarily to achieve high single-pass ammonia conversion and to meet the stringent purity requirements for hydrogen. The distinguishing feature of integrating ammonia cracking in an ammonia-fueled chemically recuperated gas turbine cycle is the moderate temperature level of the exhaust gas (approximately 600–700°C), pressure typically not exceeding 30 bar, and the objective of only partial, rather than complete, decomposition to balance hydrogen content with fuel flow and system complexity. Therefore, the key focus for the application of ammonia decomposition in this context should be directed towards the following points:

- achieving a high reaction rate and target conversion efficiency under the constrained heat source temperature of the turbine exhaust gas;
- minimizing reactor size and catalyst usage through innovative thermal integration and reactor design to enhance the economic feasibility of the process.

Ammonia decomposition is a catalytic and endothermic process. The conversion efficiency and reaction enthalpy depend on both operational parameters – such as temperature and pressure – and design parameters, including the type of catalyst, method of heat supply, and reactor configuration. Thermodynamic analysis establishes the relationship between operational parameters and the equilibrium conversion and enthalpy of the process. To account for the effects of kinetics and transport phenomena, detailed experimental and numerical studies are required. Therefore, although ammonia decomposition is a fundamentally well-understood reaction, there remains significant interest in reviewing papers focused on application in advanced power cycles with ammonia-fueled chemically recuperated gas turbines.

The results of thermodynamic analysis of ammonia decomposition process are presented in Fig. 28. This figure illustrates the equilibrium parameters governing the ammonia cracking process. Subfigures (a) and (b) depict the product composition as a function of temperature at 10 bar and 20 bar, respectively. The data confirm that increasing the temperature and decreasing the pressure promotes the complete decomposition of ammonia, with minimal unreacted ammonia present above 500°C. Under these conditions, hydrogen constitutes the dominant product, reaching approximately 75% of the resulting hydrogen-rich gas mixture. Subfigure (c) presents the corresponding reaction enthalpy at various pressures, which quantifies the thermal energy absorbed during the decomposition. The plot reveals that to achieve the equilibrium product composition at temperatures above 500°C, the reaction absorbs approximately 3100 kJ per kg of ammonia. This absorbed energy is stored as chemical energy in the hydrogen-rich gas, resulting in a combustion heat for the products that is higher than that of pure ammonia by this same amount, thereby upgrading the fuel's energy content through the application of heat.

In ammonia-fueled chemically recuperated gas turbines, the integration of an efficient ammonia cracking reformer is critical for system performance. While ammonia can decompose thermally, the use of a catalyst is essential to lower the required reaction temperature to levels compatible with gas turbine exhaust heat (typically 500–700°C). This catalytic cracking process enables effective thermochemical recuperation, converting waste heat into chemical energy by producing a hydrogen-rich fuel. Consequently, the selection of high-activity catalysts and the design of advanced reactor configurations are crucial. These factors directly influence the decomposition efficiency, reactor size, and overall energy balance, thereby determining the feasibility and economic viability of using ammonia as a carbon-free fuel in advanced power cycles with CRGTs.

Despite the fact that the first works on ammonia thermochemical decomposition were carried out in the beginning of 20th century [138], the increased interest for this topic is observed from year to year in the last decades [139–142].

The decomposition of ammonia into hydrogen-rich gas, achieved through thermal or catalytic cracking, is a key step in its utilization as an energy carrier. Catalytic cracking is favored as it reduces the reaction temperature, thereby lowering the energy consumption. Optimizing this process through the development of active catalysts and innovative reactor configurations is therefore vital for enhancing overall efficiency of ammonia-fueled CRGT.

The selection of a catalyst is a critical consideration in the ammonia cracking process, with the market segmented into several key types based on performance and cost [141,143,144]. Nickel-based, ruthenium-based, and cobalt-based catalysts represent the dominant categories, alongside an "Others" segment for emerging alternatives.

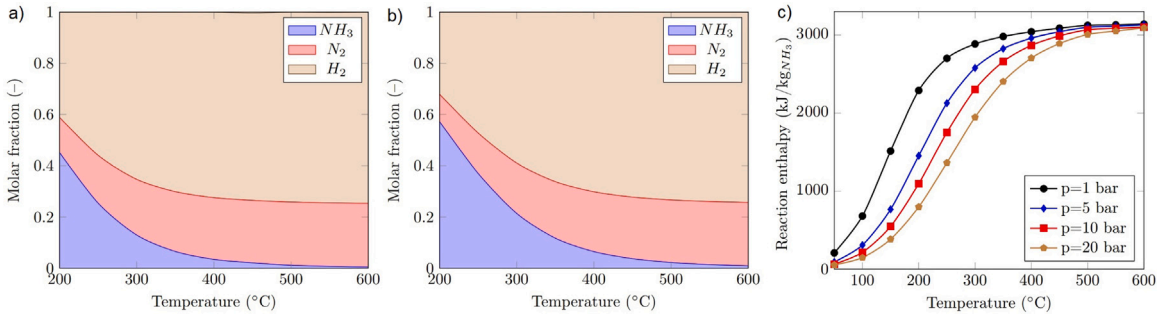


Fig. 28. Equilibrium parameters for ammonia cracking as a function of temperature: (a) product composition at 10 bar; (b) product composition at 20 bar; (c) reaction enthalpy at various pressures.

Among these, ruthenium-based catalysts are recognized as the high-performance option [145–147]. The highest catalytic activity for ammonia decomposition reported in the literature is achieved with ruthenium supported on various oxides or structured and unstructured carbon, though its main challenge is deactivation over time.

However, ruthenium is a noble metal, rare in nature, and consequently an expensive element. This high cost underscores the market's characteristic trade-off between performance (ruthenium) and cost-effectiveness (nickel) [148,149]. For this reason, significant research is dedicated to finding low-cost catalytic compositions with activity comparable to ruthenium.

Beyond the direct price of the active metal, a comprehensive assessment within the framework of environmental sustainability must also consider the environmental impact of the catalyst material itself. This includes factors such as its global warming potential and the energy demand associated with its extraction and refining. Therefore, ongoing research and development aims not only to improve the efficiency and reduce the cost of these alternative catalysts but also to enhance their overall sustainability, which could significantly shape the future landscape of the ammonia cracking catalysts market. Fig. 29a shows the market share of different types of catalysts in the global ammonia industry. This figure indicates that Ni-based catalysts are, and will remain, the primary catalyst type for thermochemical ammonia cracking. Despite progress in Ru-based catalyst development, its market share will remain at about one quarter, and its economical use in ammonia-fueled CRGT is unfavorable due to its high cost compared to Ni-based catalysts. Beyond direct material cost, a comprehensive sustainability assessment must also consider the environmental impact of the catalyst, including its global warming potential (GWP) and the energy demand of its extraction and refining. As shown in Fig. 29b, a strong linear correlation exists between the price of an element and its 100-year GWP, as both metrics are intrinsically linked to its natural abundance. Scarcer elements, such as the noble metal ruthenium (Ru), exhibit high costs and significant environmental impacts, positioning them at the top of the chart. Consequently, transition metals like Fe, Ni, Mo, and Co, located in the lower part of this relationship, are preferable from both an economic and environmental standpoint [150,151].

As mentioned above, the commercially available catalysts used for the large-scale ammonia reformers are primarily based of nickel supported on alumina oxides (Al_2O_3) [152–154]. For the application of Ni-based catalyst in ammonia-fueled CRGT, it is important to obtain high ammonia conversion at moderate low temperature. Table 9 summarizes the catalytic performance of various nickel-based catalysts used for ammonia decomposition. The table compiles data on how different catalyst parameters – including nickel loading (wt %), the type of support (e.g., Al_2O_3, CeO_2), and the use of promoters (e.g., Ce, La, Zr) – influence key performance metrics such as ammonia conversion (%) under varying operational conditions (temperature, WHSV (mL g^{-1} h^{-1}), GHSV (h^{-1})). The data illustrates that even for temperature of 600°C the high ammonia conversion can be obtained. For example, Liang et al. reported that Ni-αAl_2O_3 catalysts are highly effective for ammonia decomposition, achieving conversions exceeding 98.7% at 650°C [155]. Moreover, the data from Table 9 and overview of published papers illustrates that the support material and the addition of promoters, particularly rare-earth elements like Cerium, significantly enhance catalytic activity and stability by improving nickel dispersion, reducibility, and resistance to sintering. However, if a catalyst is intended for waste-heat recuperation, the addition of promoters can significantly increase its cost. Therefore, selecting a catalyst is a subject of detailed techno-economic analysis [156]. In Table 9 the following nomenclature is used: WHSV is weight hourly space velocity (mL g^{-1} h^{-1}); GHSV is gas hourly space velocity (h^{-1}); X_{NH3} is ammonia conversion defined as $(NH_{3,in}–NH_{3,out}) \times NH_{3,in}^{-1}$) (mol/mol)

6.3. Ammonia cracking product combustion

Ammonia synthesized from renewable hydrogen has garnered significant international interest for power generation, owing to its high gravimetric hydrogen density, ease of storage, and carbon-free structure, which prevents CO_2 emissions. However, the direct combustion of ammonia in gas turbines faces challenges such as low combustion efficiency, a consequence of its low flame speed and long quenching distance. A promising solution is the pre-combustion reforming of ammonia into a hydrogen-rich gas using thermochemical exhaust heat recuperation. In contrast to ammonia, hydrogen possesses a high flame speed and short quenching distance, leading to significantly improved and stable combustion within the combustor.

Recently, a set of papers on the combustion of ammonia decomposition products was published, with a focus on combustion stability, flame speed, and NOx emissions.

Shohdy et al. presented the results of experimental investigation into the combustion characteristics of pure ammonia and partially cracked ammonia [71]. The primary goal is to evaluate the trade-offs involved in a strategy to improve ammonia combustion: cracking a fraction of ammonia into hydrogen and nitrogen before combustion. The authors systematically examine how different levels of cracking, specifically 0%, 10%, 20%, and 28%, affect four key performance areas. These are the overall burning velocity, which is how fast the flame consumes the fuel-air mixture; the lean blow-off limit, indicating how lean a mixture can be before the flame extinguishes; pollutant emissions, specifically the production of nitrogen oxides; and flame dynamics, meaning the flame's response to acoustic perturbations, a critical factor for avoiding thermoacoustic instabilities in engines like gas turbines. The central and somewhat counterintuitive finding is that while ammonia cracking offers significant benefits in flame speed and stability, it comes with severe drawbacks in emissions and dynamic stability, even at relatively low cracking percentages like 20%. Graphical summary of results obtained by Shohdy et al. is presented in Fig. 30. This figure contains the following sub-figures: (a) schematic diagram for experiments; (b) overall flame length and overall burning velocity; (c) direct flame images for pure NH_3 and partially cracked NH_3 blends

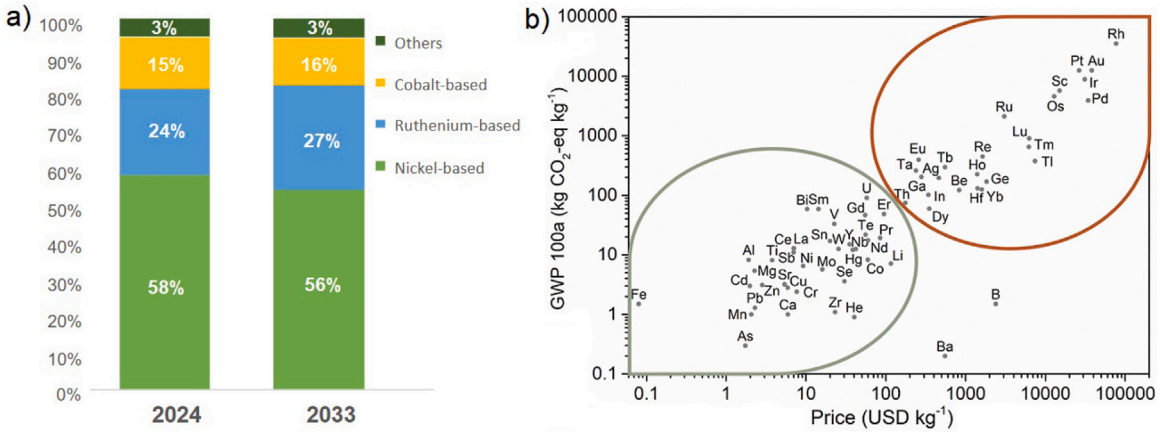


Fig. 29. Techno-economical performance of ammonia cracking catalysts: (a) Ammonia cracking catalyst global market (share of catalyst type); (b) Comparison of the price (USD/kg) and the global warming power on a 100-year basis ($kg_{CO2equiv}$/kg) of some elements.

Table 9
Catalytic performances of Ni-based catalysts at pressure of 1 bar.

wt%	Support	Prom.	WHSV	GHSV	$NH_{3,in}$ (%)	T (°C)	X_{NH3} (%)	Ref.
–	Al_2O_3	–	–	10 000	100	600	63	[157]
–	CeO_2	–	–	10 000	100	600	65	[157]
–	CeO_2/Al_2O_3	–	–	10 000	100	600	81	[157]
–	αAl_2O_3	–	–	30 000	100	450	9	[158]
–	Al_2O_3	–	–	6 000 000	15	540	30	[159]
–	Al_2O_3	–	–	30 000	100	450	15	[160]
–	Al_2O_3	–	–	30 000	100	450	15	[152]
–	Al_2O_3	–	–	60 000	100	600	29	[161]
–	Al_2O_3	–	–	60 000	100	600	17	[162]
–	Al_2O_3	–	–	1800	100	650	97	[155]
1	Al_2O_3	–	100	–	–	580	58	[163]
1.7	Al_2O_3	–	100	–	–	500	30	[164]
3.6	Al_2O_3	–	100	–	–	550	33	[165]
8.9	Al_2O_3	–	9000	100	–	500	27	[166]
10	Al_2O_3	–	6000	100	–	450	10	[167]
10	Al_2O_3	–	6000	100	–	500	35	[168]
10	Al_2O_3	–	6000	100	–	450	10	[169]
10	Al_2O_3	–	240 000	80 000	50	500	93	[170]
10	Al_2O_3	–	9000	5.9	–	400	32	[171]
10	Al_2O_3	–	13 800	3067	57	400	8	[148]
20	Al_2O_3	–	7500	100	–	500	28	[172]
20	Al_2O_3	–	2400	100	–	500	31	[173]
20	Al_2O_3	–	7500	100	–	500	27	[174]
30	Al_2O_3	–	12 000	100	–	500	42	[175]
40	Al_2O_3	–	6000	100	–	550	66	[176]
71.4	Al_2O_3	–	30 000	100	–	550	51	[177]
90	Al_2O_3	–	36 000	100	–	600	93	[178]
–	Al_2O_3	Ce	–	30 000	100	450	28	[152]
9	Al_2O_3	Ce	6000	100	–	450	15	[169]
20	Al_2O_3	Ce	7500	100	–	500	51	[172]
9	Al_2O_3	Au	6000	100	–	450	16	[169]
9	Al_2O_3	Gd	6000	100	–	450	16	[169]
9	Al_2O_3	La	6000	100	–	450	20	[169]
9	Al_2O_3	Nd	6000	100	–	450	20	[169]
9	Al_2O_3	Pr	6000	100	–	450	20	[169]
9	Al_2O_3	Sm	6000	100	–	450	18	[169]
9	Al_2O_3	Y	6000	100	–	450	19	[169]
20	Al_2O_3	Zr	7500	100	–	500	29	[174]

(10%, 20%, 28% cracking) under stable operating conditions (3.4 kW, 7 m/s bulk velocity, $\Phi = 0.95$, swirl number = 1.67).

The study reveals that cracking ammonia has a profound and dual-natured impact on combustion performance. The benefits are clear, but the costs are significant, creating a challenging optimization problem for practical applications. Regarding positive effects on reactivity and stability, cracking ammonia dramatically increases the flame's reactivity. The addition of hydrogen, a highly reactive fuel, led to a six-fold increase in the overall burning velocity. Furthermore, hydrogen-enriched flames are much more robust, as the equivalence ratio at which the flame blows out decreases almost linearly with the cracking percentage, meaning cracked ammonia flames can remain stable under much leaner conditions.

However, there are significant negative effects concerning emissions and instability risks. One of the most critical findings is the drastic increase in NOx production at higher cracking levels, more than doubling for 28% cracking. To achieve low NOx levels, the flame must be operated at progressively leaner and potentially less practical conditions. Additionally, cracked ammonia flames become highly responsive to

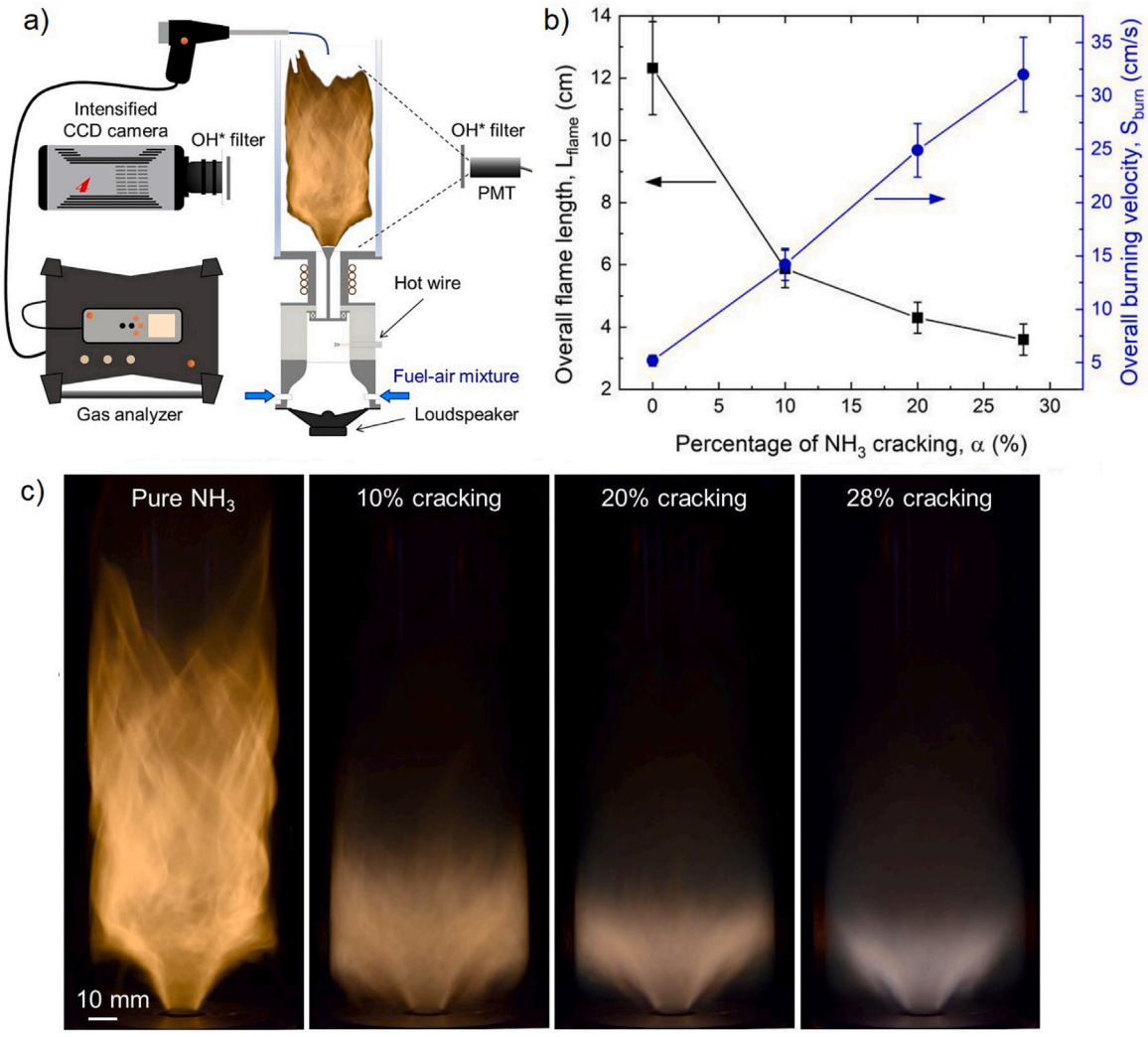


Fig. 30. Summary of paper by Shohdy et al. [71]: (a) schematic diagram for experiments; (b) overall flame length and overall burning velocity; (c) direct flame images for pure NH_3 and partially cracked NH_3 blends (10%, 20%, 28% cracking) under stable operating conditions (3.4 kW, 7 m/s bulk velocity, $\Phi = 0.95$, swirl number = 1.67).

acoustic perturbations, greatly increasing their susceptibility to dangerous thermoacoustic instabilities because the hydrogen content makes the flame more resistant to strain, allowing it to vigorously "roll up" around acoustic vortices.

The study also noted a change in the flame stabilization mechanism for pure and 10% cracked ammonia at higher flow velocities, shifting to a less efficient, long flame mode. The paper concludes that ammonia cracking is a double-edged sword. While it successfully addresses ammonia's inherent weaknesses of low burning velocity and poor lean stability, it introduces new, serious problems of sharply elevated NOx emissions and a heightened risk of thermoacoustic instabilities. The authors highlight that these detrimental effects become significant at a cracking percentage as low as 20%, suggesting an optimal level of cracking might exist below this threshold and that mitigation strategies are necessary for practical applications.

An et al. experimentally and numerically investigated the combustion characteristics of partially cracked ammonia flames in a 20 kW model gas turbine swirl combustor presented in Fig. 31a [179]. The fuel composition is simulated by mixing NH_3, H_2, and N_2 in volumetric ratios corresponding to the products of ammonia cracking. The graphical summary of paper by An et al. is presented in Fig. 31.

The cracking ratio (Cr) is varied from 0.0 (pure NH_3) to 0.6, with the fuel mixture adjusted accordingly (e.g., Cr = 0.4) corresponds to $0.43NH_3/0.43H_2/0.14N_2$ by volume as shown in Fig. 31b. All experiments are conducted at a constant equivalence ratio ($\phi = 0.9$) and atmospheric pressure. A methane-air flame at the same thermal power (≈20 kW) is also tested as a reference for comparison. Fig. 31c presents direct digital images of the flame macrostructures for different Cr values and the methane baseline. As Cr increases, the flame transitions from a long, orange-colored "V" shape (Cr = 0.0) to a shorter, stiffer, and whiter "M" shape (Cr = 0.6). The outer recirculation zone (ORZ) becomes smaller, and the flame exhibits a tendency towards flashback at higher Cr. The orange hue in pure and low-Cr ammonia flames is attributed to $NH_2\alpha$ chemiluminescence, which diminishes as ammonia is replaced by hydrogen. Notably, the Cr = 0.4 flame closely resembles the methane flame in length, ORZ structure, and flame–wall interaction, indicating similar fundamental combustion properties such as flame speed and temperature.

The OH-PLIF results in Fig. 31d reveal significant changes in flame structure with increasing Cr. For pure ammonia (Cr = 0.0), OH signals are weak and absent in the ORZ, reflecting low combustion intensity. As Cr rises, OH intensity strengthens, especially in the ORZ and shear layers, and the flame front shifts from a "V" to an "M" topology. The Cr = 0.4 condition shows OH distributions and intensities similar to those of the methane flame, confirming its structural similarity. In contrast, the Cr = 0.6 flame exhibits very high OH signals upstream, indicating

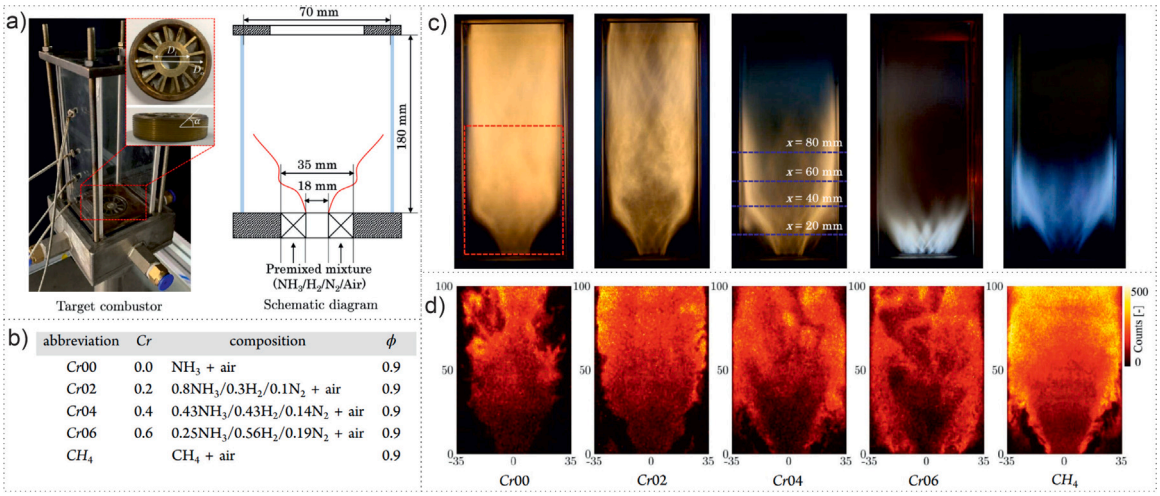


Fig. 31. Summary of paper by An et al. [179]: (a) photo of combustor and its schematic diagram; (b) experimental conditions; (c) direct digital images for cracked ammonia and methane flames; (d) instantaneous OH fields measured using PLIF for ammonia cracked flames and methane flame for reference.

intense reaction zones near the burner, but weaker signals downstream due to rapid hydrogen consumption.

In summary of paper by An et al. [179], the experimental results demonstrate that partial cracking significantly enhances flame stability, intensity, and structural compactness. The Cr = 0.4 condition emerges as a promising candidate for mimicking methane-like combustion behavior in ammonia-based systems, offering a pathway to utilize ammonia in existing gas turbine infrastructure without major modifications, while also addressing challenges related to flame stability and emissions.

The use of ammonia decomposition products leads to change in NOx emission. Meng et al. comprehensively investigated the NOx emission characteristics of partially cracked ammonia flames, revealing a strong dependence on the cracking ratio (Cr) [180]. As Cr increases from 0.0 to 0.6, the emissions of NO and NO_2 show a significant upward trend. The increase is relatively gradual for Cr < 0.2, but becomes rapid and nearly linear for higher cracking ratios. This behavior is consistent with one-dimensional flame calculations, which predict that NO emissions peak under high Cr conditions (0.6–0.9). Concurrently, the concentration of slip NH_3 in the exhaust decreases significantly as a greater proportion of the fuel is cracked and consumed. This creates a trade-off, where the optimal condition for minimizing global emissions – considering both NOx and unburnt NH_3 – is identified at Cr = 0.2.

The formation pathways of NOx shift with the cracking ratio [181–183]. At low Cr values, NO is primarily generated through the fuel-NO mechanism, where nitrogen from the ammonia itself is oxidized. However, as Cr increases, the higher hydrogen content leads to elevated flame temperatures and increased concentrations of O and OH radicals. This enhances the thermal-NO pathway via the Zeldovich mechanism [184]. Large Eddy Simulation (LES) results confirm that the contribution of thermal NO to the total NO production becomes more significant at higher Cr. The study also demonstrates that NOx emissions can be effectively mitigated by operating under fuel-rich conditions (increasing ϕ) or by enhancing heat loss through the combustor walls, with the former strategy proving particularly effective.

In addition, Gubbi et al. demonstrated that while pure ammonia combustion can achieve low NOx emissions in an optimized rich-lean staged combustor, the use of ammonia cracking products offers significant advantages, but only when cracking is nearly complete [185]. The results show a highly non-monotonic relationship between the cracking fraction (Cr) and minimum NOx emissions, where intermediate cracking levels (around Cr = 0.75–0.9) can even increase NOx compared to pure ammonia at certain pressures due to complex kinetic interactions and finite residence times. However, a major advantage is realized at very high cracking fractions; specifically, NOx emissions drop by about two orders of magnitude once the fuel is almost fully cracked (Cr > 0.999), reaching sub-30 ppm levels (at 15% O_2, dry) across all pressures, as the combustion transitions to that of a primarily H_2/N_2 mixture with minimal fuel-bound nitrogen. Therefore, the primary advantage of using cracked ammonia products is the potential for ultra-low NOx emissions, but this is critically dependent on achieving near-total cracking, which would then enable simpler lean premixed combustion strategies comparable to those used for hydrocarbon fuels.

Similarly, Ariemma et al. showed that the nitrogen dilution inherent in fuel mixtures derived from cracked ammonia does not adversely affect their combustion behavior under MILD conditions and can, in fact, be beneficial for NOx control [186]. While blending hydrogen with ammonia generally increases NOx emissions compared to pure NH_3 due to a boosted radical pool that promotes ammonia's conversion to NOx, the specific composition of the cracked mixture plays a critical role. Notably, the 100% cracked ammonia mixture (75% H_2, 25% N_2) resulted in the lowest NOx levels among all the cracked fuel blends tested across the entire range of equivalence ratios. This is attributed to the N_2 dilution promoting a more uniform distribution of reactants, which avoids the formation of high-temperature "hot-spots" within the reactor and thereby thermally suppresses NOx formation, showcasing a key advantage of using fully cracked ammonia in MILD combustion systems.

Overall, the combustion of partially cracked ammonia presents a complex trade-off between improved stability and increased NOx emissions. Research by Shohdy et al. [71] revealed that while cracking ammonia significantly enhances flame speed and extends lean blow-off limits, it comes with severe drawbacks: NOx emissions more than double at a 28% cracking ratio, and the flames become highly susceptible to thermoacoustic instabilities. Similarly, Meng et al. [180] found that NO and NO_2 emissions exhibit a strong upward trend with the cracking ratio (Cr), with a rapid, nearly linear increase for Cr > 0.2. This is due to a shift in NOx formation pathways; at higher cracking ratios, elevated hydrogen content raises flame temperatures and radical concentrations (O, OH), which promotes the thermal-NO (Zeldovich) mechanism in addition to the fuel-NO route. Consequently, a trade-off emerges where higher Cr reduces unburned NH_3 slip but drastically increases NOx, with an optimal balance for minimizing total emissions often found at relatively low cracking levels, such as Cr = 0.2.

However, the pathway to ultra-low NOx emissions using cracked ammonia appears to hinge on near-total cracking. Gubbi et al. [185] demonstrated a highly non-monotonic relationship, where intermediate cracking fractions can even increase NOx compared to pure ammonia.

The key advantage materializes only with almost complete cracking (Cr > 0.999), reducing NOx by about two orders of magnitude to sub-30 ppm levels. This is because the fuel transitions to a primarily H_2/N_2 mixture with minimal fuel-bound nitrogen. Supporting this, Ariemma et al. showed that under MILD combustion conditions, the 100% cracked ammonia mixture (75% H_2/25% N_2) resulted in the lowest NOx levels among all tested cracked blends. The inherent nitrogen dilution in these fully cracked mixtures promotes a more uniform reaction zone, suppressing hot-spot formation and thermally limiting NOx, highlighting a distinct advantage of fully utilizing the cracking process.

6.4. Summary of Section 6

This section has detailed the application of chemically recuperated gas turbine (CRGT) technology to ammonia as a carbon-free fuel. The fundamental advantage of this approach is its ability to address the inherent combustion challenges of ammonia – specifically its low flame speed, high ignition energy, and narrow flammability limits – through pre-combustion thermochemical recuperation. By utilizing gas turbine exhaust heat to drive the endothermic decomposition of ammonia into a hydrogen-rich gas, the CRGT cycle performs two critical upgrades. First, it thermochemically converts low-grade waste heat into high-grade chemical energy, thereby increasing the fuel's heating value per unit mass of primary ammonia and boosting cycle efficiency. Second, and more crucially for combustion, it transforms a low-reactivity fuel (ammonia) into a high-reactivity syngas (H_2/N_2), enabling stable, efficient combustion with a wider operational range. This makes ammonia-fueled CRGT a compelling pathway for zero-carbon power generation, leveraging ammonia's advantages as a hydrogen carrier while overcoming its direct combustion drawbacks.

Research demonstrates that ammonia-fueled CRGTs offer significant thermodynamic benefits. Simple cycles can achieve efficiency gains of up to 9 percentage points compared to conventional ammonia-fired gas turbines. Furthermore, advanced configurations integrating the CRGT with bottoming steam cycles (forming a combined cycle) or with steam injection (forming a chemically recuperated and humidified cycle) show even greater promise, with modeled efficiencies reaching 56%–60%. These high efficiencies result from the effective cascading use of exhaust heat: high-temperature heat drives the ammonia cracking reaction, while the remaining lower-temperature heat is used for steam generation or other purposes. This synergistic integration allows the system to surpass the performance of traditional combined cycles while maintaining carbon-free operation.

Despite these significant advantages, ammonia-fueled CRGTs face notable challenges that require focused research and development. A primary technical challenge is managing NO_x emissions, which exhibit a complex, non-monotonic relationship with the ammonia cracking ratio; while near-complete cracking can lead to ultra-low NO_x, intermediate cracking levels can actually increase emissions. Furthermore, the successful implementation hinges on the development of efficient, low-cost catalytic reformers capable of high conversion under the moderate temperature constraints of gas turbine exhaust, alongside combustors redesigned to handle the unique properties of hydrogen-nitrogen mixtures. Economically, the current high cost of green ammonia remains the most significant barrier to widespread adoption, underscoring the need for continued reductions in renewable hydrogen production costs to make this technology commercially competitive.

7. Methanol fed CRGT

Building upon the established principles of chemically recuperated gas turbines (CRGT), this section focuses on its specific application using methanol as a primary fuel. Methanol presents itself as a particularly suitable candidate for CRGTs due to its favorable reforming characteristics, including a relatively low reforming temperature and the absence of carbon–carbon bonds, which simplifies the reaction pathways and mitigates the risk of coke formation. The integration of methanol reforming directly into the gas turbine cycle represents a sophisticated strategy to upgrade the fuel's chemical energy and recuperated waste heat from the turbine exhaust before the combustor.

The two principal pre-combustion reforming routes employed for this purpose are methanol decomposition and methanol steam reforming (MSR). Methanol decomposition is an endothermic process that breaks down methanol into hydrogen and carbon monoxide, offering a upgrade in the fuel's heating value. Alternatively, methanol steam reforming involves the reaction of methanol with steam over a catalyst, producing a reformate rich in hydrogen and carbon dioxide, with a higher hydrogen yield compared to decomposition. The choice between these two pathways significantly influences the composition, energy content, and combustion properties of the final fuel mixture entering the combustor.

This section explores the operational principles, system configurations, and performance implications of gas turbine cycles integrated with pre-combustion methanol reforming. The discussion focuses on the comparative advantages and challenges associated with both decomposition and steam reforming pathways, examining their impact on overall system efficiency, power output, and the environmental footprint of the gas turbine plant.

7.1. Overview

In Section 3, the paper by Davies et al. [95] was identified as a pioneering work on chemically recuperated gas turbines. The authors demonstrated that pre-combustion methanol reforming increases both thermal and combustion efficiency. They also highlighted the viability of using liquid methanol as a primary fuel in existing combustors without significant modification. Subsequent research on methanol-fed CRGT has primarily focused on three key advantages:

- An increase in thermal efficiency through exhaust heat recuperation;
- Higher combustion efficiency resulting from the use of hydrogen-rich reformate as a fuel;
- The benefits of burning a hydrogen-rich gaseous fuel compared to liquid or vaporized methanol.

The patent by Basu and Rajagopal describes a method for generating power using an Advanced Thermal Recuperation Cycle, which integrates a thermochemical recuperator with a bottoming Rankine cycle [108]. The core innovation is the use of turbine exhaust heat to drive an endothermic reforming reaction in a catalytic reactor, where a raw, oxygenated fuel (such as dimethyl ether, methanol, or ethanol) is mixed with steam and converted into a hydrogen-rich reformed fuel. This reformed fuel is then combusted in the gas turbine. A key feature is the use of low-reformation-temperature fuels (300°F to 932°F), which allows the cycle to utilize the turbine exhaust heat effectively without requiring expensive high-temperature turbines or complex multi-stage expansion systems that were problematic in prior art.

The invention specifically aims to shift the majority of power generation to the more efficient and responsive gas turbine, while the bottoming Rankine cycle contributes a minor share (preferably less than 10%–30% of total power). This configuration addresses a major drawback of conventional combined-cycle plants, where the steam turbine often contributes 30%–40% of the power, making the plant difficult to control during load shifts. The patent claims demonstrate significant efficiency gains (e.g., 61.97% LHV basis in one example) compared to conventional cycles, along with simplified operation and the potential for ultra-low NOx emissions due to the reformed fuel's combustion characteristics.

While most of the known works on methanol-fed GRGT are based on the use of exhaust heat after gas turbine for methanol reforming,

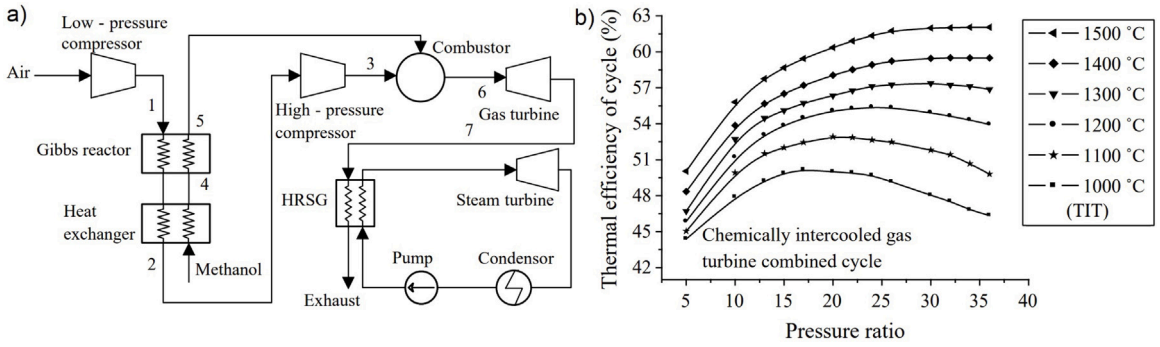


Fig. 32. Chemically intercooled gas turbine cycle by Jin et al. [187]: (a) schematic diagram: 1 - hot compressed air; 2 - cooled compressed air; 3 - high pressure air; 4 - evaporated methanol; 5 - synthesis gas; 6 - inlet working gas; 7 - exhaust gas; (b) thermal efficiency as a function on pressure ratio and turbine inlet temperature (TIT).

Jin et al. proposed the use of excess temperature potential of air after first stage of compressor for pre-combustion methanol reforming [187]. The core innovation lies in using the waste heat from the gas turbine's intercooler (at temperatures of 185–220°C) to drive the endothermic decomposition of methanol into syngas (CO and H_2), a process known as thermochemical recuperation. This strategy effectively upgrades waste heat into chemical energy, enriching the fuel before combustion. Re-print of schematic diagram of such pre-combustion methanol reforming is illustrated in Fig. 32a.

The authors demonstrate that this approach yields a substantial improvement in system performance as shown in Fig. 32b. Through thermodynamic modeling and exergetic analysis, they show that the proposed cycle achieves a thermal efficiency of 57.4% at a turbine inlet temperature of 1300°C, which is approximately 6 percentage points higher than a conventional intercooled combined cycle and 4 percentage points higher than a standard combined cycle at their respective optimum pressure ratios. This efficiency gain is attributed to two main factors: the reduction of exergy destruction in the combustor by ~7.5 percentage points due to the combustion of hydrogen-rich syngas, and the effective conversion of low-grade waste heat into high-grade chemical fuel energy.

The study by Duwig et al. demonstrates that integrating a methanol decomposition unit into a recuperated micro-gas turbine (T100) cycle yields a clear improvement in thermal efficiency [188]. Their thermodynamic analysis reveals that, across a wide range of operating conditions with pressure ratios above 6, the chemically recuperated cycle can achieve an efficiency increase of up to 5 percentage points compared to a standard recuperated cycle. For the specific T100 microturbine, retrofitting the system for methanol decomposition resulted in a more modest but still significant efficiency gain of two percentage points, raising the efficiency from 32.5% on methane to 34.3% on decomposed methanol. This enhancement is primarily attributed to the more effective utilization of exhaust gas waste heat, which is used to drive the endothermic fuel upgrading process, thereby reducing the stack temperature and lowering the required mass flow of primary methanol fuel for a constant power output.

The paper by Tola and Lonis presents a comprehensive study of a fully renewable, low-carbon power system that integrates a chemically recuperated gas turbine (CRGT) with a Power-to-Liquids (PtL) process [189]. The system's renewable character is anchored in its use of "green" methanol, synthesized from hydrogen produced via water electrolysis (using renewable electricity) and carbon dioxide captured from the CRGT's own exhaust. This design creates a nearly closed carbon loop, where the same CO_2 is continuously recycled to produce new fuel, resulting in very low net emissions and providing a long-term storage solution for intermittent renewable energy sources like solar and wind.

From a thermodynamic perspective, Tola and Lonis demonstrates that the CRGT plant itself achieves a significant efficiency gain through thermochemical recuperation, with calculated efficiencies ranging from 35.9% to 40.2% depending on the operating parameters, which is up to 10 percentage points higher than a reference gas turbine cycle without fuel reforming [189]. However, when evaluating the entire power-to-power cycle – including the energy losses from electrolysis and methanol synthesis – the overall system efficiency is considerably lower. The global power-to-power efficiency reaches a maximum of approximately 23%, which can be slightly improved by integrating an Organic Rankine Cycle (ORC) to better utilize residual exhaust heat. While this round-trip efficiency is modest, the authors position the technology not on efficiency alone, but as a commercially viable and reliable method for storing renewable energy and dispatching electricity with a minimal carbon footprint.

Yuan et al. reported that integrating thermochemical recuperation (TCR) via methanol decomposition into a Combined Cooling, Heating, and Power (CCHP) system significantly enhances thermal efficiency of gas turbine system while creating a unique chemical storage opportunity for waste heat [190,191]. The system achieves a notable increase in power generation efficiency by 7.8% to 14.83% for gas turbine and internal combustion engine scenarios, as high-temperature exhaust heat is converted into chemical energy in the form of syngas, which is then used as an upgraded fuel. This process effectively stores otherwise wasted thermal energy, allowing it to be dispatched as electricity on demand and decoupling the production of cooling and heating from the prime mover's operation. This stored chemical energy thereby provides crucial operational flexibility, enabling the system to adjust the cooling/heating-to-power output ratio over a wide range (e.g., 1.73–2.92 for a gas turbine) to better match variable user demands and avoid energy waste during off-design operation.

The study by Cocco and Tola reported that the external reforming of methanol significantly enhances the thermal efficiency of a Solid Oxide Fuel Cell–Micro-Gas Turbine (SOFC–MGT) hybrid system [192]. By situating the methanol reformer outside the SOFC stack, the system leverages the fuel's low reforming temperature (200–350°C) to optimize waste heat recovery from the turbine exhaust, which is used to drive the endothermic reforming reactions. This external configuration not only improves the overall heat integration but also boosts the SOFC's performance by supplying the anode with a reformate gas of higher hydrogen partial pressure, leading to an increased cell voltage. As a result, the hybrid plant achieves exceptional thermal efficiencies of 67%–68% at a SOFC temperature of 900°C, and 72%–73% at 1000°C, with an optimal reforming temperature identified at approximately 240°C. This integration strategy effectively decouples the reforming process from the high-temperature stack, allowing for superior thermodynamic performance compared to internally reformed systems.

This section has detailed a range of schematic configurations proposed for gas turbines utilizing pre-combustion methanol reforming, all cases demonstrate significant gains in thermal efficiency. Pioneering and subsequent studies have consistently shown that by using

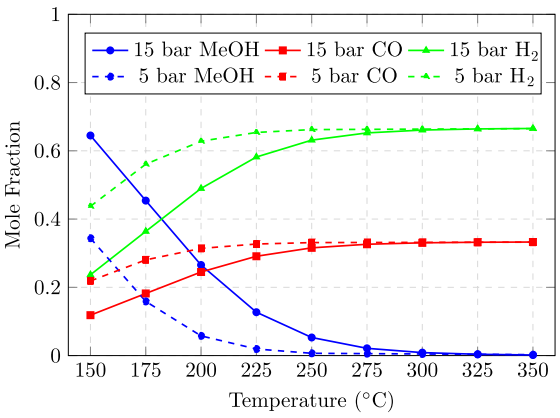


Fig. 33. Equilibrium composition of methanol decomposition at different pressures (5 bar and 15 bar). Solid lines represent 15 bar data, dashed lines represent 5 bar data.

exhaust heat to drive methanol decomposition or methanol steam reforming, the chemical energy of the fuel is upgraded, leading to efficiency improvements of several percentage points across various cycles, including simple, recuperated, intercooled, and even hybrid solid oxide fuel cell systems. The resulting shift to a hydrogen-rich reformate not only enhances the cycle's thermodynamics through better heat integration and reduced exergy destruction but also alters the fundamental combustion properties in the turbine. Consequently, the combustion performance of this reformate, characterized by its high hydrogen content, presents a compelling and complex topic for further investigation.

7.2. Methanol reforming

The thermodynamic feasibility and energy balance of pre-combustion methanol reforming in gas turbines are primarily governed by two key reactions: methanol decomposition and methanol steam reforming according the following chemical reactions, respectively:

$$CH_3OH \leftrightarrow CO + 2H_2; \quad \Delta H = +90.7\,kJ/mol; \tag{10}$$

$$CH_3OH + H_2O \leftrightarrow CO_2 + 3H_2; \quad \Delta H = +49.7\,kJ/mol. \tag{11}$$

Methanol decomposition is a moderately endothermic reaction. Its equilibrium conversion is highly favorable at the temperature range of typical gas turbine exhaust or intercooler heat sources (250–350°C), especially at lower pressures. The equilibrium gas composition for methanol decomposition reaction is presented in Fig. 33. The reaction produces a syngas with a high hydrogen content, which significantly increases the lower heating value (LHV) of the fuel per mass of initial methanol. This "fuel upgrade" is a primary source of efficiency gain, as waste heat is converted into chemical energy. Alternatively, methanol steam reforming is less endothermic and produces a reformate with a higher hydrogen-to-carbon monoxide ratio and negligible CO if coupled with an in-situ water-gas shift reaction [14,193,194]. While the net energy upgrade is lower than with decomposition, steam reforming offers advantages in managing combustion emissions and catalyst stability.

The choice between these two pathways involves a critical trade-off. Decomposition maximizes the energy recuperation potential and syngas LHV for a given heat source, leading to greater theoretical cycle efficiency improvements, as demonstrated in systems by Jin et al. [187] and Duwig et al. [188]. Conversely, steam reforming requires a lower reaction temperature for near-complete conversion and produces a cleaner syngas, which can simplify downstream combustion and emission control. The steam-to-carbon (S/C) ratio is a crucial operating parameter; while a stoichiometric S/C of 1.0 is sufficient, higher ratios (e.g., 1.5–2.0) are often used in practice to prevent carbon formation (coking) on the catalyst, albeit at the cost of diluting the reformate and slightly reducing the peak cycle efficiency.

While methanol steam reforming produces a cleaner, high-hydrogen reformate, a significant thermodynamic penalty is associated with steam production. The process requires a substantial energy input to vaporize and superheat the water feedstock, which is typically supplied by the system's own heat streams. Crucially, this water is ultimately lost with the combustion flue gases after the reformate is burned in the gas turbine. The latent heat of vaporization carried away by this steam represents an unrecovered energy loss that is not present in the dry decomposition pathway. Consequently, although the reforming reaction itself is less endothermic, the overall cycle efficiency of a steam reforming system can be negatively impacted by this parasitic energy load for steam generation and the subsequent loss of the water in the exhaust.

A notable side reaction in both methanol decomposition and steam reforming is the potential formation of methane via methanation reaction:

$$CO + 3H_2 \leftrightarrow CH_4 + 2H_2O; \quad \Delta H = -206.6\,kJ/mol. \tag{12}$$

As it was shown in Section 5.2.1, in the temperature range below 950 K the reverse reaction of steam methane reforming is favored at lower temperatures and higher pressures, which can coincide with the operating conditions of the reformer. While typically occurring in minor quantities due to kinetic limitations on standard catalysts, the presence of even small amounts of methane is undesirable. Since methane is more stable and harder to combust completely than hydrogen or carbon monoxide, its formation represents a loss of chemical energy available for immediate combustion in the turbine, effectively slightly reducing the extent of the fuel upgrade achieved by the thermochemical recuperation process.

From a system perspective, the integration of either reforming process acts to reduce the exergy destruction that is inherent in the direct combustion of a hydrocarbon fuel. By using mid-to-low-grade waste heat to drive the endothermic reactions, the system effectively performs a cascade utilization of energy, upgrading thermal exergy to chemical exergy. Ultimately, the thermodynamic analysis confirms that both methanol decomposition and steam reforming are highly effective pathways for pre-combustion methanol reforming, with the optimal configuration being dependent on the specific heat source temperature, pressure constraints, and the desired composition of the reformate for stable, low-emission combustion in the gas turbine.

Fig. 33 provides equilibrium gas composition of reformate. The real gas composition depends on operational and design parameters among type of catalyst is crucial because it affected not only on methanol conversion but also on reaction rate of side reactions. A diverse range of catalyst formulations has been developed to facilitate methanol reforming, with the selection heavily dependent on the desired reaction pathway and operating conditions. For methanol steam reforming, copper-based catalysts, particularly $Cu/ZnO/Al_2O_3$, are the industrial benchmark due to their high activity and selectivity towards CO_2 and H_2 at relatively low temperatures (200–300°C), although their susceptibility to sintering and pyrophoricity necessitates careful handling. For decomposition-focused systems, which operate at slightly higher temperatures, nickel-based catalysts offer robustness and lower cost but often produce significant CO and require strategies to mitigate carbon formation. More recently, catalysts incorporating palladium, zirconia, and ceria have been explored to enhance low-temperature activity, improve thermal stability, and suppress the formation of carbon monoxide. The ongoing optimization of these materials focuses on achieving a balance between high methanol conversion, cost efficiency, long-term stability under cyclic operation, and resistance to deactivation, thereby ensuring the practical viability in pre-combustion fuel reforming gas turbines.

Ni-based catalyst

Ni-based catalysts are recognized as a cost-effective and highly active option for methanol steam reforming (MSR), offering a promising alternative to more expensive noble metal catalysts. These catalysts demonstrate significant potential for industrial-scale application due to their favorable catalytic performance, particularly at elevated temperatures. However, a major challenge associated with Ni-based catalysts is their tendency to deactivate over time due to coke formation, which results from side reactions such as methanation and dehydration. This coking is often linked to the presence of acidic sites on the catalyst surface, which promote unwanted carbon deposition.

To mitigate these issues, researchers have explored the use of metal oxide additives such as CeO_2, which can help reduce coke formation and enhance catalytic stability. For example, studies on Ni/CeO_2 catalysts have shown that the reduction of CeO_2 facilitates the formation of formate species, which are then decomposed into carbonates in the presence of metallic Ni. This pathway promotes the dehydrogenation route and improves CO_2 selectivity, especially at higher temperatures. The enhanced water dissociation on CeO_2 surfaces and oxygen transfer from ceria to Ni further boost the catalytic performance.

The dispersion of Ni particles plays a critical role in determining catalyst efficiency. Innovative synthesis methods, such as the use of molybdenum carbide supports (e.g., Ni/α-MoC), have enabled the achievement of atomically dispersed Ni sites, leading to exceptional catalytic activity — up to six times higher than conventional Pt/Al_2O_3 catalysts at low temperatures. Additionally, the incorporation of promoters like lanthanum (La) has been shown to improve NiO dispersion and lower reaction temperatures, thereby enhancing overall catalytic performance.

Despite these advancements, challenges such as metal sintering and carbon deposition remain significant hurdles for Ni-based MSR catalysts. Future efforts are focused on optimizing support materials, incorporating synergistic promoters, and refining preparation techniques to improve metal dispersion and stability. With continued development, Ni-based catalysts hold strong potential for sustainable and efficient hydrogen production via methanol steam reforming.

Cu-based catalyst

Cu-based catalysts are among the most widely studied and commercially utilized systems for methanol steam reforming (MSR), primarily due to their high activity at relatively low temperatures (200–300°C) and their excellent selectivity towards hydrogen with low CO production [195]. The catalytic performance of copper is strongly dependent on its dispersion and the nature of its interaction with the support. Conventional catalysts like Cu/ZnO/Al_2O_3 are industry benchmarks, where ZnO plays a crucial role in facilitating the reduction of copper oxide and stabilizing the active Cu metallic sites. However, a significant drawback of Cu-based catalysts is their susceptibility to thermal sintering and deactivation at high operating temperatures, which limits their long-term stability.

To address the issue of sintering, researchers have focused on enhancing metal dispersion and strengthening metal-support interactions through various strategies. The incorporation of secondary metals such as Zr, Ga, and Ni has been shown to improve thermal stability and reduce sintering. For instance, Xu et al. demonstrated that a Cu-ZnO-Al_2O_3-ZrO_2-Ga_2O_3 catalyst prepared via a nanocasting technique exhibited superior Cu distribution and twice the metallic copper surface area compared to a catalyst made by conventional co-precipitation [195]. This led to higher methanol conversion and zero CO selectivity, underscoring the importance of synthesis methods in defining catalyst morphology and performance.

The choice of support material profoundly influences the redox properties and catalytic mechanism of Cu-based catalysts. Layered double hydroxides (LDHs) have emerged as promising precursor materials, as they transform into highly dispersed mixed metal oxides upon calcination. He et al. reported that *in situ* synthesis of ZnAl-LDHs on γ-Al_2O_3 resulted in a uniform distribution of zinc cations, which promoted copper reduction at lower temperatures and enhanced catalytic activity compared to conventionally prepared catalysts [196]. Similarly, the addition of cerium oxide CeO_2 as a promoter has been shown to create oxygen vacancies and enhance the mobility of surface oxygen species, which aids in the oxidation of CO and improves hydrogen purity [197].

Another advanced strategy to minimize CO production and enhance hydrogen yield is the sorption-enhanced steam reforming of methanol (SESRM), which integrates CO_2 capture within the reaction process. In a study by Li et al. a bifunctional Cu-MgO/Al_2O_3 catalyst-sorbent was developed via the sol–gel method [198]. The incorporation of Cu^{2+} into the MgO lattice distorted its structure, increasing the number of basic sites and accelerating CO_2 sorption-desorption kinetics. This design not only achieved 100% methanol conversion and 99.3% hydrogen purity over multiple cycles but also suppressed CO formation by promoting the formate reaction pathway and preventing methoxy decomposition.

In summary, while Cu-based catalysts are highly effective for low-temperature MSR, their practical application is constrained by thermal instability and sintering [199]. Ongoing research is focused on optimizing catalyst composition through the use of promoters like Ga, Zr, and Ce, and employing advanced synthesis methods such as nanocasting and the use of LDH precursors. Furthermore, innovative processes like SESRM represent a promising direction for producing high-purity hydrogen with minimal CO byproduct. With continued development in support engineering and bimetallic formulations, Cu-based catalysts are poised to remain at the forefront of efficient hydrogen production technology.

Summary on methanol reforming catalysts

The performance of some catalysts for methanol reforming are systematized in Table 10. The catalytic performance data reveals critical insights for selecting and designing catalysts for pre-combustion methanol reforming in gas turbines. A primary consideration is the operational temperature window, which must align with the gas turbine's waste heat availability to ensure system efficiency. Copper-based catalysts demonstrate a significant advantage here, achieving high methanol conversions (>98%) at moderate temperatures of 240–275°C. This range is ideally suited for integration with the lower-temperature exhaust streams of a gas turbine, minimizing the need for additional, energy-intensive heating. In contrast, while nickel-based catalysts like Ni/Zn/Al and Ni/La/Al_2O_3 achieve complete conversion, they require substantially higher temperatures (500°C). This necessitates sourcing heat from the turbine's higher-temperature primary zone, which is thermodynamically less efficient and complicates system design.

Beyond temperature, catalyst activity and formulation are paramount. The data highlights the superior performance of specific copper formulations, such as Cu/Cu(Al)O_x and CuZnO/γ-Al_2O_3/Al, which combine high activity with moderate operating conditions. For nickel-based systems, the table underscores a critical trade-off: while monometallic Ni/Al_2O_3 shows modest conversion (60%), the introduction of a second metal, as in the bimetallic Ni/Cu/Al_2O_3, dramatically enhances performance to 98.5% at a more manageable 325°C. This suggests that bimetallic Ni–Cu systems offer a promising compromise, leveraging nickel's high activity while using copper to lower the required operating temperature and potentially mitigate nickel's known issues with coke formation.

Finally, the reaction conditions provide practical guidance for system engineering. The successful operation of several high-performance catalysts, such as Cu/Cu(Al)O_x and Ni/Cu/Al_2O_3, at a steam-to-carbon (S/C) ratio of 2.0 indicates a viable balance between promoting the water-gas shift reaction for high hydrogen yield and minimizing the energy penalty of vaporizing excess water. The broad range of successful weight hourly space velocities (WHSV) and gas hourly space velocities (GHSV) confirms that reactor design can be flexible, allowing for compact reformers that meet the specific space and flow rate constraints of a gas turbine package. In conclusion, for pre-combustion application,

Table 10
Catalytic performance of various copper- and nickel-based catalysts in methanol steam reforming.

Catalysts	Temp. (°C)	X_{CH_3OH} (%)	Reaction conditions	Ref.
$Cu/Cu(Al)O_x$	240	99.5	Feed rate = 2.4 mL h^{-1}; H_2O/CH_3OH = 2	[200]
Cu_5–Al	240	98.0	Feed rate = 0.048 mL h^{-1}; H_2O/CH_3OH = 1.5	[201]
Cu/Al_2O_3	250	89.7	WHSV = 10.56 h^{-1}; H_2O/CH_3OH = 1	[202]
CuZnAl	350	98.0	GHSV = 15,500 h^{-1}; H_2O/CH_3OH = 2	[203]
$CuZnO/\gamma–Al_2O_3/Al$	275	100.0	GHSV = 4 mL g^{-1} h^{-1}; H_2O/CH_3OH = 2	[204]
$Cu/ZnO/Al_2O_3$	225	67.0	WHSV = 6 h^{-1}; H_2O/CH_3OH = 1.3	[205]
$Cu/ZnO/Al_2O_3$	240	90.0	GHSV = 10 000 cm^3 g^{-1} h^{-1}; H_2O/CH_3OH = 1.5	[206]
$Ni/Zn/Al_2$	500	100	GHSV = 300 kg L^{-1} s^{-1}; H_2O/CH_3OH = 1.5	[207]
Ni/Cu/Al	380	35	GHSV = 2838 h^{-1}; H_2O/CH_3OH = 3	[208]
$Ni/Cu/Al_2O_3$	325	98.5	WHSV = 7.5 h^{-1}; H_2O/CH_3OH = 2	[209]
Ni/Al_2O_3	350	60.0	WHSV = 9.2 h^{-1}; H_2O/CH_3OH = 2	[210]
Ni/Mg/Al	300	80.0	GHSV = 800 h^{-1}; H_2O/CH_3OH = 1.2	[211]
$Ni/La/Al_2O_3$	500	100.0	GHSV = 10 000 cm^3 g^{-1} h^{-1}; H_2O/CH_3OH = 1.5	[212]

advanced copper-based catalysts operating between 240–275°C are the most directly applicable, while optimized Ni–Cu bimetallic catalysts present a viable, robust alternative for systems designed to operate at slightly higher temperatures.

7.3. Syngas (reformate) combustion

The successful integration of pre-combustion reforming of hydrocarbon fuels such as methane and methanol with gas turbines hinges critically on the efficient and stable combustion of its product: synthesis gas, or reformate. Distinctly different from conventional natural gas, reformate is typically a medium-heating-value fuel (in joules per volume) composed primarily of hydrogen (H_2), carbon monoxide (CO), and diluents like carbon dioxide (CO_2) and excess steam (H_2O). This altered composition presents a unique set of challenges and opportunities for gas turbine combustors, impacting key aspects such as flame stability, ignition, CO and NOx emissions, and combustor dynamics. While a significant body of literature exists on syngas combustion derived from coal gasification, studies focusing specifically on reformate from pre-combustion of lighter hydrocarbons are more limited. Therefore, this section examines the fundamental combustion characteristics of a range of syngas compositions relevant to this context. It reviews the primary challenges encountered in their utilization and discusses the design modifications and strategies required for their successful implementation in pre-combustion fuel reforming of gas turbine systems.

Lee et al. published a paper on syngas fueled gas turbine combustor [213]. A graphical summary of some results is presented in Fig. 34. Fig. 34 shows that the direct flame photographs reveal visual differences between syngas compositions and methane flames. The methane flame shows a compact, well-defined recirculation zone near the dump plane, while syngas flames exhibit elongated structures that move farther downstream as heat input increases. Particularly notable is the CO-rich flame (20%–60% CO), which appears highly luminous due to soot radiation. The flame photo matrix demonstrates that as hydrogen content increases across the fuel blends, the flames become progressively less luminous and more elongated, indicating differences in combustion behavior and heat release patterns compared to natural gas.

The emission characteristics show a clear trade-off between NOx and CO emissions depending on syngas composition. NOx emissions increase exponentially with heat input for all fuels, following Zeldovich's thermal mechanism, with hydrogen-rich blends producing the highest NOx levels. This correlates with higher flame temperatures and equivalence ratios for hydrogen-containing fuels. Conversely, CO emissions show an inverse relationship, where fuels with higher carbon monoxide content emit significantly more CO, especially at lower heat inputs. Pure hydrogen produces virtually zero CO emissions across all operating conditions, while 100% CO fuel shows the highest CO emissions, particularly at 30 kW where it reaches approximately 13 ppm. Methane exhibits intermediate CO emission levels between hydrogen-rich and CO-rich syngas blends.

Lee et al. also reported that the comparison between methane and syngas combustion reveals crucial operational implications. While methane combustion generates combustion instability with a dominant frequency of 204 Hz, all syngas compositions exhibit nearly zero combustion pulsation. However, syngas presents other challenges: hydrogen-rich fuels risk flashback and nozzle damage due to higher burning velocities and flame temperatures, while CO-rich fuels suffer from higher radiative heat losses and CO emissions. The flame structure analysis explains the temperature distribution patterns, with syngas flames moving downstream reducing dump plane temperatures despite increasing heat input. These findings suggest that syngas can enable more stable operation than natural gas but requires careful consideration of composition balance and potentially different cooling strategies when adapting existing gas turbines.

Similarly, Lee et al. investigated the effects of diluting a baseline H_2/CO syngas (with an approximate 1:2 ratio which is corresponding to syngas composition of complete methanol decomposition) with nitrogen (N_2), carbon dioxide (CO_2), and steam on the emissions from an industrial gas turbine combustor [214]. The research demonstrates that all three diluents are highly effective at reducing NOx emissions,

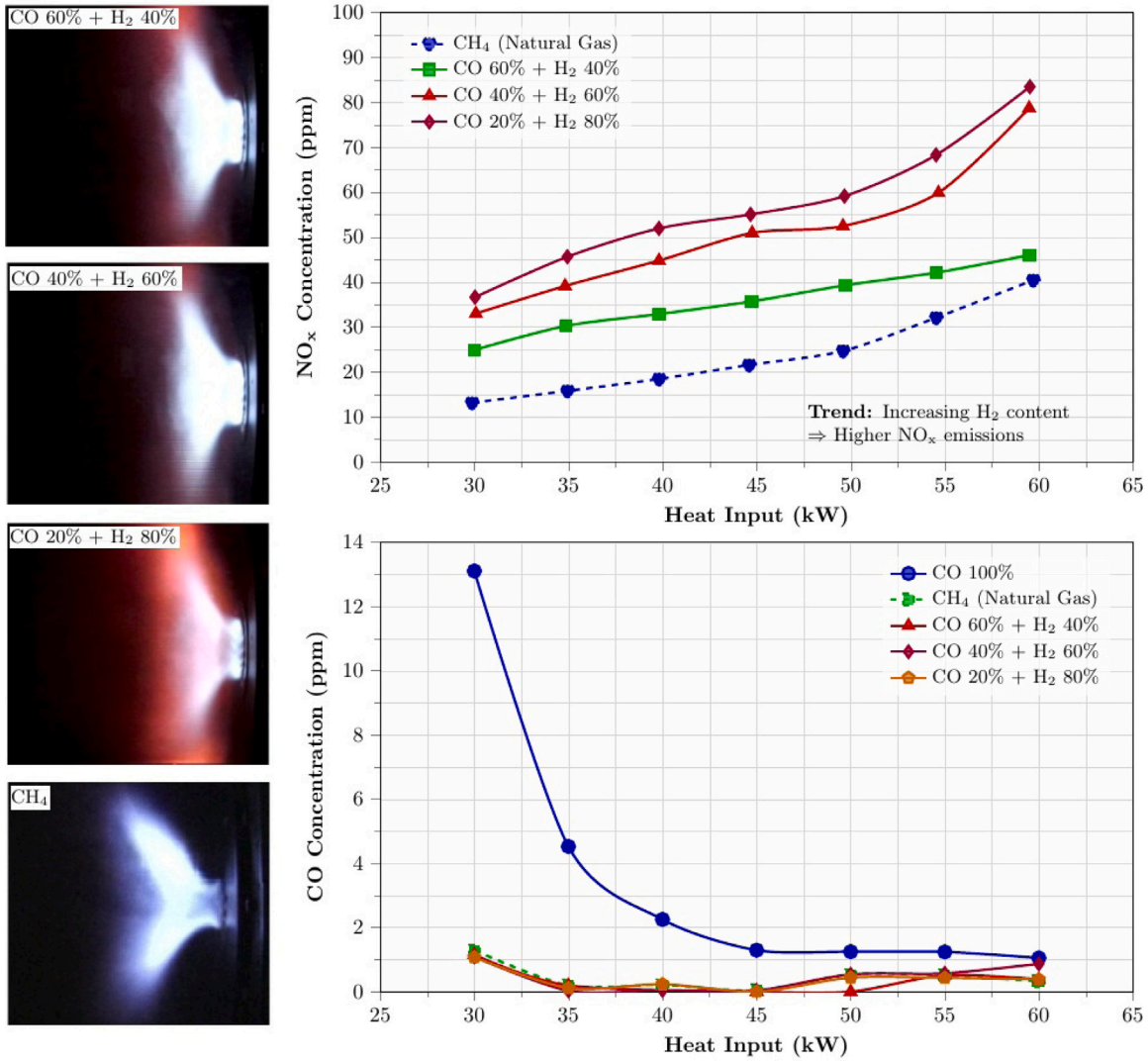


Fig. 34. A graphical summary of Lee et al. paper "Gas turbine combustion performance test of hydrogen and carbon monoxide synthetic gas" [213]: left side – direct flame photos of H_2/CO syngas and CH_4 at 55 kW heat input; right side – NO_x and CO concentrations.

with the reduction being logarithmically related to a single unifying parameter: the diluent's heat capacity (the product of its mass flow rate and specific heat). At a 40% dilution ratio and 55 kW heat input, steam was the most effective, reducing NOx by 95%, followed by CO_2 (87.7%) and N_2 (78.6%). In contrast, CO emissions remained below 10 ppm (indicating high combustion efficiency) across nearly all tested conditions, except for one critical case: at a low load of 30 kW with steam dilution, CO emissions increased exponentially due to the flame temperature dropping below the threshold for complete CO oxidation, a consequence of steam's very high heat capacity. The key finding is that while dilution is a powerful tool for NOx abatement, the choice and amount of diluent must be carefully managed, particularly at low loads, to avoid trading NOx reductions for unacceptable levels of CO emissions.

Due to the different combustion properties of syngas compared to natural gas, blending syngas with other gases is a widely used approach in gas turbines. Shi et al. published a comprehensive review that examines syngas combustion characteristics when blended with other gases, highlighting that fuel composition significantly impacts fundamental combustion properties and practical application performance, e.g. in gas turbine [215]. The analysis reveals that blending syngas with methane generally reduces laminar burning velocities due to physical dilution effects outweighing chemical enhancement, while ammonia addition shows minimal effect on ignition delay times but typically decreases flame speeds, though the magnitude depends strongly on syngas composition and operating conditions. For NOx control, the review emphasizes that diluents like nitrogen, carbon dioxide, and steam effectively reduce emissions, with steam proving most effective due to its high heat capacity, though it can increase CO emissions at low loads. The paper also discusses emerging strategies such as syngas/ammonia co-firing and flameless combustion, which offer pathways for reduced emissions but require careful management of fuel reactivity variations that may lead to thermoacoustic instabilities or combined thermal- and fuel-NOx pathways. These blending effects underscore the critical importance of understanding multi-component fuel dynamics and developing adaptive combustion technologies to optimize syngas utilization across various energy systems while maintaining stability and minimizing emissions.

Shen et al. investigated the combustion of synthesis gas obtained via methanol reforming in a gas turbine with pre-combustion fuel reforming, employing high-fidelity Large Eddy Simulation (LES) and chemical explosive mode analysis (CEMA) with the focus on flame stabilization and oxidation pathways [216]. The study systematically examines how steam addition and fuel composition – specifically reformed methanol blends ($CH_3OH/H_2/CO$) – affect combustion characteristics under conditions relevant to gas turbine operation. Key findings reveal that steam

Table 11
Summary of research on syngas ignition delay time (IDT) and laminar flame speed (LFS).

Research targets	Syngas composition	Operating conditions	Refs.
IDT of syngas	H_2/CO	1–40 bar; 470–1950°C	[217–222]
IDT of syngas with H_2O, CO_2, N_2	H_2/CO/H_2O, H_2/CO/CO_2, H_2/CO/CO_2/N_2	1–30 bar; 500–1200°C	[223–231]
IDT of syngas with methane	H_2/CO/CH_4/CO_2, H_2/CO/CH_4/CO_2/H_2O	1.5–60 bar; 600–1750°C	[232–234]
LFS of syngas	H_2/CO	0.5–40 bar; 25–170°C	[222,235–242]
LFS of syngas with H_2O, CO_2, N_2	H_2/CO/CO_2, H_2/CO/H_2O, H_2/CO/N_2, H_2/CO/CO_2/N_2	1–50 bar; 25–325°C	[243–251]
LFS of syngas with methane	H_2/CO/CH_4, H_2/CO/CH_4/CO_2, H_2/CO/CH_4/N_2, H_2/CO/CH_4/N_2/CO_2	1–11 bar; 25–370°C	[252–259]

addition promotes distributed combustion, characterized by enlarged heat release zones and reduced temperature gradients, particularly in the inner shear layer. The analysis demonstrates that auto-ignition becomes the dominant stabilization mechanism in distributed flames, facilitated by enhanced radical pools (HO_2 and OH) and altered reaction pathways, despite a slight increase in local extinction areas due to the expanded flame region.

Shen et al. highlighted that steam enrichment significantly modifies the oxidation chemistry, advancing H-atom abstraction reactions involving H radicals while suppressing those involving O radicals, thereby altering the dominant reaction pathways during ignition [216]. Through advanced diagnostic tools like conservative CEMA (CCEMA), the study provides a detailed classification of local combustion modes, showing a clear transition towards auto-ignition-dominated regimes in distributed flames, which aligns with the conditions for colorless distributed combustion (CDC). The authors conclude that wet combustion of reformed methanol blends not only enables stable and efficient operation but also supports the integration of chemical recuperation in gas turbines, offering a promising pathway for the utilization of carbon-neutral e-fuels like methanol in future energy systems.

The transition from natural gas to synthesis gas combustion leads also to changes in ignition performance. Walton et al. presented an experimental investigation into the ignition properties of hydrogen and carbon monoxide mixtures, relevant to syngas turbine applications, using a rapid compression facility [217]. The study measured ignition delay times across a wide range of conditions – including pressures from 7.1 to 26.4 atm, temperatures from 855 to 1051 K, equivalence ratios from 0.1 to 1.0, and H_2:CO ratios from 0.25 to 4.0 – and developed a correlated expression to describe the dependence of ignition delay on pressure, temperature, equivalence ratio, and oxygen concentration. The results show good agreement with predictions from a detailed H_2/CO reaction mechanism, though discrepancies in activation energy highlight uncertainties in key reactions such as $CO + HO_2 = CO_2 + OH$. This work provides the first high-pressure ignition data for syngas-relevant mixtures and offers valuable benchmarks for improving chemical kinetic models and guiding combustor design.

The gas turbines operate in a wide range of operating parameters, e.g. pressure ratio, air equivalence ratio, inlet temperature that leads to change in flame speed, ignition delay, quenching distance. Table 11 provides overview of recent papers related to study of synthesis gas and its blends combustion with the focus on ignition delay time (IDT) and laminar flame speed (LFS).

In summary, this subsection reviews the fundamental combustion characteristics of synthesis gas (syngas), including synthesis gas obtained via methanol or methane reforming, and its application in gas turbines, highlighting that despite differences in combustion performance compared to natural gas, syngas serves as a viable alternative fuel. A significant number of publications focus on adapting gas turbine systems for syngas combustion, addressing challenges such as altered flame stability, ignition behavior, and emissions profiles. Research by Lee et al. [213] demonstrates that while syngas can eliminate combustion instabilities present in methane systems, it introduces other issues: hydrogen-rich blends increase NOx emissions and flashback risk, whereas CO-rich blends elevate CO emissions and radiative heat loss. These studies confirm that, with appropriate operational adjustments and design modifications, syngas can be effectively utilized in gas turbines, underscoring its potential as a sustainable alternative to conventional natural gas.

The combustion properties of syngas, including ignition delay times (IDT) and laminar flame speeds (LFS), are systematically investigated across various compositions and conditions, as summarized in Table 11. Experimental and modeling efforts, such as those by Walton et al. [217], provide critical high-pressure IDT data and correlations essential for combustor design, while studies on diluted and methane-blended syngas explore strategies for emissions control and stability. The collective findings emphasize that syngas combustion performance is highly sensitive to its specific composition (H_2/CO ratio, diluents) and operating parameters. Nevertheless, the body of work consolidates the technical feasibility of syngas in gas turbines, proving that its combustion characteristics, though distinct, can be managed to achieve stable and efficient operation, thereby supporting its role in advanced power generation systems like gas turbines with pre-combustion fuel reforming.

7.4. Summary of Section 7

This section has detailed the application of chemically recuperated gas turbine (CRGT) technology to methanol as a primary fuel, focusing on two principal reforming pathways: methanol decomposition and methanol steam reforming. The fundamental advantage of a methanol-fed CRGT lies in its ability to efficiently upgrade low-grade turbine exhaust heat into high-grade chemical energy. By using this waste heat to drive the endothermic reforming of methanol into a hydrogen-rich syngas, the cycle achieves a significant boost in the fuel's lower heating value per unit mass. This thermochemical recuperation process directly translates to improved thermal efficiency, with studies demonstrating gains of several percentage points across various cycle configurations, from simple recuperated systems to advanced combined cycles and hybrid SOFC-MGT plants.

Beyond thermodynamic benefits, methanol-fed CRGTs offer combustion and operational advantages. The transition from liquid or vaporized methanol to a gaseous, hydrogen-rich reformate facilitates cleaner, more stable combustion with higher combustion efficiency. The reformed fuel's properties can enable operation in modes like flameless or distributed combustion, which are conducive to low NO_x emissions. Furthermore, methanol's relatively low and manageable reforming temperature (200–350°C) for steam reforming) aligns well with the temperature profiles of gas turbine exhaust or intercooler streams, simplifying thermal integration compared to fuels requiring higher-temperature reforming.

The choice between decomposition and steam reforming presents a key trade-off. Decomposition maximizes the fuel energy upgrade and cycle efficiency for a given heat input but produces a CO-rich syngas. Steam reforming, while slightly less efficient due to the energy penalty of steam generation, yields a hydrogen-richer, cleaner reformate that simplifies emissions control. The successful implementation hinges on selecting and developing robust, cost-effective catalysts – primarily copper-based for low-temperature steam reforming and nickel-based or bimetallic systems for decomposition – that maintain high activity and resist deactivation under cyclic operation. A primary challenge remains the economic production of "green" methanol from renewable sources, as its current high cost is a significant barrier to the widespread adoption of this otherwise promising carbon-neutral or carbon-recycling power generation technology.

8. Solar-hybrid gas-turbine systems with fuel reforming

The integration of solar-driven pre-combustion fuel reforming with a conventional gas turbine represents a hybrid approach to cleaner and efficient power generation. At its core, this system uses concentrated solar thermal energy – typically from a field of heliostats focusing sunlight onto a central receive – not to directly generate steam [260–263], but to drive an endothermic chemical reaction [264–266]. In a solar reformer, a fuel endothermically decomposed by this high-temperature solar flux (500–900°C) to undergo fuel reforming process. This process converts the fuel into a hydrogen-rich gas, or syngas in case of use hydrocarbon fuel. In effect, solar energy is chemically stored within the bonds of this new fuel mixture.

This hydrogen-enriched gas is then routed to the combustor of a standard gas turbine. The combustion of this reformed fuel offers significant advantages over burning pure primary fuel, e.g. ammonia. The higher hydrogen content allows for a faster combustion kinetics, which can increase the turbine's combustion efficiency [267,268]. The primary benefit of this configuration is its unique method of storing and utilizing intermittent solar energy. Unlike photovoltaic systems that require batteries, or solar thermal plants that rely on molten salt, this system stores solar heat chemically within the syngas. This decouples solar energy collection from power generation, allowing the gas turbine to operate at full capacity even when the sun is not shining, using fuel that has been "solar-upgraded". It effectively transforms the gas turbine plant into a dispatchable, partially renewable asset, enhancing grid stability while reducing its carbon footprint.

However, the path to commercialization is not without hurdles. The technology faces challenges related to high capital costs for the solar concentrating field and the complex high-pressure reformer, as well as the need for durable materials that can withstand daily cycling between extreme temperatures. Despite these barriers, gas turbines with solar pre-combustion reforming offer a compelling transitional pathway. They leverage existing natural gas infrastructure and turbine technology while integrating a significant portion of renewable solar energy, providing a tangible step towards decarbonizing the power sector.

8.1. Natural gas fed solar-hybrid gas-turbine systems

Solar-hybrid gas-turbine systems that utilize natural gas represent a significant pathway for the near-term decarbonization of power generation and industrial heat. A key enabling technology within this concept is Solar Steam Methane Reforming (SSMR), which uses concentrated solar thermal energy to drive the endothermic reforming reaction, thereby producing a hydrogen-rich syngas from natural gas and steam. This process directly reduces the fossil fuel consumption and associated carbon emissions of conventional reforming by supplying the high-temperature process heat (typically 800–1000°C) with solar energy [269–272].

Pregger et al. highlighted that while SSMR is not a zero-carbon process – it still produces direct CO_2 emissions from the chemical reaction – it offers a substantial reduction in emissions compared to conventional steam methane reforming [269]. A key advantage is its potential for near-term commercial integration, as it can be added to existing industrial plants. The study references the successful testing of a 400 kW thermal solar receiver-reformer prototype within the EU SOLASYS project [273], which demonstrated the feasibility of using a solar volumetric reactor to generate syngas for power generation.

The economic analysis presented in the paper by Pregger et al. suggests that SSMR could achieve hydrogen production costs in the range of 4.9–7.0 €ct/kWh, making it potentially competitive with conventional fossil-based reforming, especially in regions with high direct solar radiation and rising natural gas prices. The authors argue that the most immediate and realistic driver for developing SSMR is not the distant future hydrogen fuel economy, but the existing and growing industrial hydrogen demand in sun-rich regions (e.g., for refineries, ammonia, and methanol production). This industrial application provides a clear market for pilot and commercial plants within the coming decades, serving as a crucial stepping stone for the technology's development while offering a tangible path to reduce the carbon footprint of major chemical and refining industries.

Despite the industrial applications of natural gas fed solar-hybrid gas-turbine systems with pre-combustion fuel reforming are not described in literature, there are several prospective works on the use such approach highlighting its benefits and advantages. The integrated solar-gas turbine system with pre-combustion natural gas reforming was proposed by Bianchini et al. [274]. The proposed schematic diagram combines a parabolic trough Concentrating Solar Power (CSP) plant with a Steam Injected Gas Turbine (STIG) cycle via a solar methane steam reformer (MSR). The core innovation is the creation of a synergistic water and fuel loop. A portion of the natural gas feed (10% of the nominal turbine fuel) is diverted and mixed with a large excess of steam generated in the plant's Heat Recovery Steam Generator (HRSG). This mixture is heated by the CSP field to approximately 600°C at 30 bar in the solar reformer, where it undergoes conversion to a hydrogen-rich syngas. This solar syngas, along with the remaining steam not consumed in the reforming reaction, is then injected directly into the combustion chamber of the gas turbine (a modified GE LM2500 with Cheng Low NOx technology) alongside the remaining natural gas. This configuration cleverly uses excess reformer steam for turbine injection, overcoming the traditional steam-to-fuel ratio limits for both the reforming reaction and the STIG cycle.

The performance analysis indicates that this integration yields significant benefits despite the sub-optimal reforming conditions (lower temperature and higher pressure). By employing a very high steam-to-methane molar ratio (~40), a conversion efficiency (η_{MSR}) of about 85% is achieved. The system demonstrates a 10%–15% increase in power output (from 26.4 MWe to over 30 MWe) due to the increased mass flow from additional steam and syngas injection. Furthermore, it achieves a 1.7% net saving in natural gas consumption and an associated reduction in CO_2 emissions of approximately 685 tons per year, as the solar-derived syngas displaces fossil fuel. Critically, the hydrogen content in the fuel mixture allows for a higher steam injection

ratio without increasing CO emissions, maintaining low NO_x levels. The study posits that the system functions as a solar fuel generator, where the saved natural gas in the pipeline acts as a virtual storage medium for solar energy, enhancing the plant's role in a smart energy grid.

Based on the detailed economic assessment by the same team [274], the integration of solar steam reforming of natural gas (SSMR) with a Steam Injected Gas Turbine (STIG) plant presents a favorable economic case for deployment in sun-rich regions with access to natural gas grids. The analysis for a hybrid plant using parabolic trough CSP to reform 10% of the turbine's natural gas feed shows a reduction in the Levelized Cost of Energy (LCOE) by 2.7% (from 73.5 to 71.5 €/MWh) compared to a standard STIG plant, achieved through a 5.1% increase in annual electricity generation and concomitant fuel savings. Despite a higher initial investment (~€ 2 million additional), the project has a payback period of 6 years and yields an 11% higher Net Present Value (NPV) over a 20-year lifespan under baseline assumptions (electricity price: 90 €/MWh, gas price: 0.129 €/N m^3). A sensitivity analysis identifies natural gas and electricity prices as the most influential parameters for NPV, while the technology remains economically attractive in markets like Tunisia and Morocco. However, its competitiveness is limited in countries with heavily subsidized fossil fuels (e.g., Libya, Algeria), unless higher levels of solar integration or export-oriented strategies are pursued.

Author's previous paper presents a comparative thermodynamic analysis of two configurations of an Integrated Solar Combined Cycle System (ISCCS) [18]. The first system integrates solar energy specifically for steam methane reforming (SMR), using the produced hydrogen-rich syngas as fuel in a gas turbine. In contrast, the conventional ISCCS utilizes solar energy directly for steam generation in the steam turbine cycle. The key distinction lies in the pathway of solar energy integration: the SMR-based system channels high-temperature solar heat into the fuel cycle on the gas turbine side, while the conventional system applies solar heat to the lower-temperature steam turbine bottoming cycle.

The performance analysis reveals that the SMR-integrated ISCCS achieves superior overall efficiency. When the steam used in reforming is condensed and recovered from the exhaust gases, the efficiency gains are particularly significant — reaching up to 6.2% and 8.9% higher than the conventional ISCCS for steam-to-methane ratios (β) of 1.0 and 2.0, respectively. Furthermore, the SMR configuration shifts the power output distribution towards the more efficient gas turbine cycle, which contributes approximately two-thirds of the total power, compared to an almost equal split in the conventional system. This demonstrates that using solar energy for high-temperature fuel reforming, rather than for direct steam production, more effectively leverages the high-grade solar heat and enhances the thermodynamic performance of the combined cycle plant.

8.2. Ammonia fed solar-hybrid gas-turbine systems

The recent progress in ammonia-fed gas turbine technologies opens significant prospects for the use of pre-combustion ammonia reforming powered by solar energy. Ammonia-fed solar-hybrid gas turbine systems represent a relatively new direction in sustainable power generation. However, despite the novelty of this approach, only a few studies highlight the synergistic effect of integrating ammonia-fed gas turbines with concentrated solar systems for solar-driven, pre-combustion ammonia decomposition.

Recently author of present paper introduced a novel Integrated Solar Combined Cycle (ISCC) system fueled by ammonia, whose core innovation is captured in a schematic diagram integrating three subsystems as shown in Fig. 35: a solar-driven ammonia reformer, a gas turbine, and a bottoming steam turbine cycle [275]. Concentrated solar energy supplies high-temperature heat (≥500°C) to drive the endothermic decomposition of ammonia ($NH_3 \rightarrow N_2+3H_2$) in a catalytic reformer, producing a hydrogen-rich gas that is used as the primary fuel in the gas turbine's combustion chamber.

In Pashchenko's paper, a key thermodynamic analysis compares this solar-ammonia reforming system with a conventional solar-ammonia ISCC system where solar energy is used only for steam generation. Modeling shows the reforming system achieves higher overall thermal efficiency, with gains of up to 3.5%, because high-grade solar heat is used more effectively within the higher-efficiency gas turbine cycle instead of the lower-efficiency steam cycle. This shifts the power output distribution: in the reforming system, ~75% of the power is generated in the gas turbine, compared to ~60% in the conventional system.

A major challenge for ammonia-fed gas turbines is its low laminar flame speed (~7 cm/s), which leads to unstable combustion. The paper's solution – pre-combustion solar reforming – directly addresses this, as the produced hydrogen-rich fuel (up to 75% H_2) has a significantly higher flame speed. Analysis shows that under typical gas turbine conditions (10–20 bar, TIT up to 1400°C), the laminar flame speed for the reformate ($H_2 : N_2 = 3 : 1$) ranges from 20 to 30 cm/s, making it suitable for stable combustion in existing gas turbine hardware with minimal modification.

The synergistic integration of concentrated solar systems with ammonia-fed gas turbines offers a synergistic pathway to efficient, dispatchable, and carbon-free power generation. By using solar heat to drive the on-site decomposition of ammonia into a hydrogen-rich gas, the system fundamentally improves combustion stability and efficiency, overcoming the inherent drawbacks of pure ammonia such as low flame speed. This solar thermochemical recuperation effectively upgrades low-potential solar energy into high-grade chemical energy stored in hydrogen, which is then utilized in the high-efficiency gas turbine cycle. Consequently, the overall plant efficiency increases by up to 3.5%, while the incorporation of hydrogen storage elegantly resolves the issue of solar intermittency, ensuring a continuous and reliable fuel supply. This approach not only enhances thermal and combustion performance but also leverages existing gas turbine infrastructure, facilitating a practical transition towards sustainable, ammonia-based energy systems.

8.3. Summary of Section 8

Solar-hybrid chemically recuperated gas turbines (CRGTs) integrate concentrated solar thermal energy to drive endothermic fuel reforming, providing a unique pathway for decarbonized, dispatchable power. This approach upgrades intermittent solar heat into chemical energy stored within a hydrogen-rich syngas — produced via solar steam methane reforming (SSMR) of natural gas or solar decomposition of ammonia. This solar-derived reformate is then combusted in the gas turbine, effectively decoupling electricity generation from direct solar availability and enabling the plant to operate continuously. The system leverages existing turbine infrastructure while significantly increasing the renewable share of the fuel input, offering a practical transition towards low-carbon energy systems.

The thermodynamic and economic benefits of this integration are substantial. For natural gas systems, solar reforming boosts power output by 10%–15%, reduces fossil fuel consumption and CO_2 emissions, and enhances combustion stability and emissions control due to the hydrogen-enriched fuel. Economic analyses indicate competitiveness in sun-rich regions, with improved levelized cost of energy and favorable payback periods. For ammonia systems, solar-driven decomposition directly overcomes ammonia's poor combustion properties – such as low flame speed – by producing a stable H_2/N_2 mixture, leading to efficiency gains of up to 3.5% in combined-cycle configurations. In both cases, utilizing high-grade solar heat for fuel reforming within the gas turbine cycle proves more efficient than applying it to a bottoming steam cycle.

Despite these advantages, key challenges remain, including the high capital cost of solar concentrators and high-pressure solar reformers, as

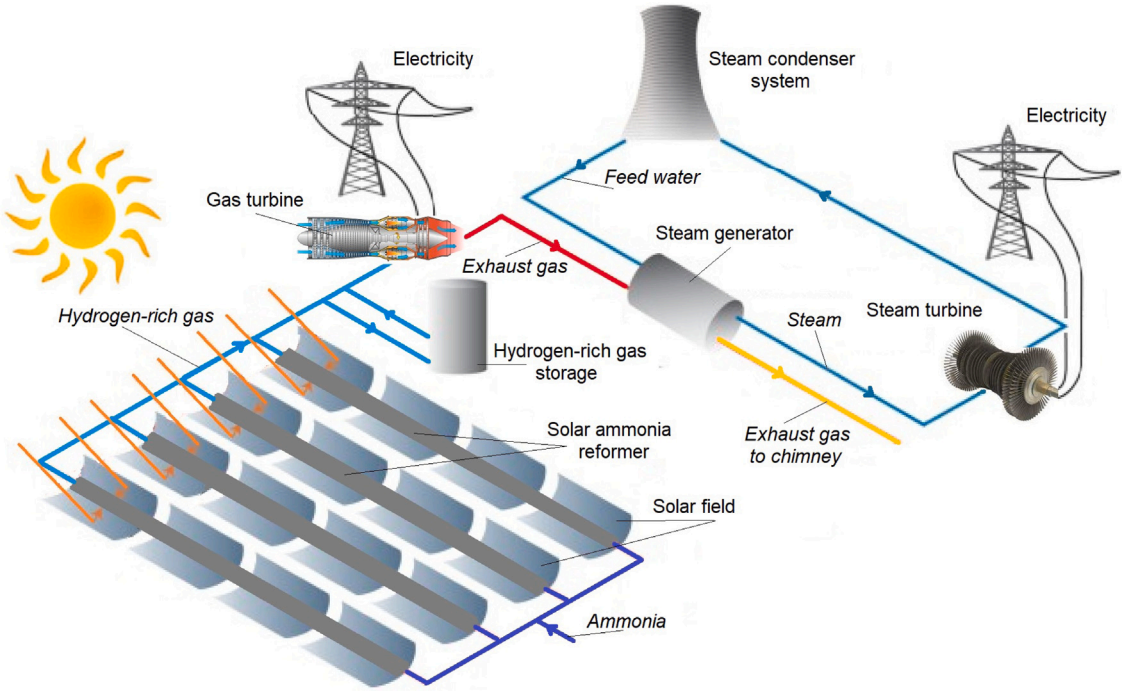


Fig. 35. Schematic diagram of integrated solar ammonia-fed gas turbine with bottom steam turbine proposed by Pashchenko [275].

well as material durability under cyclic thermal stress. However, solar-hybrid CRGTs represent a compelling near-to-mid-term solution. They enhance grid stability through inherent solar energy storage, reduce the carbon intensity of conventional gas turbine power, and pave the way for fully renewable systems using carbon-neutral fuels like green ammonia or synthetic methanol.

9. Hybrid fuel cell gas turbines (FC-GT)

Another direction of pre-combustion fuel reforming provides hybrid fuel cell gas turbines (FC/GT). The concept of integrating a fuel cell with a gas turbine first emerged in the mid-1970s [276–278]. By 1998, more than ten different hybrid configurations had been patented, varying in fuel cell type, component layout within the integrated system, and operating pressure.

Recently, several comprehensive reviews were published on hybrid fuel cell gas turbine systems. Buonomano et al. systematized the possible layout configurations for hybrid power plants integrating solid oxide fuel cells (SOFC) and gas turbines (GT), focusing on methane-fed, pressurized arrangements that offer higher efficiency and lower capital costs [279]. It also examines complex system configurations, control strategies, part-load performance limitations, and the use of alternative fuels such as coal and biomass in SOFC/GT hybrid systems. Azizi and Brouwer reviews the system design, transient control, and optimization of solid oxide fuel cell-gas turbine (SOFC-GT) hybrid systems, focusing on control strategies such as variable/fixed-speed shaft operation and bypass valves to manage compressor surge and stack inlet temperatures [280]. It also evaluates the impacts of key parameters (e.g., steam-to-carbon ratio, fuel utilization) on performance, compares net efficiencies across configurations, and discusses applications including CO_2 capture and alternative fuels. Bao et al. provided a comprehensive review of macroscopic solid oxide fuel cell (SOFC) models and model-based control strategies for SOFC/gas turbine (GT) hybrid systems, covering topics from electrochemical and transport phenomena (e.g., multi-component mass transfer, H_2/CO co-oxidation) to system-level issues such as safe operation, load following, and decentralized control [281]. It also presents analytical solutions, tabulated data, and perspectives on advanced modeling to bridge the gap between mechanistic understanding and system-level control.

System studies for hybrid power systems with capacities up to 300 MWe, based on 40 MWe power modules, have been conducted for the U.S. Department of Energy and other organizations [282–284]. Around the year 2000, the first experimental demonstrations of hybrid configurations integrating fuel cells with gas turbines commenced through initiatives in both the United States and Japan. Prototype systems employing both molten carbonate fuel cells (MCFC) and solid oxide fuel cells (SOFC) have been constructed and tested. These demonstrations have confirmed the capability of hybrid systems to deliver high electrical efficiency and very low emissions when operating on natural gas. To date, several distinct hybrid gas turbine fuel cell systems have been tested, each embodying a unique design approach [285,286].

Fig. 36 illustrates the fundamental layout of a hybrid fuel cell gas turbine system, in which the fuel cell integrates with a standard Brayton cycle [287]. In this arrangement, the fuel cell directly converts fuel into electricity – replacing the chemical-to-thermal conversion that normally occurs in a combustor – while its waste heat drives the compressor and generates additional power via a turbo-generator. Because electrochemical power generation reduces emissions and improves efficiency, approximately 80% of the total electricity is produced electrochemically in the fuel cell, with the remaining 20% coming from the turbo-generator.

Hybrid systems integrating a fuel cell and a gas turbine can be arranged in numerous ways, leading to a wide variety of potential cycle configurations. Four fundamental parameters serve to characterize these hybrid system layouts: (1) fuel cell topping cycle [288], (2) fuel cell bottoming cycle [289], (3) direct hybrid cycle [290], and (4) indirect hybrid cycle [291]. The vast majority of hybrid cycles proposed and analyzed in the literature can be described using these four categories.

A fuel cell topping cycle treats the gas turbine as the balance of plant, with the turbomachinery placed downstream of the fuel cell as shown in Fig. 36a. This configuration essentially substitutes the fuel cell for the combustor in a conventional Brayton cycle. The turbine expands the exhaust gases leaving the fuel cell to produce compression power

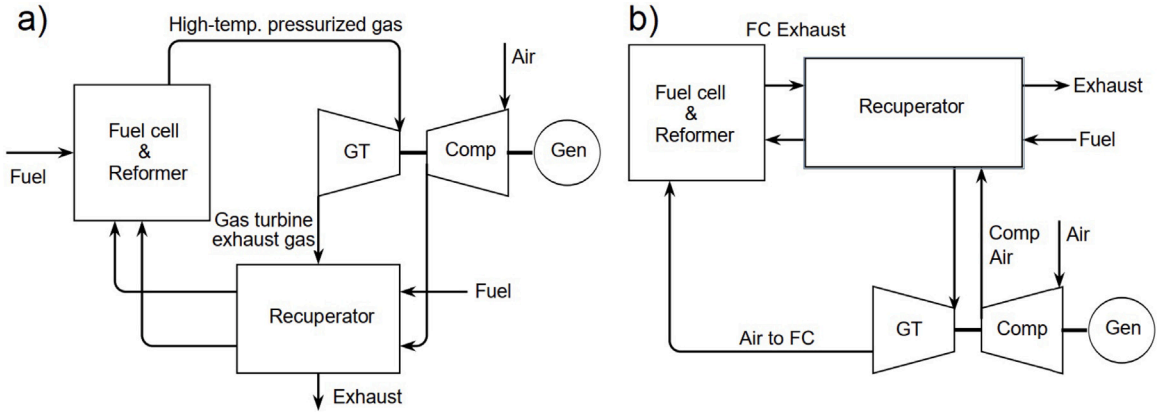


Fig. 36. Schematic design concept of a hybrid fuel cell gas turbine system: (a) gas turbine is topping cycle; (b) gas turbine is bottoming cycle.

and additional electricity, while the fuel cell serves as the primary electricity generator.

In contrast, a fuel cell bottoming cycle positions the gas turbine turbomachinery upstream of the fuel cell presented in Fig. 36b. Here, the fuel cell is located downstream of the turbine and receives the gas turbine exhaust as its air supply. The fuel cell remains the main power producer. This bottoming configuration is especially well suited to molten carbonate fuel cells (MCFCs), because MCFCs require carbon dioxide in the oxidant stream to form carbonate ions — and this CO_2 can be supplied directly by the combustor of an upstream gas turbine engine.

A direct hybrid cycle is defined by the direct use of flow from upstream components in downstream components without intermediate devices to decouple the two cycles. Heat exchangers are not required to separate the cycles, though they may be employed for other purposes such as preheating. In a direct hybrid cycle, the fuel cell typically operates under pressurized conditions — specifically at the pressure existing between the gas turbine compressor and turbine. While this arrangement presents greater challenges for system control, fuel cell operation, and long-term degradation, direct hybrid cycles generally achieve higher efficiency than their indirect counterparts.

An indirect hybrid cycle, by contrast, employs heat exchangers or similar devices to decouple the gas turbine and fuel cell components so that the flow leaving upstream elements does not pass directly into downstream elements. Thermal integration in an indirect cycle incurs additional losses due to the extra heat exchangers, and the fuel cell typically operates at atmospheric pressure. This configuration is less challenging to control and operate, and it imposes fewer difficulties related to fuel cell degradation. However, indirect hybrid cycles tend to be less efficient than direct cycles, and the cost and physical size of the required heat exchanger components can be substantial.

9.1. Difference between FC-GT and CRGT

The fundamental distinction between a fuel cell gas turbine hybrid system (FC-GT) and a chemically recuperated gas turbine (CRGT) lies in how chemical energy is converted into electrical power and how waste heat is recovered.

The overarching objective of any hybrid energy system is to combine multiple energy conversion devices – or multiple fuels for a single device – into an integrated configuration that yields advantages in fuel adaptability, efficiency, operational reliability, economic performance, or environmental sustainability that no single device could achieve on its own [289]. Such systems are often described as synergistic because the integrated components complement one another, producing overall performance characteristics that exceed the sum of their individual contributions.

In the specific context of a fuel cell gas turbine hybrid, four primary design principles guide system development. First, the majority of the fuel is converted through electro-oxidation within the fuel cell, which results in low emissions of conventional pollutants and relatively high fuel-to-electricity conversion efficiency [292]. Second, the heat generated by both the fuel cell and the turbine exhaust is recovered elsewhere in the system – for applications such as fuel processing, reactant preheating, or supplying compression power – in a manner that enhances overall efficiency [293]. Third, the elevated pressure produced by the gas turbine compressor is utilized to improve the output and efficiency of the fuel cell, where feasible. Fourth, the separated fuel and oxidant streams characteristic of fuel cell operation enable additional features of the hybrid cycle, such as carbon dioxide capture or fuel production.

Synergy in hybrid cycles can be realized in numerous ways, as illustrated by the latter two principles. Operating a fuel cell at the high-pressure conditions existing between the compressor and turbine of a gas turbine engine accomplishes two beneficial effects: it increases the power output of a given fuel cell stack, and it reduces certain electrochemical losses – most notably by improving electrochemical kinetics – thereby leading to higher efficiency [294]. Furthermore, because a fuel cell operates with physically separated fuel and oxidant streams, multiple opportunities arise for enhanced hybrid cycle performance. These include easier sequestration of a more concentrated carbon dioxide stream, as well as significant potential for producing hydrogen or synthetic hydrocarbon fuels. These synergistic aspects of hybrid gas turbine fuel cell systems have attracted considerable research interest and are a major reason why analyses and experimental testing to date have concluded that such hybrid systems hold great promise.

In contrast, a chemically recuperated gas turbine relies on thermochemical conversion followed by thermal conversion through combustion. The gas turbine remains the prime mover [110]. Turbine exhaust heat drives an endothermic reforming reaction – such as steam methane reforming, ammonia cracking, or methanol reforming – that converts the primary fuel into a hydrogen-rich gas before combustion. This upgraded fuel is then burned in the combustor. Thus, the gas turbine exhaust heat is recuperated back into the same gas turbine cycle by being stored as chemical energy in the fuel. This process increases overall cycle efficiency and improves combustion stability, particularly for low-reactivity fuels like ammonia. Nearly all power comes from the gas turbine; there is no fuel cell in the cycle (see Table 12).

In summary, a fuel cell gas turbine hybrid recovers waste heat from an upstream fuel cell that electrochemically converts most of the fuel's energy, whereas a chemically recuperated gas turbine uses exhaust heat to reform the fuel before combustion, upgrading waste heat into chemical energy within the same gas turbine cycle.

Table 12
Comparison of fuel cell gas turbine hybrids and chemically recuperated gas turbines.

Feature	FC-GT	CRGT
Primary conversion principle	Electrochemical	Thermochemical + combustion
Role of gas turbine	Bottoming/topping cycle	Topping cycle
Waste heat utilization	Expanded through turbine	Drives fuel reforming
Fuel before combustion	Unreformed (to fuel cell)	Reformed (hydrogen-rich)
Typical electrical efficiency	60%–70%	45%–55% Simple GT; 60%–64% CCPP
Emissions	Extremely low	Low, but combustion-dependent
Fuel flexibility	Moderate	High (natural gas, ammonia, methanol)
Typical power split	~80% fuel cell, ~20% gas turbine	~100% Simple GT; ~75% GT/~25% CCPP

9.2. Constraints for FC-GT applications

Fuel cell gas turbine (FC-GT) hybrid systems face several significant constraints that arise from the technical challenges of integrating two fundamentally different technologies with distinct operating characteristics. These challenges lead to high capital costs, durability issues, and complex control requirements. While FC-GT hybrids offer very high electrical efficiency exceeding 60%–70% combined with ultra-low emissions, they are primarily suited for base-load operation and encounter substantial difficulties in transient load following and part-load performance.

System integration and component mismatch

A primary constraint is the inherent mismatch between existing gas turbines and high-temperature fuel cells such as solid oxide fuel cells (SOFCs). Gas turbines are typically designed for specific pressure ratios, mass flows, and thermal dynamics that often do not align with the optimal operating conditions of the fuel cell [295]. Forcing these components to work together frequently requires operating them under off-design conditions, which reduces overall system efficiency and increases mechanical stress on both subsystems.

High capital costs and limited durability

Fuel cells remain expensive due to the cost of specialized materials, including precious metals and ceramic components, and they possess a limited operational lifetime compared to conventional power generation systems [294]. Thermal cycling – which is unavoidable during load following and start-stop operations – significantly increases thermal stress on the fuel cell stack, leading to accelerated deterioration of stack components and an increased risk of premature failure. This limited durability directly impacts the economic viability of FC-GT hybrids for applications requiring frequent load changes.

Thermal and dynamic limitations

Fuel cells exhibit relatively slow dynamic responses compared to gas turbines [296]. The fuel cell stack requires strict thermal management to maintain uniform temperature distribution and prevent localized hot spots or thermal gradients that can cause degradation. This requirement complicates rapid power adjustment, as sudden changes in fuel flow or load can disrupt the thermal balance of the stack. Consequently, FC-GT hybrids are better suited for steady-state, base-load operation rather than grid services requiring fast ramping capabilities.

Compressor surge risk

Pressurized hybrid systems, particularly those operating at part-load conditions or during transient events, face an increased risk of compressor stall or surge [297]. The close coupling between the gas turbine compressor and the pressurized fuel cell means that any disturbance in the fuel cell's backpressure or exhaust flow can propagate back to the compressor, pushing it towards unstable operating regions. Compressor surge can cause severe mechanical damage to blades and bearings, making surge avoidance a critical design and control consideration for pressurized FC-GT configurations.

Fuel purity requirements

Fuel purity represents a critical constraint for FC-GT hybrid systems, though the severity of this constraint depends heavily on the type of fuel cell employed [298,299]. High-temperature fuel cells such as SOFCs, which are the standard choice for gas turbine integration due to their high operating temperatures and compatibility with pressurized operation, are generally more tolerant of fuel impurities than low-temperature systems. Nevertheless, even SOFCs face significant performance degradation from specific contaminants, which imposes strict limits on the quality of the gas supplied to the fuel cell.

The sensitivity to contaminants varies significantly between fuel cell types. Proton exchange membrane fuel cells (PEMFCs) are extremely sensitive, requiring hydrogen purity of 99.999% or higher because even trace amounts of carbon monoxide or other impurities can poison the platinum catalyst and cause permanent failure. In contrast, SOFCs are substantially more tolerant. They can utilize carbon monoxide as a direct fuel and can even internally reform simple hydrocarbons, which relaxes some purity requirements compared to PEMFCs.

The fuel purity requirement creates a direct economic trade-off between system cost and long-term durability. Achieving the high purity levels necessary to prevent degradation adds significant capital expenditure (CAPEX) and operating expenditure (OPEX). Purification technologies such as pressure swing adsorption (PSA) and activated carbon filters are effective but expensive and can result in some loss of fuel during processing. Furthermore, storing and transporting ultra-pure fuel is more complex, as standard pipelines, storage tanks, and distribution infrastructure can introduce new contaminants from seals, liners, or corrosion products. Consequently, system operators must choose between investing in expensive upstream purification equipment or accepting more frequent and costly fuel cell stack replacements due to gradual poisoning.

Comparison of FC-GT with CRGT on constraints

In contrast to FC-GT hybrid systems, chemically recuperated gas turbines (CRGTs) are more tolerant to gas quality after the reformer. This fundamental difference arises because the reformate produced in a CRGT – whether from natural gas, ammonia, or methanol reforming – is directly combusted in the gas turbine combustor. Combustion processes are inherently tolerant to a wide range of fuel compositions and impurities. Trace contaminants that would poison a fuel cell catalyst, such as sulfur compounds, siloxanes, or halogens, are simply oxidized in the high-temperature flame and do not cause long-term degradation of the turbine components, provided that appropriate materials are selected to resist corrosion from combustion byproducts.

Furthermore, CRGT systems offer significantly greater fuel flexibility compared to FC-GT hybrids. A chemically recuperated gas turbine can operate on a variety of primary fuels – including natural gas, ammonia, methanol, and even low-grade or impure fuel streams – without requiring the stringent gas purification steps necessary for fuel cell protection. The reforming process itself can accommodate certain levels of impurities, and any remaining contaminants are safely oxidized during combustion. This flexibility makes CRGT technology particularly attractive for applications where fuel quality cannot be

guaranteed or where the cost of fuel purification would be prohibitive. Additionally, CRGT systems exhibit better dynamic response characteristics than FC-GT hybrids because they lack the thermal inertia and slow electrochemical kinetics of fuel cell stacks, enabling more rapid load following and improved grid support capabilities.

9.3. Summary of Section 9

This section has provided a comparative analysis of fuel cell-gas turbine (FC-GT) hybrid systems and chemically recuperated gas turbines (CRGT), highlighting their distinct principles of operation, performance characteristics, and practical constraints.

FC-GT hybrids represent a synergistic integration where the fuel cell serves as the primary power generator, electrochemically converting 60%–70% of the fuel's chemical energy into electricity, while the gas turbine recovers waste heat from the fuel cell exhaust to produce additional power. These systems achieve very high electrical efficiencies (60%–70%) and ultra-low emissions, making them attractive for base-load power generation. However, they face significant constraints including component mismatch between the fuel cell and turbine, high capital costs, limited durability due to thermal cycling, slow dynamic response, compressor surge risk in pressurized configurations, and stringent fuel purity requirements that necessitate expensive purification systems.

In contrast, CRGT systems maintain the gas turbine as the prime mover. Turbine exhaust heat drives endothermic reforming reactions that convert the primary fuel into a hydrogen-rich reformate before combustion. This thermochemical recuperation upgrades waste heat into chemical energy, improving cycle efficiency and combustion stability. While CRGTs achieve lower electrical efficiencies than FC-GT hybrids in simple cycle configurations (45%–55%), they offer several practical advantages: superior fuel flexibility, greater tolerance to fuel impurities, better dynamic load-following capabilities, and simpler integration with existing gas turbine infrastructure. Combined cycle CRGT configurations (CCCRGT) can achieve efficiencies of 60%–64%, approaching those of FC-GT systems.

The fundamental distinction lies in the energy conversion pathway: FC-GT hybrids rely on electrochemical conversion with the gas turbine as a bottoming or topping cycle, while CRGTs employ thermochemical conversion with the gas turbine as the sole prime mover. For near-term decarbonization of existing turbine fleets, CRGT technology offers a more versatile and economically accessible pathway, whereas FC-GT hybrids represent a longer-term solution for achieving the highest possible efficiencies in new-build power plants, provided that the technical and economic constraints can be overcome.

10. Summary and perspectives

This paper provides a comprehensive review of pre-combustion fuel reforming in gas turbines, focusing on its potential to enhance thermal efficiency and enable the transition to low-carbon fuels such as natural gas, ammonia, and methanol. By converting waste heat from turbine exhaust – or external sources such as solar thermal energy – into chemical energy through endothermic reforming reactions, this technology effectively upgrades the fuel's energy content and improves combustion stability. The review systematically examines the historical development, thermodynamic principles, system configurations, and combustion characteristics of chemically recuperated gas turbine (CRGT) cycles. It also evaluates performance across different fuel types and integration schemes, including combined cycles and solar-hybrid systems.

While distinct from fuel cell-gas turbine (FC-GT) hybrids – where the fuel cell is the primary power source and the turbine recovers residual heat – the CRGT maintains the gas turbine as the prime mover. FC-GT systems face critical constraints such as component mismatch, slow dynamic response, and stringent fuel purity requirements, whereas CRGT technology offers superior fuel flexibility, greater tolerance to impurities, and better dynamic load-following capabilities, making it a more versatile near-term solution for decarbonizing existing turbine fleets.

Despite the demonstrated benefits in efficiency and emissions reduction, several technical and economic hurdles must be addressed for widespread commercial adoption of CRGT technology. The primary technical challenges include: (i) catalyst deactivation through sintering and carbon deposition under moderate-temperature, high-pressure operating conditions; (ii) increased NO_x emissions when combusting hydrogen-rich reformates, requiring advanced combustor designs and operational strategies; (iii) the energy penalty associated with steam generation for steam methane and methanol reforming; (iv) material durability issues in high-temperature reformers and heat exchangers subjected to thermal cycling; and (v) the thermodynamic limitation imposed by turbine exhaust gas temperature, which constrains methane conversion and heat recuperation efficiency. Economically, the high capital cost of reformer components and, for carbon-neutral pathways, the currently high cost of green ammonia and methanol remain significant barriers to competitiveness with conventional natural gas combined cycles. These challenges underscore the need for continued research in catalyst development, combustor design, and system integration to realize the full potential of CRGT technology.

The following research and development priorities are identified:

- System optimization and integration: Further work is needed to optimize the coupling between reformers and gas turbines under variable load conditions. This includes dynamic modeling and control strategies to maintain high efficiency and stable operation during transients.
- Catalyst development and durability: For natural gas and methanol reforming, catalysts must maintain high activity and resistance to coking and sintering under moderate-temperature, high-pressure conditions typical of turbine exhaust. For ammonia cracking, cost-effective Ni-based catalysts with improved low-temperature activity and longevity are essential to reduce system costs.
- Combustion stability and emission control: While reforming improves flame speed and stability, particularly for ammonia, it can also increase NO_x emissions. Research is needed to develop combustor designs and operational strategies – such as staged combustion, dilution, and exhaust gas recirculation – that minimize NO_x while maintaining stable combustion of hydrogen-rich reformates.
- Material and component reliability: High-temperature reformers and heat exchangers must withstand thermal cycling and corrosive environments. Advanced materials and coatings are required to extend component life and reduce maintenance costs.
- Economic viability and scalability: Although CRGT cycles show higher efficiency and lower emissions, their economic competitiveness depends on fuel prices, capital costs, and operational flexibility. Pilot and demonstration projects are needed to validate techno-economic models and de-risk scale-up.
- Integration with renewable energy: Solar-driven reforming offers a pathway to store intermittent solar energy chemically, enabling dispatchable power with reduced carbon footprint. Further research should focus on reactor design, thermal management, and system integration for hybrid solar-CRGT plants.
- Hybrid and polygeneration systems: CRGT can be combined with bottoming cycles, fuel cells, or carbon capture systems to further improve efficiency and sustainability. Research into such integrated energy systems could unlock new applications in industry and power generation.

Pre-combustion reforming provides distinct thermodynamic and combustion benefits depending on the fuel pathway, with each pre-

Table 13
Comparative assessment of pre-combustion reforming pathways for gas turbines.

Criterion	Natural gas (SMR)	Ammonia cracking	Methanol reforming
Primary benefit	High efficiency gain (4%–7% in CCPP)	Enables combustion of carbon-free fuel	Low reforming temperature (200–350°C)
Thermodynamic benefit	Significant (60%–64% CCCRGT efficiency)	Moderate (up to 9% efficiency gain)	Moderate (2%–5% efficiency gain)
Combustion benefit	Improved flame speed and stability	Critical for flame stabilization	Improved combustion of liquid fuel
Technology maturity	High (industrial SMR well established)	Medium (catalysts under development)	High (well-established for H_2 production)
Main technical barrier	Catalyst coking at moderate T	NO_x emissions trade-off	Steam generation energy penalty
Catalyst challenge	Sintering, carbon deposition	Low-temperature activity, cost	Thermal stability (Cu-based)
Reforming temperature	600–750°C	500–650°C	240–350°C
CO_2 emissions	Reduced (not eliminated)	Zero (carbon-free fuel)	Potentially zero (green methanol)
Economic viability	Competitive with natural gas	Limited by green NH_3 cost	Limited by green methanol cost
Readiness level	Near-term deployment	Pilot/demonstration	Near-term deployment

senting unique challenges and varying levels of technological maturity. Table 13 summarizes these comparative aspects.

Natural gas reforming via SMR offers the most significant thermodynamic benefits, with combined cycle CRGT configurations achieving efficiencies of 60%–64%, and is the most technologically mature pathway. The primary barrier is catalyst durability, specifically coking and sintering under the moderate-temperature, high-pressure conditions of gas turbine exhaust. This pathway is the closest to commercial deployment, particularly for retrofitting existing natural gas-fired plants.

Ammonia cracking provides a critical enabling function for carbon-free power generation, as the hydrogen-rich reformate overcomes the poor combustion properties of pure ammonia. While efficiency gains are moderate (up to 9%), the combustion benefits are transformative. The main challenges include NO_x emissions control (which exhibits a complex, non-monotonic relationship with cracking ratio) and the development of cost-effective, low-temperature Ni-based catalysts. Pilot-scale demonstrations are underway, but widespread adoption depends on reductions in green ammonia production costs.

Methanol reforming offers the lowest operating temperature (240–350°C), enabling integration with lower-temperature heat sources and simplifying thermal management. The thermodynamic benefits are moderate (2%–5% efficiency gain), but the reforming chemistry is well-established with commercially available Cu-based catalysts. The primary challenge is the energy penalty associated with steam generation for methanol steam reforming, and, for carbon-neutral operation, the high cost of green methanol. This pathway is technologically mature and suitable for near-term deployment where methanol infrastructure exists.

Future development priorities differ across pathways. For natural gas, research should focus on coke-resistant catalysts and optimization of combined cycle integration. For ammonia, the priorities are low-cost, high-activity catalysts and NO_x mitigation strategies. For methanol, efforts should concentrate on process intensification to minimize the steam generation penalty. Across all pathways, system-level optimization, dynamic modeling, and techno-economic analysis are essential to de-risk scale-up and demonstrate commercial viability. The choice of the most promising pathway will depend on regional fuel availability, existing infrastructure, and the specific decarbonization goals of the power generation system. Pre-combustion fuel reforming represents a versatile and promising technology that can help modernize existing gas turbine fleets and support the transition to a low-carbon energy future. By leveraging waste heat to produce hydrogen-rich fuels, it addresses both efficiency improvement and fuel flexibility — key challenges in decarbonizing gas turbine power generation. Continued interdisciplinary efforts in thermodynamics, catalysis, combustion, and system engineering will be essential to overcome current barriers and realize the full potential of chemically recuperated gas turbines in sustainable energy systems.

Nomenclature

Abbreviations

CCGT	Combined Cycle Gas Turbine
CCCRGT	Combined Cycle Chemically Recuperated Gas Turbine
CCPP	Combined Cycle Power Plant
CET	Coefficient of Exergy Transformation
CHGT	Chemical-recuperated and Humidified Gas Turbine
CHT	Coefficient of Heat Transformation
CRGT	Chemically Recuperated Gas Turbine
CSP	Concentrating Solar Power
EGR	Exhaust Gas Recirculation
FC-GT	Fuel Cell-Gas Turbine hybrid
GHSV	Gas Hourly Space Velocity
GT	Gas Turbine
HRSG	Heat Recovery Steam Generator
HTC	Heat Transformation Coefficient
ICE	Internal Combustion Engine
IDT	Ignition Delay Time
ISCCS	Integrated Solar Combined Cycle System
LCOE	Levelized Cost of Electricity
LFS	Laminar Flame Speed
LHV	Lower Heating Value
MCFC	Molten Carbonate Fuel Cell
MSR	Methanol Steam Reforming
NGCC	Natural Gas Combined Cycle
ORC	Organic Rankine Cycle
PCACC	Partially Cracked Ammonia Combined Cycle
PEMFC	Proton Exchange Membrane Fuel Cell
PtL	Power-to-Liquids
S/C	Steam-to-Carbon ratio
SCR	Selective Catalytic Reduction
SMR	Steam Methane Reforming
SOFC	Solid Oxide Fuel Cell
SSMR	Solar Steam Methane Reforming
STIG	Steam-Injected Gas Turbine
TCR	Thermochemical Recuperation
TEGT	Turbine Exhaust Gas Temperature
TIT	Turbine Inlet Temperature
WHSV	Weight Hourly Space Velocity
WOT	Wide-Open Throttle

Symbols

CAF_T	Total cooling air fraction (–)
Ex_{chem}	Chemical exergy (MJ/kg)
Ex_q	Thermal exergy of supplied heat (MJ/kg)
ΔEx_{fuel}	Increase in fuel exergy (MJ/kg)
G^{sys}	Total Gibbs free energy of system (kJ/mol)
H_i	Enthalpy of ith stream (MJ/kg)

ΔH^0_{298}	Standard reaction enthalpy at 298 K (kJ/mol or MJ/kg)
ΔH_{gas}	Sensible enthalpy of gas mixture (MJ/kg)
$K_{eq,i}$	Equilibrium constant for reaction i (–)
LHV	Lower heating value (MJ/kg)
P	Pressure (bar or kPa)
Q	Heat stream (MJ/kg or MW)
R	Universal gas constant (kJ/mol K)
T	Temperature (K or°C)
W	Specific work (kJ/kg$_{air}$)
f_i	Fugacity of component i (bar)
$k_{ex,q}$	Coefficient of exergy transformation (–)
$k_{q(I)}$	Coefficient of heat transformation based on LHV (–)
$k_{q(II)}$	Coefficient of heat transformation based on GAH (–)
m	Mass flow rate (kg/s)
n_i	Number of moles of component i (mol)
q_{rec}	Recuperated heat per unit mass (kJ/kg)
y_i	Mole fraction of component i (–)

Greek Symbols

β	Steam-to-methane ratio (–)
η	Efficiency (–)
λ	Excess air ratio (–)
μ	Chemical potential (kJ/mol)
π	Pressure ratio (–)
Φ	Equivalence ratio (–)

Subscripts

0	Reference (ambient) condition
eq	Equilibrium
exh	Exhaust
fuel	Primary fuel
in	Inlet
out	Outlet
pf	Primary fuel
prod	Reaction products
rec	Recuperated
sol	Solar

CRediT authorship contribution statement

Dmitry Pashchenko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dmitry Pashchenko reports financial support was provided by Guangdong Technion-Israel Institute of Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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